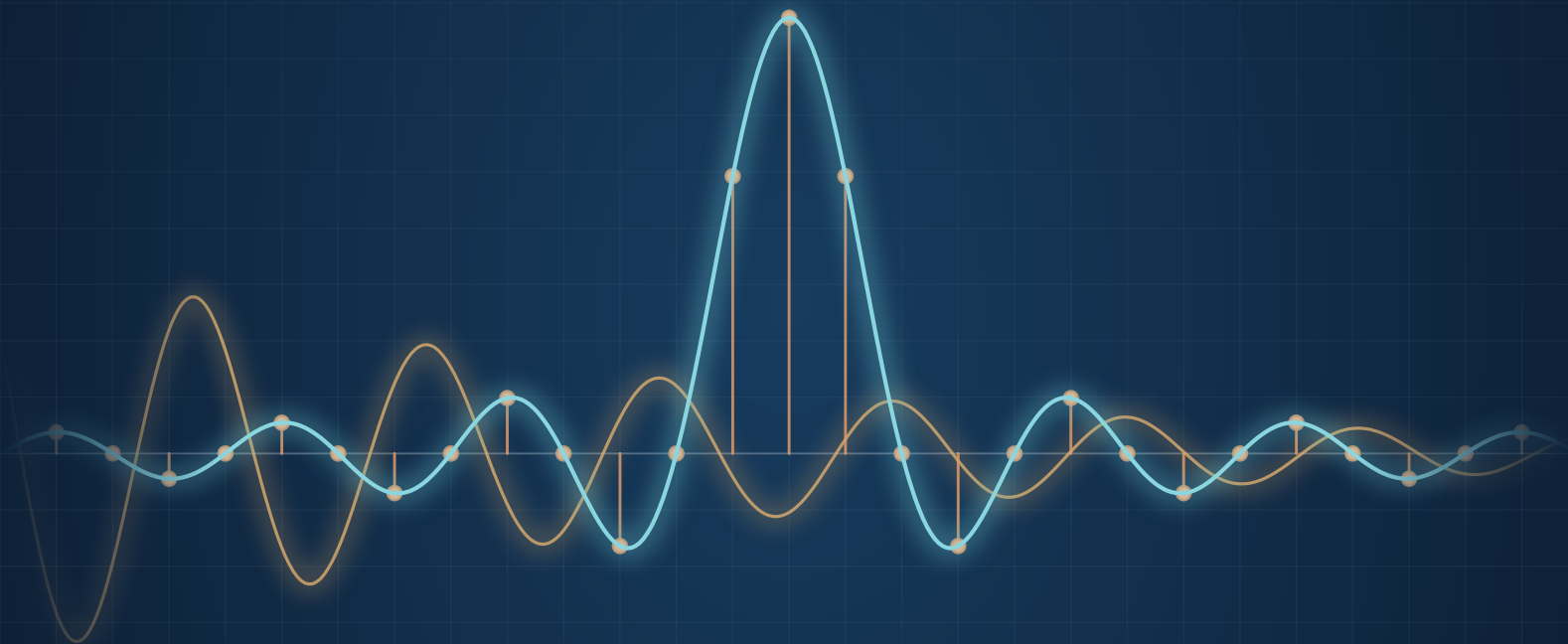




SIGNALS AND SYSTEMS

Signals, Systems and Frequency-Domain Analysis

Lecture Notes



Contents

1	Signals	OW CH1.1–1.4
	Energy and power. Shifting, reversal and scaling. Periodicity. Impulses and steps. Complex exponentials.	
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7	Sampling and aliasing	OW CH7.1–7.3
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	Everything from Chapters 1 to 7 on two pages.	

How to read these notes

These notes teach the full course at undergraduate level. They assume calculus, complex numbers and basic circuits. They define signals, classify systems and show why one function describes a linear time-invariant system. They then represent signals with complex exponentials and explain how sampling changes a signal.

Read the chapters in order because each chapter uses the one before it. First, a signal is defined as a function. Next, a system is defined as a map on functions. A linear time-invariant system is then described by one function. The frequency-domain chapters use this function to turn convolution into multiplication.

Each topic follows the same teaching order. A picture introduces the purpose of the idea. A definition and an equation state it exactly. A short derivation names each step. A worked example applies the method and ends with a check. Each worked example uses five headings: Given, Find, Method, Solution, Check. Try the Check step before you read it.

Two conventions apply throughout the notes. Energy and power are **normalised**: the resistance is $1\ \Omega$, so instantaneous power is $|x|^2$. The imaginary unit is j .

The third contents column gives textbook references. For example, **OW CH1.1–1.4** points to the matching material in Oppenheim and Willsky, *Signals and Systems*, second edition. The **OW** mark separates a textbook address from a course address. The numbering systems differ: these notes introduce the continuous-time Fourier transform in chapter 5, while the textbook introduces it in chapter 4.

Signals

A signal represents information as a function. This chapter defines continuous-time and discrete-time signals. It then shows how to calculate energy and power, change the time axis, test periodicity and use impulses and complex exponentials.

1.1 What a signal is

A signal is a physical quantity that varies and carries information. Mathematically, it is a function of one or more independent variables.

The independent variable in this course is usually time. An image instead uses two space variables. The same signal operations apply to those variables.

Continuous-time and discrete-time signals use different domains. Keep their notation separate.

CONTINUOUS TIME

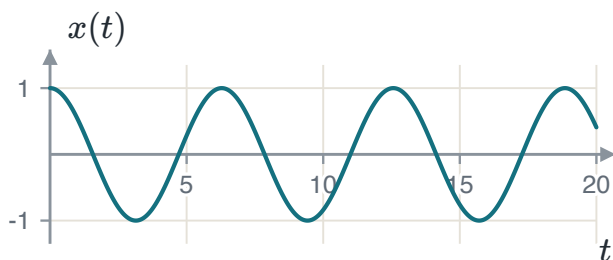
$$x(t), \quad t \in \mathbb{R}$$

Round brackets. The signal has a value at every real instant.

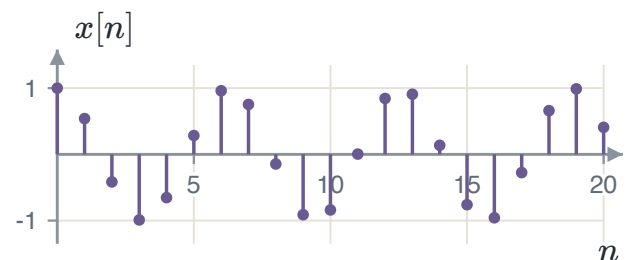
DISCRETE TIME

$$x[n], \quad n \in \mathbb{Z}$$

Square brackets. The signal has a value only at integer indices. Here n is an integer **index**, not a time in seconds.



A continuous-time signal is drawn as an unbroken curve.



A discrete-time signal is drawn with stems. The dots are the signal values. No signal value exists between adjacent integer indices.

NOTATION

Writing $x[t]$ or $x(n)$ states the wrong domain. The domain determines the periodicity test, the convolution limits and the transform properties.

1.2 Signal energy and power

Signal energy and power come from circuit power. If $v(t)$ is the voltage across a resistance R , use Ohm's law to write the instantaneous power as

$$p(t) = v(t) i(t) = v(t) \left(\frac{v(t)}{R} \right) = \frac{1}{R} v^2(t).$$

To find the energy over a time interval, integrate the instantaneous power:

$$E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t) dt.$$

From this point onward, set $R = 1 \Omega$ and use $|x(t)|^2$. This is the normalised convention. Restore the factor $1/R$ when a physical calculation uses a different resistance.

We use the modulus because a signal may be complex valued, and $|x(t)|^2 = x(t) x^*(t)$ is the quantity that is real and never negative. Writing $x^2(t)$ instead is correct only for real signals.

Total energy

TOTAL ENERGY OVER ALL TIME

$$E_{\infty} = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E_{\infty} = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The integral or sum may not converge. Because E_{∞} is a limit, non-convergence means that the signal has no finite total energy. Calculate average power next before classifying the signal.

Average power

Average power measures energy per unit time. For each finite interval, divide its energy by the interval length. Then let the interval grow.

AVERAGE POWER OVER ALL TIME

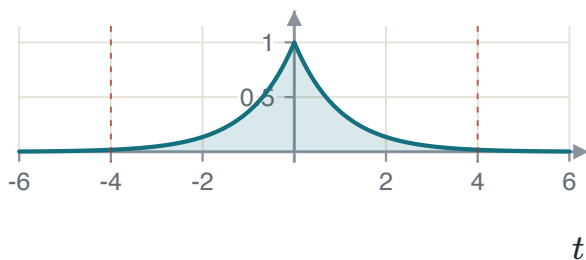
$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

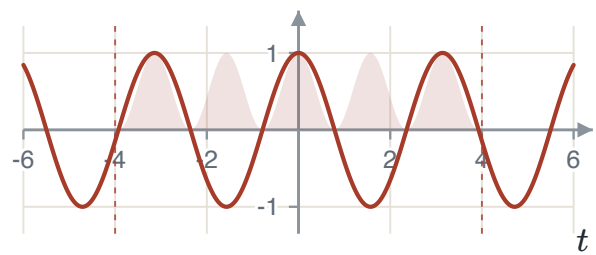
In discrete time the divisor is $2N + 1$, because that is the number of samples between $-N$ and $+N$ inclusive.

TAKE THE LIMIT OF THE RATIO

Calculate the energy-to-duration ratio for each finite interval. The ratio may approach any non-negative value, including zero.



Converging. As the window grows, the shaded area approaches a finite limit.



Diverging. Every period adds the same area, so $E_{\infty} \rightarrow \infty$ and only the average power is meaningful.

1.3 Energy signals, power signals, and neither

TYPE	CONDITION	TYPICAL SIGNALS
Energy signal	$E_{\infty} < \infty$ and $P_{\infty} = 0$	Finite pulses and decaying responses
Power signal	$E_{\infty} \rightarrow \infty$ and $0 < P_{\infty} < \infty$	Constants, sinusoids and other continuing signals
Neither	$E_{\infty} \rightarrow \infty$ and $P_{\infty} \rightarrow \infty$	Signals that grow without bound, such as $t u(t)$

The two energy-signal conditions are not independent. If E_{∞} is finite, the energy in every finite window is at most E_{∞} . Dividing this bound by $2T$ makes P_{∞} approach zero. The power condition follows from the energy condition.

$$\begin{aligned}
 E_T &= \int_{-T}^T |x(t)|^2 dt \leq E_\infty \\
 0 \leq P_\infty &= \lim_{T \rightarrow \infty} \frac{E_T}{2T} \leq \lim_{T \rightarrow \infty} \frac{E_\infty}{2T} = 0 \\
 \therefore P_\infty &= 0.
 \end{aligned}$$

EXAMPLE 1.1

GIVEN $x(t) = 1$ for $0 \leq t \leq 1$, and $x(t) = 0$ otherwise.

FIND Is this an energy signal or a power signal?

METHOD Find E_∞ first because finite energy implies zero average power. This result then determines the class.

SOLUTION Start with the definitions and then use the support of the signal:

$$\begin{aligned}
 E_\infty &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\
 &= \int_0^1 1^2 dt \\
 &= t \Big|_0^1 = 1 < \infty.
 \end{aligned}$$

For every $T \geq 1$, the window contains the whole pulse. Therefore

$$\begin{aligned}
 P_\infty &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^1 1 dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} = 0.
 \end{aligned}$$

So $x(t)$ is an **energy signal**, with $E_\infty = 1$ J.

CHECK Halving the amplitude must divide the energy by four. Direct calculation gives $\int_0^1 (1/2)^2 dt = 1/4$, so the result has the required quadratic dependence on amplitude.

EXAMPLE 1.2

GIVEN $x[n] = 4$ for every integer n .

FIND Is this an energy signal or a power signal?

METHOD Each sample contributes the same positive energy, so E_∞ diverges. Calculate P_∞ next and use the exact count $2N + 1$.

SOLUTION Each sample contributes $|4|^2 = 16$, so

$$\begin{aligned} E_\infty &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\ &= \sum_{n=-\infty}^{\infty} |4|^2 \\ &= \sum_{n=-\infty}^{\infty} 16 \rightarrow \infty. \end{aligned}$$

The finite window from $-N$ to N contains $2N + 1$ samples. Hence

$$\begin{aligned} P_\infty &= \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N |x[n]|^2 \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N 16 \\ &= \lim_{N \rightarrow \infty} \frac{(2N + 1)16}{2N + 1} \\ &= \lim_{N \rightarrow \infty} 16 = 16. \end{aligned}$$

So $x[n]$ is a **power signal**, with $P_\infty = 16$.

CHECK A constant of amplitude A must have $P_\infty = A^2$. The factor $2N + 1$ cancels exactly, which confirms the sample count was right.

DO NOT INFER POWER FROM ENERGY ALONE

Infinite energy does not imply infinite average power. For the constant sequence, both the energy sum and the sample count grow at the same rate. Their ratio approaches 16.

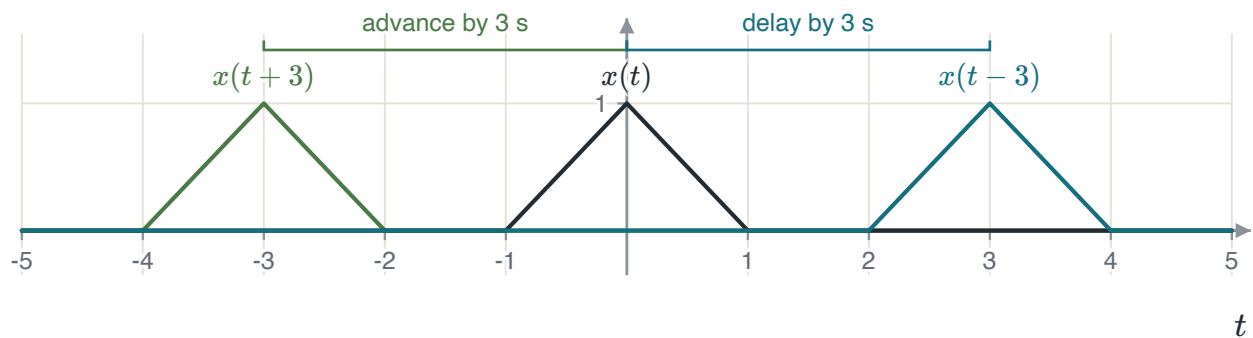
1.4 Time shifting

TIME SHIFT

$$x(t) \rightarrow x(t - t_0)$$

If $t_0 > 0$ the signal is **delayed** and moves right. If $t_0 < 0$ it is **advanced** and moves left.

Read the argument before moving the graph. At time t , the shifted signal uses the original value at $t - t_0$. For $t_0 > 0$, it reaches each original value t_0 seconds later. The graph therefore moves right.



A time shift changes the signal position but not its shape.

Discrete time is identical: $x[n] \rightarrow x[n - n_0]$ with n_0 an integer. A one-sample delay, $x[n - 1]$, is the basic memory element of every difference equation in Chapter 3.

1.5 Time reversal and time scaling

TIME REVERSAL

$$x(t) \rightarrow x(-t) \quad (x[n] \rightarrow x[-n])$$

Reflect the signal about the vertical axis.

TIME SCALING

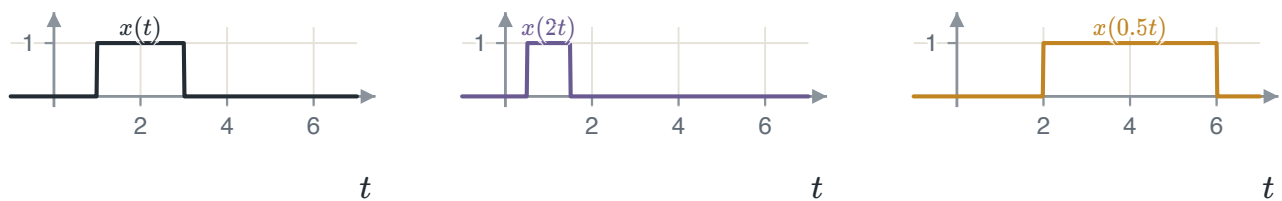
$$y(t) = x(at), \quad a > 0$$

If $a > 1$ the signal is compressed and speeded up. If $0 < a < 1$ it is stretched and slowed down.

Write the scaled signal as a new signal $y(t)$. Writing $x(t) = x(at)$ would force $a = 1$.

Find the new support by solving $at \in [\alpha, \beta]$. If x is non-zero only on $[\alpha, \beta]$, then $x(at)$ is non-zero only on $[\alpha/a, \beta/a]$. The width is divided by a .

$$at \in [\alpha, \beta] \iff \alpha \leq at \leq \beta \iff \frac{\alpha}{a} \leq t \leq \frac{\beta}{a}, \quad a > 0.$$



DISCRETE-TIME SCALING NEEDS SEPARATE RULES

$x[2n]$ keeps only the even-indexed samples and discards the odd-indexed samples. This operation is decimation and may lose information. The expression $x[n/2]$ is undefined at odd n unless an interpolation rule supplies those values. Continuous-time scaling does not have this integer-index restriction.

1.6 Combining a shift and a scale

To construct $x(at - b)$ from $x(t)$, use an intermediate signal so that each operation is explicit.

SHIFT, THEN SCALE

$$(1) \quad v(t) = x(t - b) \qquad (2) \quad y(t) = v(at) = x(at - b)$$

Shift by b first. Then scale the result by a .

WHY THE ORDER MATTERS

Scaling first gives $w(t) = x(at)$. Shifting that by b gives $w(t - b) = x(a(t - b)) = x(at - ab)$. Unless $a = 1$ this is a different signal: it is shifted by b instead of by b/a .

EXAMPLE 1.3

GIVEN $x(t)$ is zero for $t < -2$, equal to 1 on $[-2, 0]$, equal to 2 on $[0, 2]$, and falls linearly from 2 to 0 on $[2, 4]$.

FIND Plot $x(3t - 5)$.

METHOD Here $a = 3$ and $b = 5$. Shift right by 5, then compress by 3.

SOLUTION For the shift, each original corner c moves to $c + 5$, so $-2, 0, 2, 4$ become $3, 5, 7, 9$. For the final signal, solve the argument equation for each original corner:

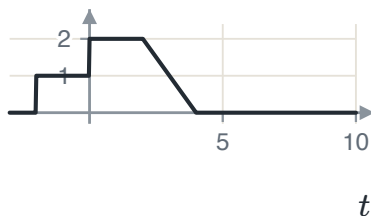
$$3t - 5 = c \implies t = \frac{c + 5}{3}.$$

Thus

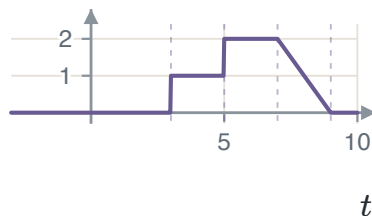
$$c = -2, 0, 2, 4 \implies t = 1, \frac{5}{3}, \frac{7}{3}, 3.$$

These are the four corners of $y(t) = x(3t - 5)$.

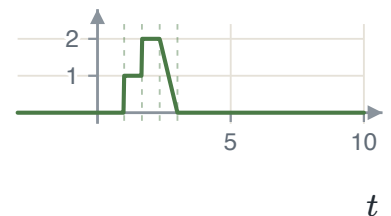
CHECK The original support has width 6. Compression by $a = 3$ must give width 2, and the result has support $[1, 3]$. Also set $at - b = 0$. This gives $t = b/a = 5/3$, where $y(t)$ must equal $x(0) = 2$.



$x(t)$



$v(t) = x(t - 5)$



$y(t) = x(3t - 5)$

1.7 Periodic signals

PERIODICITY

$$x(t) = x(t + T) \quad \text{for all } t \in \mathbb{R}, \text{ for some } T > 0$$

$$x[n] = x[n + N] \quad \text{for all } n \in \mathbb{Z}, \text{ for some integer } N > 0$$

A signal that is not periodic is **aperiodic**.

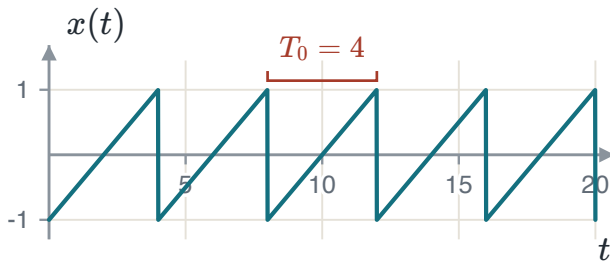
The definition has two requirements. A positive period must exist, and the equality must hold for every t or n . In continuous time, T may be any positive real number. In discrete time, N must be a positive integer because the sequence exists only at integer indices. A value such as 3.5 samples cannot be a period.

If T is a period then so is $2T$, $3T$, and so on. The **fundamental period** T_0 is the smallest positive period. The same applies to N_0 in discrete time.

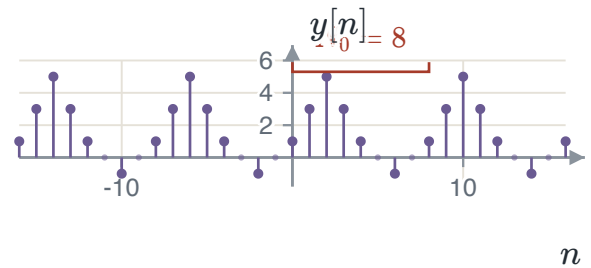
FUNDAMENTAL FREQUENCY

$$\omega_0 = \frac{2\pi}{T_0} \quad \text{and} \quad \omega_0 = \frac{2\pi}{N_0}$$

Without the word *smallest*, ω_0 would not be well defined.



Periods are 4, 8, 12, ... The fundamental period is $T_0 = 4$.



Periods are 8, 16, 24, ... The fundamental period is $N_0 = 8$.

1.8 Even and odd signals

A signal is **even** if $x(t) = x(-t)$, and **odd** if $x(t) = -x(-t)$. The same definitions apply to $x[n]$ with n in place of t .

To test an odd signal at the origin, set $t = 0$ in the definition. This gives $x(0) = -x(0)$, so $x(0) = 0$. Therefore any signal with $x(0) \neq 0$ is not odd.

Every signal splits into an even part and an odd part, in exactly one way:

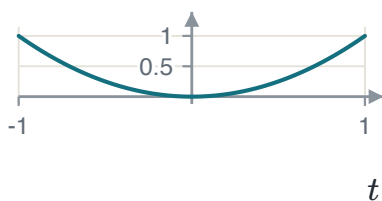
EVEN AND ODD PARTS

$$\mathcal{E}\{x(t)\} = \frac{1}{2}[x(t) + x(-t)], \quad \mathcal{O}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$$

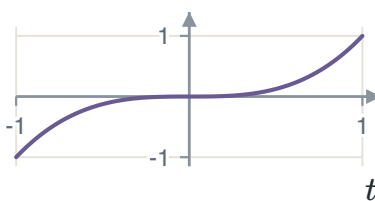
$$x(t) = \mathcal{E}\{x(t)\} + \mathcal{O}\{x(t)\}$$

Adding the two definitions returns x exactly.

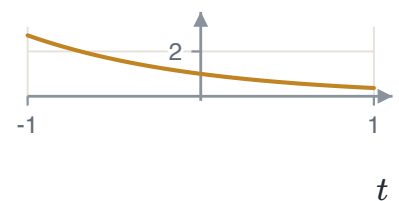
$$\begin{aligned} \mathcal{E}\{x(t)\} + \mathcal{O}\{x(t)\} &= \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)] \\ &= \frac{1}{2}[2x(t)] = x(t). \end{aligned}$$



t^2 is even.



t^3 is odd.



e^{-t} is neither.

1.9 The unit impulse and the unit step in discrete time

DEFINITIONS

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

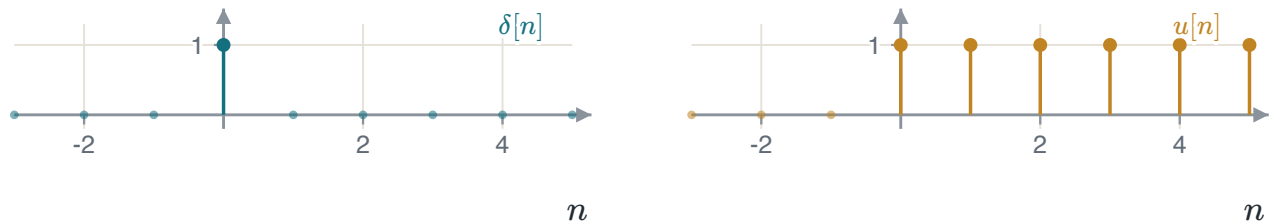
Both are ordinary sequences. Nothing infinite happens here.

A first difference converts the step into the impulse. A running sum converts the impulse back into the step. These operations are the discrete-time forms of differentiation and integration.

FIRST DIFFERENCE AND RUNNING SUM

$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots$$



Sampling and sifting

Sampling and sifting both use a shifted impulse, but they return different types of result.

SAMPLING PROPERTY

$$x[n] \delta[n - n_0] = x[n_0] \delta[n - n_0]$$

Both sides are **sequences**. Multiplying by a shifted impulse keeps one sample and deletes the rest.

SIFTING PROPERTY

$$x[n_0] = \sum_{n=-\infty}^{\infty} x[n] \delta[n - n_0]$$

The right-hand side is a **number**. Sifting is sampling followed by a sum.

EXAMPLE 1.4

GIVEN $x[0] = 1$, $x[1] = 2$, $x[2] = 3$, and $x[n] = 0$ elsewhere. Take $n_0 = 2$.

FIND The sampled sequence and the sifted value.

METHOD Multiply by $\delta[n - 2]$ to keep the sample at $n = 2$. Then sum the sampled sequence to sift out its value.

SOLUTION Sampling: $x[n] \delta[n - 2] = x[2] \delta[n - 2] = 3 \delta[n - 2]$, a single stem of height 3 at $n = 2$.

Sifting: $\sum_n x[n] \delta[n - 2] = x[0] \cdot 0 + x[1] \cdot 0 + x[2] \cdot 1 = 3$.

CHECK The sampled result is a sequence. The sifted result is the number 3.

Apply the sampling property at every integer shift and add the results. This produces the representation used to derive convolution in Chapter 3.

REPRESENTATION PROPERTY

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Any sequence is a sum of weighted, shifted impulses. The weights are the sample values themselves.

1.10 The unit impulse and the unit step in continuous time

The continuous-time step is $u(t) = 1$ for $t \geq 0$ and 0 otherwise. The continuous-time impulse is not an ordinary function.

DEFINE THE IMPULSE BY ITS ACTION

An ordinary function that is zero except at one point has integral 0, not 1. The impulse is therefore a **distribution**, also called a generalized function. A distribution is defined by how it acts inside an integral.

DEFINING PROPERTY (SIFTING)

$$x(t_0) = \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt$$

This is the definition. Everything else about δ follows from it.

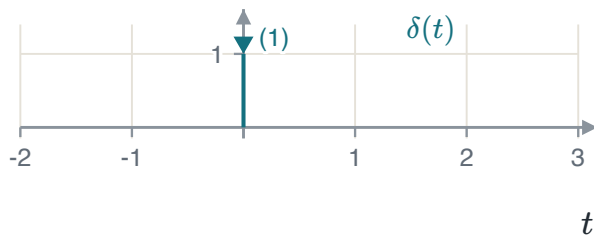
To picture the impulse, use a rectangle of width ε and height $1/\varepsilon$. Its area is 1. Then let $\varepsilon \rightarrow 0$. The rectangles do not approach an ordinary function at each fixed t . Their integrals against a continuous test function do approach the sifting result.

The impulse and the step are related by differentiation and integration:

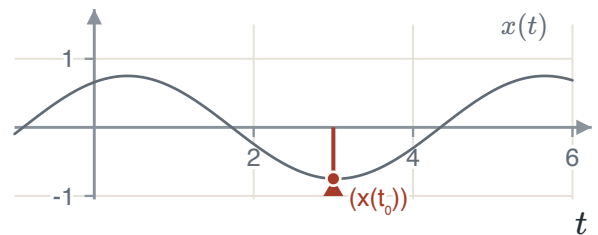
$$\delta(t) = \frac{d}{dt}u(t), \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau.$$

SAMPLING PROPERTY, CONTINUOUS TIME

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$



The arrow height shows the **area**, never a value of a function.



Sifting: the impulse at t_0 is scaled by the value of x there, and integrating returns that number.

1.11 Complex exponentials in continuous time

DEFINITION

$$x(t) = C e^{at}, \quad C, a \in \mathbb{C}$$

Complex exponentials describe growth, decay and oscillation in one form. The real and imaginary parts of a determine which behaviour occurs.

Case 1: C and a real

If $a < 0$ the signal decays. If $a > 0$ it grows. If $a = 0$ it is the constant C . A larger $|a|$ makes the decay or growth faster.

Case 2: $a = j\omega_0$ purely imaginary

Write $C = Ae^{j\theta}$ and use Euler's relation $e^{jx} = \cos x + j \sin x$:

$$\begin{aligned} x(t) &= C e^{j\omega_0 t} \\ &= A e^{j\theta} e^{j\omega_0 t} \\ &= A e^{j(\omega_0 t + \theta)} \\ &= A \cos(\omega_0 t + \theta) + j A \sin(\omega_0 t + \theta). \end{aligned}$$

Here A is the amplitude, ω_0 is the angular frequency in rad/s, and θ is the phase in radians.

To test periodicity, impose the condition $x(t) = x(t + T)$ and solve for T :

$$\begin{aligned}
x(t+T) &= x(t) \\
Ae^{j[\omega_0(t+T)+\theta]} &= Ae^{j(\omega_0 t + \theta)} \\
e^{j\omega_0 T} &= 1 \\
\omega_0 T &= 2\pi k, \quad k \in \mathbb{Z} \\
T &= \frac{2\pi k}{\omega_0}.
\end{aligned}$$

RESULT

Taking $k = 1$ gives the fundamental period $T_0 = 2\pi/\omega_0$. Every continuous-time complex exponential with $\omega_0 \neq 0$ is periodic. There is no extra condition.

EXAMPLE 1.5

GIVEN $x(t) = e^{j0.5\pi t}$.

FIND The fundamental period.

METHOD Use $T_0 = 2\pi/\omega_0$ because the exponent is purely imaginary and $\omega_0 \neq 0$.

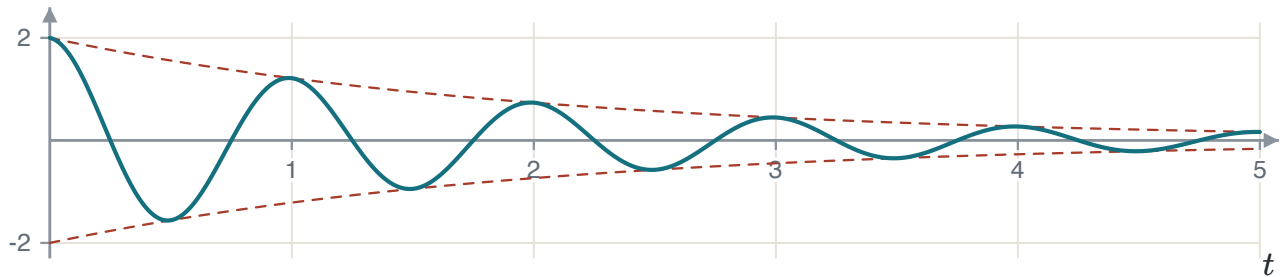
SOLUTION $T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{0.5\pi} = 4$ seconds.

CHECK $0.5\pi \times 4 = 2\pi$, so the phase increases by one full turn over the calculated period.

Case 3: $a = r + j\omega_0$

$$\begin{aligned}
x(t) &= Ae^{j\theta} e^{(r+j\omega_0)t} \\
&= Ae^{rt} e^{j(\omega_0 t + \theta)} \\
&= Ae^{rt} \cos(\omega_0 t + \theta) + jAe^{rt} \sin(\omega_0 t + \theta).
\end{aligned}$$

The curves $\pm Ae^{rt}$ form an envelope that bounds the sinusoid. If $r < 0$, the oscillation is damped. If $r > 0$, it grows. If $r = 0$, it is sustained.



A damped sinusoid, $\text{Re}\{2e^{-0.5t}e^{j2\pi t}\}$, with its envelope shown dashed.

1.12 Complex exponentials in discrete time

DEFINITION

$$x[n] = C e^{\beta n} = C \alpha^n, \quad \alpha = e^\beta$$

Use the power form in discrete time because solutions of difference equations have this form.

For real C and α : if $0 < \alpha < 1$ the sequence decreases; if $\alpha > 1$ it increases. For complex $C = |C|e^{j\theta}$ and $\alpha = |\alpha|e^{j\omega_0}$:

$$\begin{aligned}
x[n] &= C\alpha^n \\
&= |C|e^{j\theta} (|\alpha|e^{j\omega_0})^n \\
&= |C||\alpha|^n e^{j(\omega_0 n + \theta)} \\
&= |C||\alpha|^n \cos(\omega_0 n + \theta) + j|C||\alpha|^n \sin(\omega_0 n + \theta).
\end{aligned}$$

THE BOUNDARY MOVES

In continuous time the growth-decay boundary is $\operatorname{Re}\{a\} = 0$, the imaginary axis. In discrete time it is $|\alpha| = 1$, the unit circle. The map between them is $\alpha = e^\beta$.

1.13 When is a discrete-time exponential periodic?

A discrete-time period must be an integer. Apply $x[n] = x[n + N]$ to $x[n] = Ce^{j\omega_0 n}$ and solve for that integer:

$$\begin{aligned} x[n + N] &= x[n] \\ Ce^{j\omega_0(n+N)} &= Ce^{j\omega_0 n} \\ e^{j\omega_0 N} &= 1 \\ \omega_0 N &= 2\pi k, \quad k \in \mathbb{Z} \\ N &= \frac{2\pi k}{\omega_0}. \end{aligned}$$

PERIODICITY CONDITION

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q}$$

N must be an integer. This is possible only when $\omega_0/2\pi$ is rational. If the ratio is irrational, no integer N satisfies the periodicity condition and the sequence is aperiodic.

EXAMPLE 1.6

GIVEN $x[n] = e^{j(3\pi/5)n}$.

FIND The fundamental period N_0 .

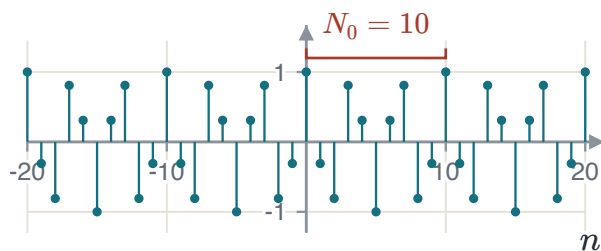
METHOD Use $N = 2\pi k/\omega_0$ and take the smallest k that makes N an integer.

SOLUTION Substitute the frequency before choosing k :

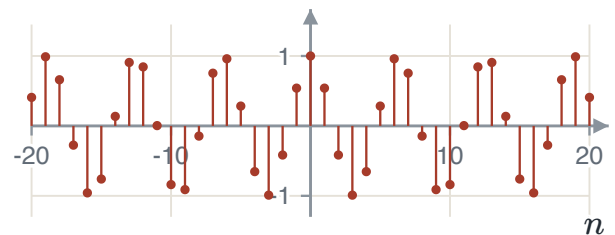
$$\begin{aligned} N &= \frac{2\pi k}{\omega_0} \\ &= \frac{2\pi k}{3\pi/5} \\ &= \frac{10}{3}k. \end{aligned}$$

The smallest positive integer k giving an integer N is $k = 3$, so $N_0 = 10$.

CHECK $\omega_0 N_0 = \frac{3\pi}{5} \times 10 = 6\pi = 2\pi \times 3$. The phase therefore changes by three full turns in ten samples.



$\operatorname{Re}\{e^{j3\pi n/5}\}$ repeats every 10 samples.



$\cos(n)$ has $\omega_0 = 1$, so $\omega_0/2\pi$ is irrational. The samples never repeat exactly.

FREQUENCIES SEPARATED BY 2π GIVE THE SAME SEQUENCE

$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$ for every integer n . Therefore discrete-time frequencies that differ by 2π produce identical samples. This identity makes the discrete-time Fourier transform 2π -periodic and leads to aliasing.

1.14 Chapter summary

1. E_∞ and P_∞ are limits. Finite energy forces zero average power. Finite non-zero average power forces infinite energy. Unbounded growth gives neither.
2. To build $x(at - b)$, shift by b first, then scale by a . The other order gives $x(at - ab)$.
3. T_0 and N_0 are the smallest positive periods, and $\omega_0 = 2\pi/T_0 = 2\pi/N_0$.
4. $\delta[n]$ is an ordinary sequence. $\delta(t)$ is a distribution defined by sifting.
5. Every signal is a sum of weighted, shifted impulses: $x[n] = \sum_k x[k]\delta[n - k]$.
6. A discrete-time exponential is periodic only when $\omega_0/2\pi$ is rational. A continuous-time one always is.

Exercises

1.1 Classify $x(t) = e^{-3t}u(t)$ as an energy signal, a power signal, or neither. Give the value of whichever quantity is finite.

Answer: Energy signal, $E_\infty = 1/6$.

1.2 For $x[n]$ non-zero only on $-1 \leq n \leq 1$ with values 1, 2, 1, sketch $x[n + 4]$ and $x[n - 5]$ and give the support of each.

Answer: Supports $-5 \leq n \leq -3$ and $4 \leq n \leq 6$.

1.3 A signal $x(t)$ is non-zero only on $[-1, 5]$. On what interval is $x(2t + 3)$ non-zero?

Answer: $[-2, 1]$.

1.4 Find the fundamental period of $x[n] = \cos(4\pi n/7)$, or show that it is aperiodic.

Answer: $N_0 = 7$.

1.5 Show that if $x(t)$ is odd then $\int_{-a}^a x(t) dt = 0$ for every $a > 0$.

1.6 Evaluate $\int_{-\infty}^{\infty} (t^2 + 1) \delta(t - 2) dt$ and $\sum_{n=-\infty}^{\infty} 2^{-n} \delta[n - 3]$.

Answer: 5 and $1/8$.

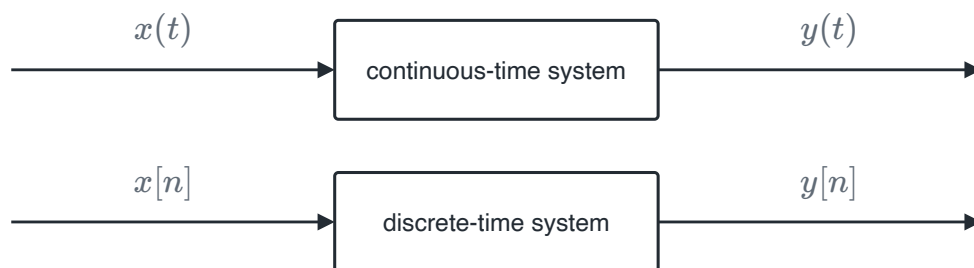
Systems and their properties

This chapter gives tests for memory, invertibility, causality, stability, time invariance and linearity. Each test describes a system through its input and output signals.

2.1 What a system is

A system is a rule that turns an input signal into an output signal. The rule is deterministic: the same input always gives the same output.

Write the rule as an operator, $y = S\{x\}$. The operator acts on the whole input signal. This form lets us test which input times affect an output and how the system responds to a time shift.



The operator description applies in continuous time and discrete time.

2.2 Memory

CRITERION

A system is **memoryless** if the output at time t (or n) depends only on the input at that same time.

SYSTEM	VERDICT	REASON
$y(t) = [2x(t) - x^2(t)]^2$	memoryless	Only $x(t)$ appears. There is no $x(t+1)$ or $x(t-2)$ term.
$y[n] = x[n]$	memoryless	The identity system.
$y[n] = x[n-1]$	has memory	The output at n uses the sample at $n-1$.
$y[n] = x[n] + y[n-1]$	has memory	See the derivation below.

For the last system, assume initial rest and expose the memory by substituting the feedback relation repeatedly:

$$\begin{aligned}
 y[n] &= x[n] + y[n-1] \\
 &= x[n] + x[n-1] + y[n-2] \\
 &= x[n] + x[n-1] + x[n-2] + y[n-3] \\
 &\vdots \\
 &= x[n] + x[n-1] + x[n-2] + \dots \\
 &= \sum_{k=0}^{\infty} x[n-k].
 \end{aligned}$$

The result uses $x[n-k]$ for every $k \geq 0$, so the output depends on the whole input history. Output feedback can therefore give a system memory.

CIRCUIT EXAMPLES

A resistor, $v(t) = Ri(t)$, is memoryless because the voltage at t uses only the current at t . A capacitor is not memoryless because $v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$ uses the current history.

2.3 Invertibility

CRITERION

A system is **invertible** if distinct inputs always produce distinct outputs. The map must be one-to-one.

To prove invertibility, find a formula that recovers every input from its output. To disprove invertibility, find two distinct inputs that give the same output.

EXAMPLE 2.1

GIVEN $y(t) = [\cos(t) + 2]x(t)$.

FIND Is the system invertible?

METHOD Solve the system rule for $x(t)$ and verify that the divisor never vanishes.

SOLUTION First bound the gain:

$$-1 \leq \cos(t) \leq 1 \implies 1 \leq \cos(t) + 2 \leq 3.$$

It is never zero. Now solve the system equation for the input:

$$y(t) = [\cos(t) + 2]x(t) \implies x(t) = \frac{y(t)}{\cos(t) + 2}.$$

The input is recovered uniquely, so the system is invertible.

CHECK

Substituting back gives $[\cos t + 2] \frac{y(t)}{\cos t + 2} = y(t)$. Had the gain been $\cos(t)$ alone, the system would fail at every zero of the cosine.

EXAMPLE 2.2

GIVEN $y(t) = x^2(t)$.

FIND Is the system invertible?

METHOD Search for two distinct inputs that lose the same information under squaring.

SOLUTION No. Take $x_1(t) = 1$ and $x_2(t) = -1$ for all t . Then

$$S\{x_1\} = 1^2 = 1, \quad S\{x_2\} = (-1)^2 = 1.$$

The inputs are different, but their outputs are equal.

CHECK

One counterexample is enough. The sign information is destroyed and cannot be recovered from y alone.

NOT AN INVERSION FORMULA

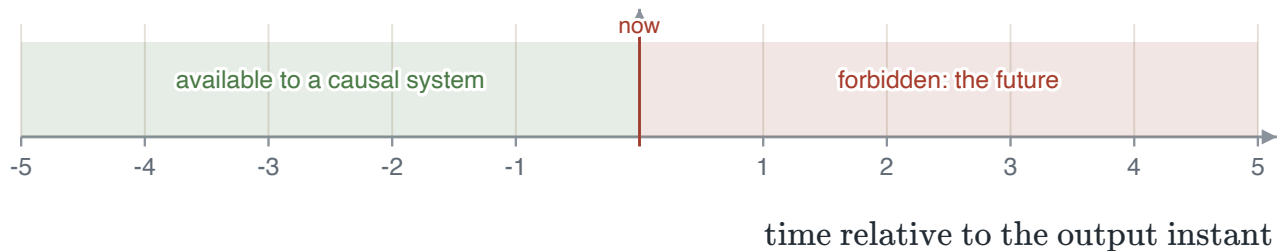
Writing $x(t) = \sqrt{y(t)}$ does not invert $y = x^2$. It picks one of two possible inputs by convention. An inversion formula must return the actual input, for every admissible input.

2.4 Causality

CRITERION

A system is **causal** if the output at time t (or n) depends only on inputs at times $\leq t$: the present and the past.

SYSTEM	VERDICT	REASON
$y[n] = x[n - 1]$	causal	Uses only a past sample.
$y[n] = x[n] + x[n + 1]$	not causal	$x[n + 1]$ is a future sample.
$y(t) = \int_{-\infty}^t x(\tau) d\tau$	causal	The upper limit is t , so only $\tau \leq t$ contributes.
$y[n] = x[-n]$	not causal	$y[-1] = x[1]$. The output at $n = -1$ needs a future input.
$y(t) = x(t) \cos(t + 1)$	causal	Uses only $x(t)$. The factor $\cos(t + 1)$ is a known function of t , fixed in advance, not a future input.



Causality restricts which part of the input axis the output is allowed to consult.

A real-time system must be causal because future input values are not available. A non-causal system can be used when the complete signal has already been stored, as in offline audio or image processing.

2.5 Stability

A signal is **bounded** if there is a constant $B < \infty$ with $|x(t)| < B$ for all t .

CRITERION

A system is **BIBO stable** if every bounded input produces a bounded output. Bounded Input, Bounded Output.

PROOF AND COUNTEREXAMPLE

To prove stability, derive an output bound for every bounded input. To disprove stability, find one bounded input that produces an unbounded output. One bounded example cannot prove stability.

EXAMPLE 2.3

GIVEN $y(t) = 2x^2(t - 1) + x(3t)$.

FIND Is the system stable?

METHOD Assume $|x(t)| \leq B < \infty$ and bound $|y|$ with the triangle inequality.

SOLUTION Use the system rule, then apply the triangle inequality, then use the input bound:

$$\begin{aligned}
 |y(t)| &= |2x^2(t - 1) + x(3t)| \\
 &\leq |2x^2(t - 1)| + |x(3t)| \\
 &= 2|x(t - 1)|^2 + |x(3t)| \\
 &\leq 2B^2 + B < \infty.
 \end{aligned}$$

The output therefore has the finite bound $2B^2 + B$, so the system is stable.

CHECK Shifting and scaling the time axis cannot change the set of values a signal takes, so $x(t - 1)$ and $x(3t)$ are bounded by the same B .

EXAMPLE 2.4

GIVEN $y[n] = \sum_{k=-\infty}^n x[k]$, the accumulator.

FIND Is the system stable?

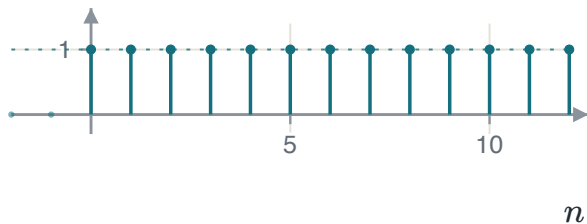
METHOD Look for one bounded input with an unbounded output.

SOLUTION Take $x[n] = u[n]$, so $|x[n]| \leq 1$. For $n \geq 0$,

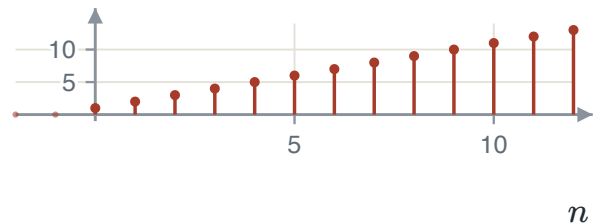
$$\begin{aligned} y[n] &= \sum_{k=-\infty}^n u[k] \\ &= \sum_{k=0}^n 1 \\ &= n + 1 \rightarrow \infty. \end{aligned}$$

The input is bounded and the output is unbounded. The system is **not** stable.

CHECK The input never exceeds 1 and the output grows without bound. That is exactly the failure BIBO stability forbids.



Bounded input $u[n]$.



Unbounded output $n + 1$.

CAUSAL DOES NOT MEAN STABLE

The accumulator is causal and unstable. The reversal $y[n] = x[-n]$ is stable and not causal. The six properties are independent except where a proof links them.

2.6 Time invariance

CRITERION

If $x(t) \rightarrow y(t)$, the system is **time invariant** when $x(t - t_0) \rightarrow y(t - t_0)$ for every t_0 .

The test is a comparison of two computed signals.

1. **Path 1.** Shift the input, then run the system: $y_2 = S\{x(t - t_0)\}$.
2. **Path 2.** Run the system, then shift the output: $y_1(t - t_0)$.

The system is time invariant exactly when the two agree, for every input and every t_0 .

EXAMPLE 2.5

GIVEN $y(t) = \sin(x(t))$.

FIND Is the system time invariant?

METHOD Compute the response to a shifted input and compare it with the shifted response.

SOLUTION Path 1 shifts the input and then applies the system:

$$x_2(t) = x_1(t - t_0) \implies y_2(t) = \sin(x_2(t)) = \sin(x_1(t - t_0)).$$

Path 2 applies the system and then shifts its output:

$$y_1(t) = \sin(x_1(t)) \implies y_1(t - t_0) = \sin(x_1(t - t_0)).$$

Thus $y_2(t) = y_1(t - t_0)$ for every input and every t_0 . The system is time invariant.

CHECK Both paths give the same expression for every input and every t_0 .

EXAMPLE 2.6

GIVEN $y[n] = n x[n]$.

FIND Is the system time invariant?

METHOD Look for a counterexample using an impulse.

SOLUTION Take $x_1[n] = \delta[n]$. The original output and its one-sample delay are

$$y_1[n] = n\delta[n] = 0, \quad y_1[n - 1] = 0.$$

Now delay the input first:

$$\begin{aligned} x_2[n] &= \delta[n - 1], \\ y_2[n] &= n\delta[n - 1] \\ &= 1 \cdot \delta[n - 1] = \delta[n - 1]. \end{aligned}$$

The identity $n\delta[n - 1] = \delta[n - 1]$ follows because the impulse is non-zero only at $n = 1$. Therefore $y_2[n] \neq y_1[n - 1]$, so the system is **not** time invariant.

CHECK The impulse isolates one index, so the unequal outputs form a complete counterexample.

WHEN TO APPLY THE TEST FIRST

An explicit time-dependent coefficient such as n or $\cos(t)$ usually breaks time invariance. A scaled input argument such as $x(3t)$ also changes input shifts. In either case, use the two-path test.

2.7 Linearity

CRITERION

$$a x_1 + b x_2 \longrightarrow a y_1 + b y_2 \quad \text{for all } a, b \in \mathbb{C}$$

where $x_1 \rightarrow y_1$ and $x_2 \rightarrow y_2$. This is additivity and homogeneity in one statement.

EXAMPLE 2.7

GIVEN $y(t) = 2\pi x(t)$.

FIND Is the system linear?

METHOD Apply the system to a weighted sum and compare the result with the same weighted sum of the two outputs.

SOLUTION Let $x_3 = ax_1 + bx_2$. Apply the system and distribute the constant:

$$\begin{aligned} S\{x_3\} &= S\{ax_1 + bx_2\} \\ &= 2\pi(ax_1 + bx_2) \\ &= a(2\pi x_1) + b(2\pi x_2) \\ &= a y_1 + b y_2. \end{aligned}$$

The required superposition equality holds, so the system is linear. It is also time invariant, so it is LTI.

CHECK The final expression is exactly $a S\{x_1\} + b S\{x_2\}$ for arbitrary inputs and scalars.

EXAMPLE 2.8

GIVEN $y[n] = (x[2n])^2$.

FIND Is the system linear?

METHOD Apply superposition and compare the powers and cross term with the required linear result.

SOLUTION Let $x_3[n] = ax_1[n] + bx_2[n]$. Combining first and then applying the system gives

$$\begin{aligned} y_3[n] &= (x_3[2n])^2 \\ &= (ax_1[2n] + bx_2[2n])^2 \\ &= a^2 x_1^2[2n] + 2ab x_1[2n]x_2[2n] + b^2 x_2^2[2n]. \end{aligned}$$

Applying the system first and then combining gives

$$a y_1[n] + b y_2[n] = a x_1^2[2n] + b x_2^2[2n].$$

The expressions differ: the first has a cross term and different powers of a and b . The system is **not** linear.

CHECK Scaling the input by a scales this output by a^2 , so homogeneity also fails.

ZERO-INPUT TEST

$y(t) = x(t) + 5$ is not linear because the zero input does not give the zero output. Every linear system must satisfy $S\{0\} = 0$.

2.8 How to classify an unfamiliar system

1. **Test time invariance.** Apply the two-path test when the rule contains an explicit t or n , a time-dependent gain, or a scaled argument.
2. **Test linearity.** Look for squares, products, $\sin(x)$, absolute values or saturation. Test homogeneity with one scalar first.
3. **Inspect every input argument.** An argument different from t or n gives memory. An argument later than the output time gives non-causality.

4. **Test stability.** Assume $|x| \leq B$ and derive a finite output bound. To disprove stability, choose a bounded input that makes the output unbounded.
5. **Test invertibility.** Find an inverse formula or two distinct inputs with the same output.

ONE GENERAL IMPLICATION

Memoryless implies causal. Establish the other properties independently. A causal system need not be stable, a linear system need not be time invariant, and a stable system need not be memoryless.

SYSTEM	MEMORYLESS	INVERTIBLE	CAUSAL	STABLE	TIME INV.	LINEAR
$y(t) = 2\pi x(t)$	yes	yes	yes	yes	yes	yes
$y[n] = x[n - 1]$	no	yes	yes	yes	yes	yes
$y(t) = x^2(t)$	yes	no	yes	yes	yes	no
$y[n] = n x[n]$	yes	no	yes	no	no	yes
$y[n] = \sum_{k \leq n} x[k]$	no	yes	yes	no	yes	yes
$y[n] = x[-n]$	no	yes	no	yes	no	yes
$y[n] = (x[2n])^2$	no	no	no	yes	no	no

Read each property down its column. The examples show why the six properties require separate tests.

Exercises

2.1 Classify $y(t) = x(t)u(t)$ against all six properties.

Answer: Memoryless, not invertible, causal, stable, not time invariant, linear.

2.2 Show that $y[n] = x[n] - x[n - 1]$ is invertible on inputs that are zero for $n < 0$, and give the inverse.

Answer: $x[n] = \sum_{k=0}^n y[k]$.

2.3 Give a system that is linear and causal but not stable, and one that is stable and time invariant but not linear.

2.4 Is $y(t) = x(t/2)$ time invariant? Prove or give a counterexample.

Answer: No. Path 1 gives $x(t/2 - t_0)$ and path 2 gives $x((t - t_0)/2)$.

Linear time-invariant systems

This chapter develops a direct way to find the output of a linear time-invariant system. Its impulse response describes the system, and convolution uses that response to find the output for any input.

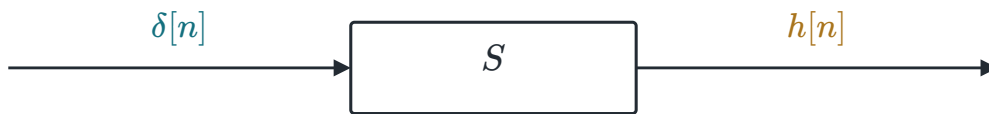
3.1 The impulse response

DEFINITION

The **impulse response** is the output when the input is a unit impulse: $x[n] = \delta[n]$ gives $y[n] = h[n]$. In continuous time, $x(t) = \delta(t)$ gives $y(t) = h(t)$.

For a general system, this experiment gives only one input-output pair. It does not determine the response to another input.

For a linear time-invariant system, one impulse response is enough. Time invariance gives the response to every shifted impulse. Linearity gives the response to every weighted sum of those impulses. Chapter 1 showed that every discrete-time signal can be written as such a sum. Therefore $h[n]$ determines the response to every input.



For an LTI system, one impulse experiment determines the response to every input.

3.2 Deriving the convolution sum

The purpose of this derivation is to obtain the output from $h[n]$. Start with the representation property from Chapter 1, then apply the system to both sides.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \longrightarrow S \longrightarrow y[n] = ?$$

1. **Time invariance.** Since $\delta[n] \rightarrow h[n]$, we also have $\delta[n-k] \rightarrow h[n-k]$.
2. **Homogeneity.** The number $x[k]$ does not depend on n , so $x[k]\delta[n-k] \rightarrow x[k]h[n-k]$.
3. **Additivity.** The response to the sum is the sum of the responses.

CONVOLUTION SUM

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Next, replace the summation index by $m = n - k$. This gives an equivalent form and shows that convolution is commutative:

$$\sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m] h[m].$$

CHECK THE SYSTEM BEFORE USING CONVOLUTION

Convolution gives the system output only when the system is linear and time invariant. The derivation uses time invariance to shift h and linearity to scale and add the responses. If either property fails, the convolution result is not the output of that system.

3.3 How to compute a convolution

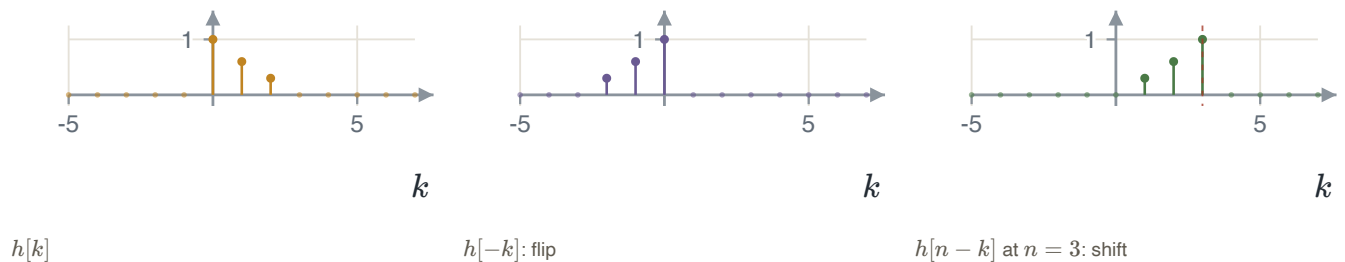
1. **Flip.** Reverse $h[k]$ to get $h[-k]$.

2. **Shift.** Move it by n to get $h[n - k]$.
3. **Multiply and add.** Form $x[k]h[n - k]$ and sum over k .
4. Repeat for every n .

Construct $h[n - k]$ in a fixed order. Treat k as the variable, shift $h[k]$ by n , and then reverse it in k . This is the shift-then-scale procedure from Chapter 1.

DO NOT OMIT THE REVERSAL

Using an unreversed h gives $\sum_k x[k]h[k - n]$, which is the cross-correlation of x and h , not their convolution. A symmetric h can hide this error because reversal does not change it. Use an asymmetric example to check the construction.



EXAMPLE 3.1

GIVEN $x[n] = \{1, 2, 1, 2\}$ for $n = 0, 1, 2, 3$, and $h[n] = \{1, 1\}$ for $n = 0, 1$. Both zero elsewhere.

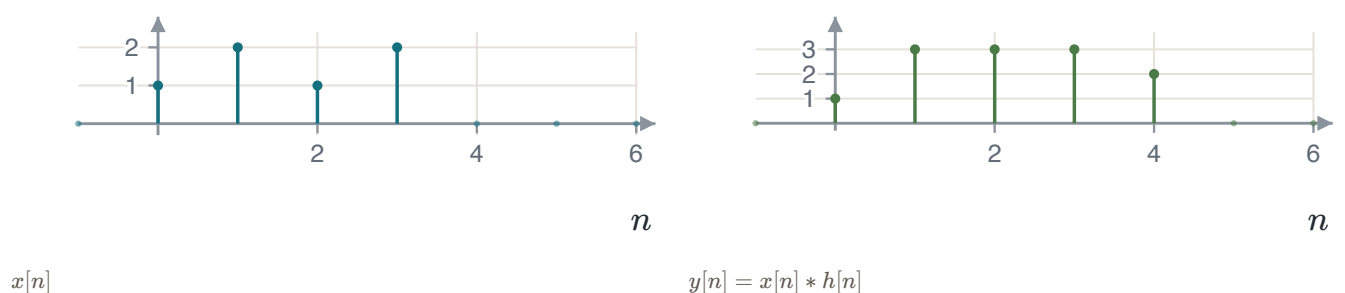
FIND $y[n] = x[n] * h[n]$.

METHOD Because x has four non-zero samples, expand the convolution sum into four terms. Each term is a shifted copy of h weighted by the corresponding sample of x .

SOLUTION $y[n] = x[0]h[n] + x[1]h[n - 1] + x[2]h[n - 2] + x[3]h[n - 3] = h[n] + 2h[n - 1] + h[n - 2] + 2h[n - 3]$

Add these four copies at each value of n . This gives $y = \{1, 3, 3, 3, 2\}$ on $n = 0, \dots, 4$.

CHECK The support must contain $4 + 2 - 1 = 5$ samples and run from 0 to 4, which the result does. The total must satisfy $\sum_n y[n] = (\sum_n x[n])(\sum_n h[n])$, and $12 = 6 \times 2$. Also, $h = \{1, 1\}$ forms a two-point moving sum, so each output is the sum of two adjacent input samples. This gives the same values.



EXAMPLE 3.2

GIVEN $x[n] = \left(\frac{1}{2}\right)^n u[n]$ and $h[n] = u[n]$.

FIND $y[n] = x[n] * h[n]$.

METHOD Use the support of $h[n - k]$ to set the summation limits: it equals 1 exactly when $k \leq n$. The overlap changes when n changes sign, so treat $n < 0$ and $n \geq 0$ separately.

SOLUTION **Case 1**, $n < 0$. The supports do not overlap: $x[k]$ needs $k \geq 0$ and $h[n - k]$ needs $k \leq n < 0$. So $y[n] = 0$.
Case 2, $n \geq 0$. The overlap is $0 \leq k \leq n$, so

$$y[n] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 - \left(\frac{1}{2}\right)^n.$$

Together, $y[n] = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n]$.

CHECK At $n = 0$, the formula gives $y[0] = 1$, and direct substitution gives $x[0]h[0] = 1$. As $n \rightarrow \infty$, $y[n] \rightarrow 2$, which equals the total sum of the input. This agrees with the interpretation of $h[n] = u[n]$ as an accumulator.

GEOMETRIC SUM

$\sum_{k=m}^n a r^k = \frac{a(r^m - r^{n+1})}{1 - r}$. The finite sum needs only $r \neq 1$. The condition $|r| < 1$ is needed only when the sum runs to infinity.

3.4 The convolution integral

The continuous-time derivation has the same purpose and uses the same three LTI moves. First, the sifting property represents x as continuously indexed weighted impulses:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau,$$

The equality uses $\delta(t) = \delta(-t)$. Now apply time invariance, homogeneity, and additivity to obtain:

CONVOLUTION INTEGRAL

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = x(t) * h(t)$$
$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

The two forms are equal. Choose the form that makes the support conditions easier to write.

BUILD THE REVERSED AND SHIFTED RESPONSE EXPLICITLY

Use τ as the variable and apply the shift before the reversal. For $\delta(-t + 5)$, first form $v(t) = \delta(t + 5)$ and then form $y(t) = v(-t) = \delta(-t + 5)$. The result is an impulse at $t = +5$. Writing both moves prevents a sign error in the integration limits.

EXAMPLE 3.3

GIVEN $x(t) = e^{2t}u(-t)$ and $h(t) = u(t - 3)$.

FIND $y(t) = x(t) * h(t)$.

METHOD Use the two support conditions to set the integration limit. $x(\tau)$ is non-zero for $\tau \leq 0$, and $h(t - \tau) = u(t - \tau - 3)$ is non-zero for $\tau \leq t - 3$. The overlap ends at $\min(0, t - 3)$, so the active upper limit changes at $t = 3$.

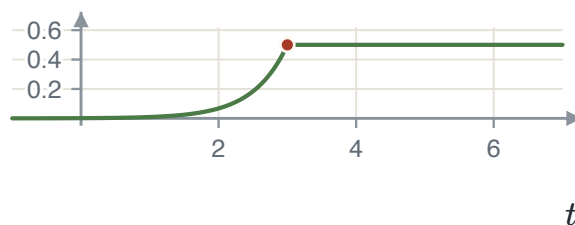
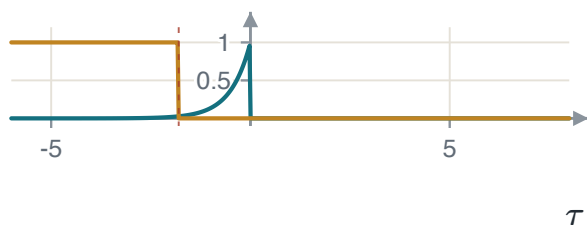
SOLUTION **Case 1, $t < 3$.** The binding limit is $t - 3$:

$$y(t) = \int_{-\infty}^{t-3} e^{2\tau} d\tau = \frac{1}{2}e^{2(t-3)}.$$

Case 2, $t > 3$. The binding limit is 0, so the whole of x is covered:

$$y(t) = \int_{-\infty}^0 e^{2\tau} d\tau = \frac{1}{2}.$$

CHECK Both branches give 0.5 at $t = 3$, so they join continuously. Also, $y(\infty)$ equals the total area of x , which is $1/2$. This agrees with the system interpretation: the delayed step makes the output the area of x collected up to three seconds earlier.



Case 1: $t < 3$. The shaded area is $y(t)$.

The result $y(t)$, continuous at $t = 3$.

EXAMPLE 3.4

GIVEN $x(t) = 1$ on $0 < t < 1$ and zero elsewhere. $h(t) = t$ on $0 < t < 2$ and zero elsewhere.

FIND $y(t) = x(t) * h(t)$.

METHOD Use $y(t) = \int h(\tau)x(t - \tau) d\tau$. This form reverses the rectangle, whose constant height makes the overlap limits easy to identify. Its support becomes $[t - 1, t]$, a window of width 1 moving across the ramp on $[0, 2]$.

SOLUTION The moving window has edges $t - 1$ and t , and the fixed ramp has edges 0 and 2. Set each moving edge equal to each fixed edge. This gives $t = 0, 1, 2, 3$, which divides the calculation into five cases.

$t < 0$: there is no overlap, so $y = 0$.

$0 < t < 1$: the overlap is $0 < \tau < t$, so $y = \int_0^t \tau d\tau = \frac{1}{2}t^2$.

$1 < t < 2$: the overlap is $t - 1 < \tau < t$, so $y = \int_{t-1}^t \tau d\tau = t - \frac{1}{2}$.

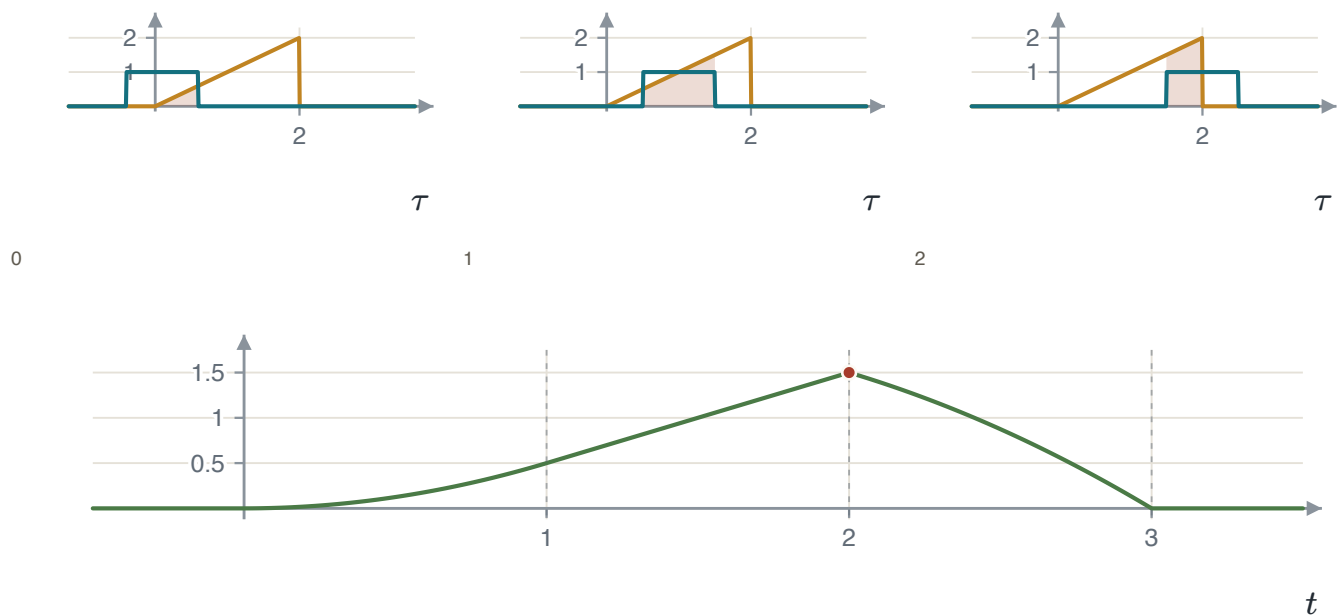
$2 < t < 3$: the overlap is $t - 1 < \tau < 2$, so $y = \int_{t-1}^2 \tau d\tau = -\frac{1}{2}t^2 + t + \frac{3}{2}$.

$t > 3$: there is no overlap, so $y = 0$.

CHECK Three checks. **Continuity:** at $t = 1$ both branches give 0.5; at $t = 2$ both give 1.5; at $t = 3$ both give 0. **Support:** $[0, 1]$ and $[0, 2]$ give $[0, 3]$, because supports add. **Area:** $\int y = (\int x)(\int h) = 1 \times 2 = 2$, and $\frac{1}{6} + 1 + \frac{5}{6} = 2$.

FIND THE BOUNDARIES BEFORE INTEGRATING

List the moving edges and the fixed edges, then set each moving edge equal to each fixed edge. The resulting values divide the calculation into all required cases.



The complete result. The peak is 1.5 at $t = 2$. A unit-width rectangle acts as a one-second moving sum, so the output rises while the window fills, grows linearly while it is full, and falls as it leaves the ramp.

3.5 Properties of convolution

PROPERTY	STATEMENT	WHAT IT MEANS FOR INTERCONNECTIONS
Commutative	$x * h = h * x$	Input and impulse response play symmetric roles in the algebra.
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Two systems in parallel , outputs added, equal one system with impulse response $h_1 + h_2$.
Associative	$x * (h_1 * h_2) = (x * h_1) * h_2$	Two systems in cascade equal one system with impulse response $h_1 * h_2$.

Associativity and commutativity show that the order of cascaded LTI systems does not change their combined impulse response. This conclusion requires both systems to be linear and time invariant. For example, a saturating amplifier followed by a filter is not generally equivalent to the filter followed by the amplifier.

3.6 System properties in terms of h

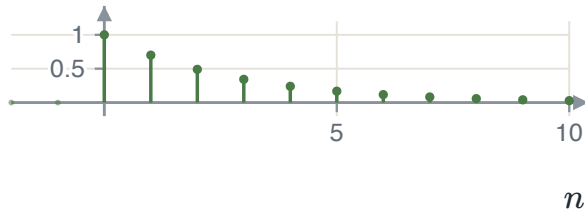
For an LTI system, each property test from Chapter 2 can be applied directly to the impulse response h .

PROPERTY	CRITERION	REASON
Memoryless	$h(t) = a \delta(t)$, or $h[n] = a \delta[n]$	Then $y[n] = a \sum_k x[k] \delta[n - k] = a x[n]$, a pure gain.
Invertible	$h * g = \delta$ for some g	g is the impulse response of the inverse system, and δ is the identity system.
Causal	$h(t) = 0$ for $t < 0$, or $h[n] = 0$ for $n < 0$	The sum collapses to $y[n] = h[0]x[n] + h[1]x[n-1] + \dots$, which uses only present and past inputs.
BIBO stable	$\sum_k h[k] < \infty$, or $\int h(t) dt < \infty$	See the proof below.

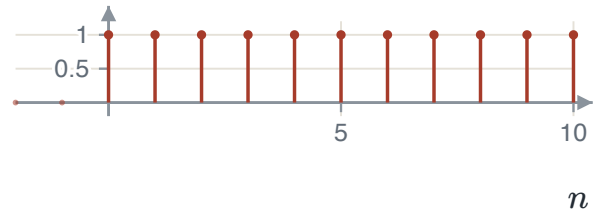
To show why absolute summability is sufficient for stability, assume $|x[n]| \leq B$ for all n . Then apply the triangle inequality:

$$|y[n]| = \left| \sum_k h[k] x[n-k] \right| \leq \sum_k |h[k]| |x[n-k]| \leq B \sum_k |h[k]| < \infty.$$

This proves sufficiency: an absolutely summable $h[n]$ maps every bounded input to a bounded output. Necessity follows by choosing the bounded input $x[n] = \text{sgn } h[-n]$, which gives $y[0] = \sum_k |h[k]|$.



$h[n] = 0.7^n u[n]$. Here $\sum |h| = 1/0.3 < \infty$: stable and causal.



$h[n] = u[n]$, the accumulator. Here $\sum |h| \rightarrow \infty$: causal but not stable.

3.7 A convolution checklist

1. Confirm that the system is linear and time invariant. Convolution gives the system output only under this condition.
2. Choose the factor whose reversal gives simpler support conditions.
3. Write the support of each factor as an inequality in the dummy variable.
4. Set the moving edges equal to the fixed edges to list every case boundary, before integrating anything.
5. Integrate or sum, case by case.
6. Check continuity at every boundary, check that the supports add, and check that the total area or total sum multiplies.

USE EACH FINAL CHECK FOR A SPECIFIC PURPOSE

A mismatch at a case boundary indicates an incorrect limit. An incorrect output support indicates an error in a shift, reversal, or support condition. An incorrect total area or sum indicates an error in the integrand or summand.

Exercises

3.1 Compute $\{1, 2, 3\} * \{1, -1\}$ with both sequences starting at $n = 0$, and check your answer with the sum rule.

Answer: $\{1, 1, 1, -3\}$; the sums are $6 \times 0 = 0$.

3.2 Let $h[n] = (\frac{1}{3})^n u[n]$. Is the system stable? Is it causal? Is it memoryless?

Answer: Stable ($\sum |h| = 3/2$), causal, not memoryless.

3.3 Find $y(t) = u(t) * u(t)$ and sketch it.

Answer: $y(t) = t u(t)$.

3.4 Two systems with $h_1[n] = \delta[n] - \delta[n - 1]$ and $h_2[n] = u[n]$ are cascaded. Find the impulse response of the cascade and identify the resulting system.

Answer: $\delta[n]$: the two systems are inverses.

3.5 $x(t) = 1$ on $0 < t < 3$ and $h(t) = t$ on $0 < t < 2$. List the case boundaries before computing anything, then find $y(t)$.

Answer: Boundaries at $t = 0, 2, 3, 5$; four cases.

Fourier series

Fourier series are used to calculate the response of a linear time-invariant system to a periodic signal. Complex exponentials are eigenfunctions of every LTI system, and periodic signals can be represented as sums of these exponentials. This representation replaces convolution by one multiplication at each harmonic frequency.

4.1 Eigenfunctions of a linear time-invariant system

DEFINITION

A signal is an **eigenfunction** of a system when the output is that same signal multiplied by a constant. The constant is the **eigenvalue**.

Put $x(t) = e^{st}$, with s any complex number, into the convolution integral. Write both limits; they are $-\infty$ and ∞ throughout this chapter.

$$y(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau.$$

The second factor does not depend on t . It is a number attached to the value of s , and it is called the transfer function.

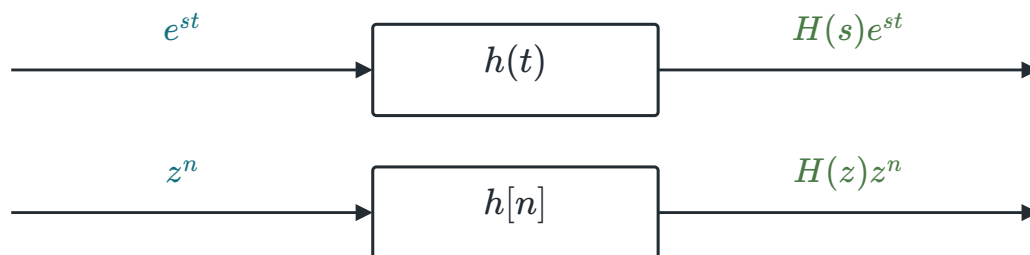
EIGENFUNCTION PROPERTY

$$e^{st} \rightarrow h(t) \rightarrow H(s)e^{st}, \quad H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

$$z^n \rightarrow h[n] \rightarrow H(z)z^n, \quad H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

The discrete-time output is a sequence, $y[n] = H(z)z^n$. Nothing in that line is a function of a continuous variable, and writing the continuous-time result there is the commonest slip in this section.

Setting $s = j\omega$ and $z = e^{j\omega}$ with ω real gives the two **frequency responses**, $H(j\omega)$ and $H(e^{j\omega})$. The discrete-time one repeats every 2π in ω , because $e^{j(\omega+2\pi)k} = e^{j\omega k}$ for integer k . The continuous-time one does not repeat at all.



The two statements, side by side, so that neither can borrow the other's variables.

EXAMPLE 4.1 – A PURE DELAY

GIVEN An LTI system with $y(t) = x(t - 3)$.

FIND The output for $x(t) = e^{j2t}$, and for $x(t) = \cos(4t) + \cos(7t)$, using eigenfunctions only.

METHOD The inputs are complex exponentials, so the eigenfunction method applies. Write h from the delay definition, integrate to obtain $H(s)$, and multiply each exponential by the corresponding value of H .

SOLUTION $h(t) = \delta(t - 3)$, so by sifting $H(s) = \int_{-\infty}^{\infty} \delta(\tau - 3)e^{-s\tau} d\tau = e^{-3s}$. Then $e^{j2t} \rightarrow e^{-6j}e^{j2t} = e^{j2(t-3)}$. Expanding each cosine into two exponentials and repeating the step gives

$$y(t) = \cos(4(t - 3)) + \cos(7(t - 3)).$$

CHECK Apply $y(t) = x(t - 3)$ directly to both inputs. The answers agree. Note also that $|H(j\omega)| = 1$ and $\angle H(j\omega) = -3\omega$: a delay changes no amplitude and adds a phase proportional to frequency.

This result gives the method used in the rest of the chapter. If $x = \sum_k a_k e^{s_k t}$, multiply each term by its eigenvalue to obtain $y = \sum_k a_k H(s_k) e^{s_k t}$. The next sections show how to write periodic signals in this form.

4.2 Existence: the Dirichlet conditions

Not every signal can be decomposed into complex exponentials. Periodic signals can, provided they satisfy three conditions. Let $x(t) = x(t + T)$ for every t .

1. Over one period, x is absolutely integrable: $\int_T |x(t)| dt < \infty$.
2. In any finite interval, x has only a finite number of maxima and minima.
3. In any finite interval, x has only a finite number of discontinuities, and each is finite.

SUFFICIENT, NOT NECESSARY

A signal meeting all three has a Fourier series that converges to it wherever it is continuous, and to the midpoint of the jump at a discontinuity. A signal failing one of them may still have a series. The conditions guarantee; they do not exclude.

The three are independent. The signal $1/t$ on each period fails the first while the other two hold. The signal $\sin(2\pi/t)$ has finite area but oscillates infinitely often, failing only the second. A staircase whose steps halve in height and width satisfies the first two and fails the third.

4.3 The synthesis and analysis equations

CONTINUOUS-TIME FOURIER SERIES

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

The first equation is synthesis, the second is analysis. One period symbol is used throughout: T_0 , with $\omega_0 = 2\pi/T_0$ stated once here. The sign of the exponent is the only thing that separates the pair.

Proof of the analysis equation

Multiply the synthesis equation by $e^{-jn\omega_0 t}$ and integrate over one period:

$$\int_{T_0} x(t) e^{-jn\omega_0 t} dt = \sum_{k=-\infty}^{\infty} a_k \int_{T_0} e^{j(k-n)\omega_0 t} dt.$$

Integrating from $-T_0/2$ to $T_0/2$ and using $\omega_0 T_0 = 2\pi$ gives $T_0 \sin((k-n)\pi) / ((k-n)\pi)$. For $k \neq n$ the sine of an integer multiple of π is zero. For $k = n$ the expression is $0/0$, and one application of l'Hôpital's rule in the variable $(k - n)$ gives T_0 .

ORTHOGONALITY OF COMPLEX EXPONENTIALS

$$\int_{T_0} e^{j(k-n)\omega_0 t} dt = \begin{cases} T_0, & k = n \\ 0, & k \neq n \end{cases}$$

Only the term with $k = n$ survives, leaving $\int_{T_0} x(t) e^{-jn\omega_0 t} dt = T_0 a_n$.

THE DC TERM

Setting $k = 0$ makes the exponential 1, so

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt,$$

the average value of the signal over one period. Use this fact to check a coefficient calculation: calculate or estimate the mean directly from the signal and compare it with a_0 .

Fundamental period of a sum

A sum of periodic signals is periodic when the ratio of every pair of component periods is rational. The fundamental period of the sum is the smallest positive number that is a whole multiple of each component period. Write each period as a fraction in lowest terms:

PERIOD OF A SUM

$$T_0 = \text{LCM}\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots\right) = \frac{\text{LCM}(p_1, p_2, \dots)}{\text{GCD}(q_1, q_2, \dots)}$$

The result is a least common multiple of fractions, so it need not be an integer. A constant term has no period and is left out.

WHERE THIS GOES WRONG

Putting every period over one common denominator first, then taking the least common multiple of the numerators, is the same rule in disguise — because for equal denominators $\text{LCM}(d, d) = \text{GCD}(d, d) = d$. Applying LCM to the denominators *without* that rewrite is wrong: for periods $2/9$ and $8/21$ it gives $8/63$, which is a multiple of neither. The correct answer is $\text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$. Always divide the answer by each period and confirm both results are whole numbers.

EXAMPLE 4.2 – COEFFICIENTS OF A SUM OF SINUSOIDS

GIVEN $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$.

FIND Every Fourier series coefficient, with magnitude and phase.

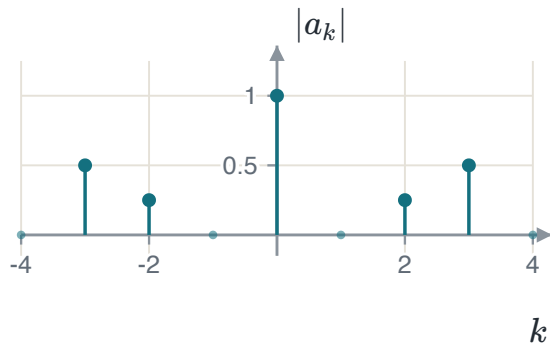
METHOD Find T_0 , hence ω_0 . Expand with Euler's relations and match each exponent to $jk\omega_0 t$.

SOLUTION The component periods are 1 and $2/3$ s, so $T_0 = \text{LCM}(1, 2) / \text{GCD}(1, 3) = 2$ s and $\omega_0 = \pi$ rad/s. Then

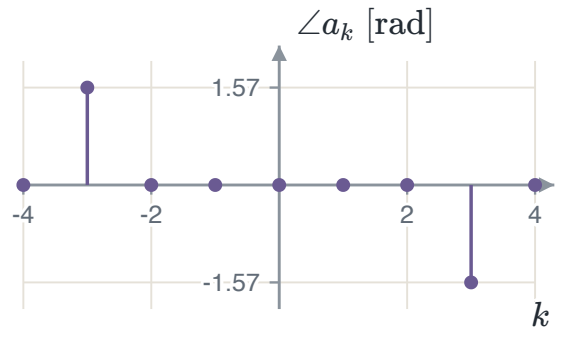
$$x(t) = 1e^{j0\omega_0 t} + \frac{1}{4}e^{j2\omega_0 t} + \frac{1}{4}e^{-j2\omega_0 t} + \frac{1}{2j}e^{j3\omega_0 t} - \frac{1}{2j}e^{-j3\omega_0 t},$$

so $a_0 = 1$, $a_{\pm 2} = 1/4$, $a_3 = 1/(2j)$, $a_{-3} = -1/(2j)$, and every other coefficient is zero.

CHECK $a_0 = 1$ is also the average of the signal: both sinusoids complete whole cycles in one period and average to zero, leaving the constant. Magnitudes: $|a_0| = 1$, $|a_{\pm 2}| = 1/4$, $|a_{\pm 3}| = 1/2$. Phases: $\angle a_3 = \angle 1 - \angle 2j = -\pi/2$ and $\angle a_{-3} = \angle(-1) - \angle 2j = +\pi/2$. The signal is real, so magnitudes are even in k and phases odd — which they are.



Magnitudes, even in k .



Phases, odd in k . A value of ± 1.57 rad is $\pm \pi/2$.

4.4 Two standard waveforms

EXAMPLE 4.3 – THE PERIODIC RECTANGULAR WAVE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 for $T_1 < |t| < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD Apply the analysis equation over $-T_0/2$ to $T_0/2$. The signal is 1 only on $[-T_1, T_1]$, so those become the limits. The case $k = 0$ is separate, because the antiderivative divides by k .

SOLUTION For $k \neq 0$,

$$a_k = \frac{1}{T_0} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T_0} = \frac{\sin(2\pi k T_1 / T_0)}{\pi k},$$

and for $k = 0$, $a_0 = 2T_1/T_0$.

CHECK As $k \rightarrow 0$ the second branch tends to $2T_1/T_0$, because $\sin \theta / \theta \rightarrow 1$. It therefore agrees with the branch for $k = 0$. The signal is real and even, so the coefficients must be real and even. The formula has this property because $\sin$ and k are both odd. The zeros occur when k is a non-zero multiple of $T_0/(2T_1)$.

The envelope, and the samples taken from it

These coefficients are values of one continuous function of ω , read at equally spaced points. Write the function first:

ENVELOPE OF A SINGLE PULSE

$$E(\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = \frac{2 \sin(\omega T_1)}{\omega}$$

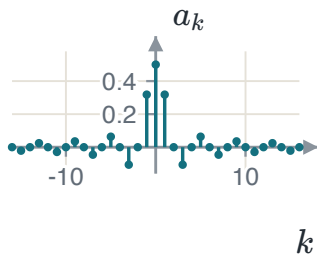
$$a_k = \frac{1}{T_0} E(k\omega_0) = \frac{\sin(2\pi k T_1 / T_0)}{\pi k}$$

E depends on the shape of one pulse and not on how often it repeats. In the unnormalised convention $\text{sinc}(\theta) = \sin \theta / \theta$, it is $E(\omega) = 2T_1 \text{sinc}(\omega T_1)$.

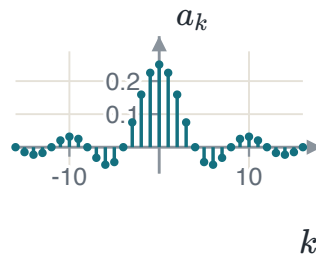
TWO OBJECTS, NOT ONE

$E(\omega)$ is a function of a continuous variable; a_k is a sequence indexed by an integer. Evaluating the envelope at the single point $\omega = \omega_0$ gives $T_0 a_1$, the first harmonic only. As a function of the harmonic index the coefficient is $T_0 a_k = T_0 \sin(2\pi k T_1 / T_0) / (\pi k)$, with k present in both places. Never join the tops of coefficient stems with a line: it claims the signal contains frequencies between the harmonics, which a periodic signal does not.

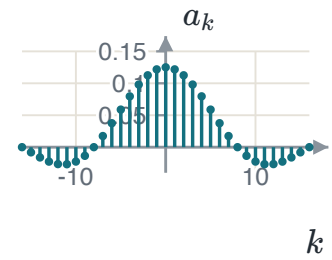
Increase T_0 while keeping the pulse fixed. The envelope remains the same. The spacing $\omega_0 = 2\pi/T_0$ decreases, and every coefficient decreases as $1/T_0$. The next chapter uses this limiting process to define the Fourier transform.



$$T_0 = 4T_1.$$



$$T_0 = 8T_1.$$



$$T_0 = 16T_1.$$

EXAMPLE 4.4 – THE SAWTOOTH WAVE

GIVEN $x(t) = t$ for $-T_0/2 < t < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD The signal is odd, so its DC term is zero. For $k \neq 0$, the integrand contains t times an exponential, so use integration by parts: $\int_b^c t e^{at} dt = \frac{1}{a^2} [(at - 1)e^{at}]_b^c$.

SOLUTION $a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} t dt = 0$. For $k \neq 0$, using $\omega_0 T_0/2 = \pi$ and $\sin(k\pi) = 0$,

$$a_k = \frac{jT_0}{2k\pi} \cos(k\pi) = \frac{jT_0(-1)^k}{2k\pi}.$$

CHECK The signal is real and odd, so the coefficients must be purely imaginary — and they are. For $T_0 = 1$ the largest is $|a_{\pm 1}| = 1/(2\pi) \approx 0.159$. Every phase is $+\pi/2$ or $-\pi/2$, with the sign alternating in k and flipping again with the sign of k .

EXAMPLE 4.5 – THE PERIODIC IMPULSE TRAIN

GIVEN $x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT_0)$.

FIND Every coefficient.

METHOD Take the period from $-T_0/2$ to $+T_0/2$. Exactly one impulse lies inside it, so sifting finishes the integral.

SOLUTION $a_k = \frac{1}{T_0} \int_{-T_0/2}^{+T_0/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0}$ for every k .

CHECK Both limits must be written and must differ. An interval with equal endpoints has zero length, so every integral over it is zero. The result is real, positive, and equal at every harmonic. At $t = 0$, every harmonic has value 1 and their contributions add. At other times, the positive and negative contributions cancel.

4.5 How many harmonics are enough

TRUNCATION AND ITS ERROR

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}, \quad e_N(t) = x(t) - x_N(t)$$

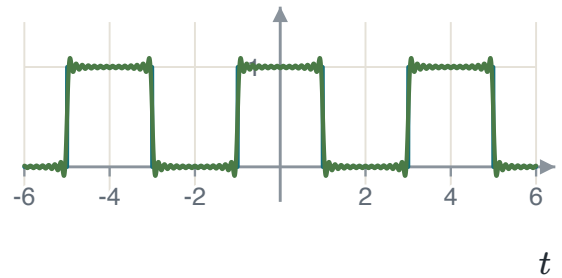
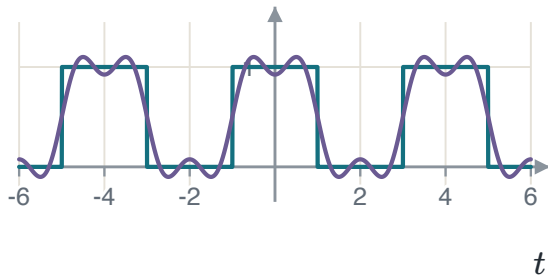
$$\text{MSE} = \frac{1}{T_0} \int_{T_0} |e_N(t)|^2 dt = \sum_{|k| > N} |a_k|^2$$

The second form follows from Parseval's relation in Section 4.7. It turns a question about a difference of signals into a question about a tail of a series.

THE GIBBS PHENOMENON

Near a jump the partial sum overshoots, and the overshoot does not shrink as N grows: its height settles at about **8.95% of the size of the jump**. What does shrink is its width. That is how the mean-square error can go to zero while the largest error does not.

N	RECTANGULAR WAVE, $T_0 = 4T_1$	SAWTOOTH, $T_0 = 1$
3	0.025	0.0144
9	0.010	0.0053
27	0.004	0.0018
81	0.001	0.0006



$N = 3$.

$N = 27$. The spike beside each jump is no shorter.

The rate at which the error falls is set by the smoothness of the signal. A jump forces the coefficients to decay like $1/k$, so the tail $\sum_{|k|>N} |a_k|^2$ behaves like $1/N$ and doubling N roughly halves the error. A signal that is continuous but has a corner — a triangular wave — has coefficients decaying like $1/k^2$, a tail like $1/N^3$, and needs far fewer harmonics.

4.6 The discrete-time Fourier series

DISCRETE-TIME FOURIER SERIES

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

$\langle N \rangle$ means any N consecutive values of the index. Both sums are finite and there are exactly N distinct coefficients.

THE COEFFICIENTS REPEAT

$a_{k+N} = a_k$. Replacing k by $k + N$ in the analysis sum multiplies the summand by $(e^{-j2\pi})^n$, which is 1 because n is an integer. The same substitution in continuous time produces $e^{-j2\pi t}$, which is 1 only when t is an integer — and it is not. Continuous-time coefficients therefore do not repeat.

A FINITE, EXACT IDENTITY

A period- N sequence is described by N numbers and its series has N terms. There is no truncation, no limit, no convergence question, and no Gibbs phenomenon anywhere in discrete time.

EXAMPLE 4.6 – A SUM OF TWO SEQUENCES

GIVEN $x[n] = \sin\left(\frac{5\pi}{6}n\right) + \cos\left(\frac{3\pi}{4}n + \frac{\pi}{5}\right)$.

FIND The fundamental period and every coefficient.

METHOD A discrete-time sinusoid repeats only if $N = (2\pi/\omega)k$ is a positive integer for some integer k . Get each component period, take the least common multiple, then expand.

SOLUTION $N = 12k/5$ gives $N = 12$ at $k = 5$; $N = 8k/3$ gives $N = 8$ at $k = 3$. So $N_0 = \text{LCM}(12, 8) = 24$ and $\omega_0 = 2\pi/24$. Since $5\pi/6 = 10\omega_0$ and $3\pi/4 = 9\omega_0$,

$$a_{10} = \frac{1}{2j}, \quad a_{-10} = -\frac{1}{2j}, \quad a_9 = \frac{1}{2}e^{j\pi/5}, \quad a_{-9} = \frac{1}{2}e^{-j\pi/5},$$

with every other coefficient in one period equal to zero.

CHECK $|a_{\pm 9}| = |a_{\pm 10}| = 1/2$; $\angle a_9 = \pi/5$, $\angle a_{-9} = -\pi/5$, $\angle a_{10} = -\pi/2$, $\angle a_{-10} = +\pi/2$. The sequence is real, so magnitudes are even and phases odd — and $a_k = a_{k+24}$ extends the answer to every k .

EXAMPLE 4.7 – THE DISCRETE-TIME SQUARE WAVE

GIVEN $x[n] = 1$ for $|n| \leq N_1$ and 0 for $N_1 < |n| \leq N/2$, repeated with period N .

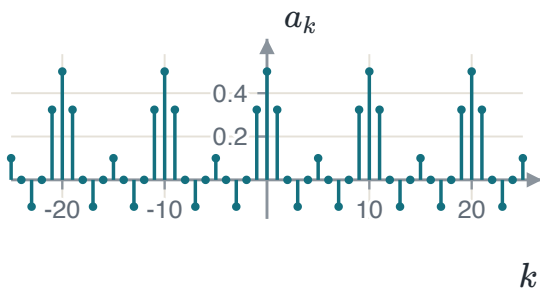
FIND Every coefficient.

METHOD The analysis sum reduces to a finite geometric sum with ratio $r = e^{-jk(2\pi/N)}$. Use $\sum_{n=m}^p r^n = (r^m - r^{p+1})/(1 - r)$, which requires only $r \neq 1$.

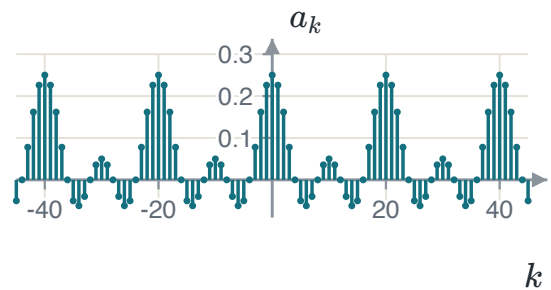
SOLUTION The lower limit is $n = -N_1$, so the first numerator term is $r^{-N_1} = e^{+jk(2\pi/N)N_1}$, with a positive exponent. Factoring $e^{-jk\pi/N}$ out of numerator and denominator turns each bracket into $2j \sin(\cdot)$, and these cancel, leaving

$$a_k = \frac{1}{N} \frac{\sin\left(\frac{2\pi k}{N} \left(N_1 + \frac{1}{2}\right)\right)}{\sin\left(\frac{\pi k}{N}\right)}, \quad a_k = \frac{2N_1 + 1}{N} \text{ when } k = 0, \pm N, \pm 2N, \dots$$

CHECK Here $|r| = 1$ at every k , so a condition $|r| < 1$ would rule the sum out altogether. What matters is $r \neq 1$, and $r = 1$ happens exactly when k is a multiple of N — which is the second branch. For $N = 10$, $N_1 = 2$ the peak is $(2N_1 + 1)/N = 0.5$, and any plot of these coefficients must be scaled so that 0.5 falls inside the labelled range.



$N = 10$, $N_1 = 2$: peak 0.5, repeating every 10.



$N = 20$: peak 0.25, repeating every 20.

EXAMPLE 4.8 – A DISCRETE SAWTOOTH

GIVEN $x[n] = n$ for $-5 \leq n \leq 5$, repeated with period $N = 11$.

FIND Every coefficient.

METHOD The sequence is odd, so pair the term at n with the term at $-n$ before expanding: $ne^{-jk\omega_0 n} + (-n)e^{+jk\omega_0 n} = -2jn \sin(k\omega_0 n)$.

SOLUTION $a_0 = \frac{1}{11} \sum_{n=-5}^5 n = 0$, and

$$a_k = -\frac{2j}{11} \sum_{m=1}^5 m \sin\left(\frac{2\pi km}{11}\right).$$

CHECK Every coefficient is purely imaginary, as a real odd sequence requires. The largest is $|a_{\pm 1}| = 1.7747$, and $a_k = a_{k+11}$, so the eleven coefficients from $k = -5$ to $k = 5$ are the whole answer.

4.7 Properties

PROPERTY	SIGNAL	COEFFICIENTS
Linearity	$Ax_1 + Bx_2$	$Aa_k + Bb_k$
Time shift, continuous	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0}$
Time shift, discrete	$x[n - n_0]$	$a_k e^{-jk\omega_0 n_0}$
Time reversal	$x(-t)$ or $x[-n]$	a_{-k}
Conjugation	$x^*(t)$ or $x^*[n]$	a_{-k}^*
Frequency shift, continuous	$e^{jM\omega_0 t} x(t)$	a_{k-M} , M an integer
Frequency shift, discrete	$e^{jM(2\pi/N)n} x[n]$	a_{k-M} , M an integer
Time scaling, continuous	$x(\alpha t)$, $\alpha > 0$	a_k , with period T_0/α
Time scaling, discrete	$x_{(m)}[n]$, m a positive integer	$\frac{1}{m} a_k$, with period mN
Periodic convolution, continuous	$\int_{T_0} x(\tau)y(t - \tau)d\tau$	$T_0 a_k b_k$
Periodic convolution, discrete	$\sum_{r=\langle N \rangle} x[r]y[n - r]$	$N a_k b_k$
Multiplication, continuous	$x(t)y(t)$	$\sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$
Multiplication, discrete	$x[n]y[n]$	$\sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Differentiation	$dx(t)/dt$	$jk\omega_0 a_k$
Integration	$\int_{-\infty}^t x(\tau)d\tau$	$a_k/(jk\omega_0)$, only if $a_0 = 0$
First difference	$x[n] - x[n - 1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running sum	$\sum_{r=-\infty}^n x[r]$	$a_k/(1 - e^{-jk(2\pi/N)})$, only if $a_0 = 0$

Three properties need further explanation. **Time scaling** in continuous time does not change the coefficient values, but it changes their frequencies. In discrete time, $x_{(m)}[n]$ places the original samples m indices apart and inserts $m - 1$ zeros after each sample. The factor $1/m$ appears because the average is taken over m times as many samples. **Periodic convolution** includes the factor T_0 or N before the coefficient product. The analysis equation divides by the period, and the convolution adds an operation over one period. Omitting this factor gives the wrong amplitude. **Integration** and the **running sum** divide by a factor that is zero at $k = 0$. A signal with non-zero mean therefore produces a non-periodic term. Subtract the mean, apply the rule to the zero-mean part, and then add the term $a_0 t$.

The time-shift factor has modulus 1, so $|b_k| = |a_k|$ at every k : a shift never changes a magnitude, only a phase, and it changes that phase by $-k\omega_0 t_0$, proportionally to the harmonic index. Two signals differing only by a delay have identical magnitude plots, which is why a phase plot is not decoration.

THE DISCRETE-TIME PRODUCT

The multiplication property in discrete time is a **periodic** convolution: the sum runs over one period of ℓ . Discrete-time coefficients repeat, so an infinite sum would add each of the N distinct products over and over and diverge. The product $x[n]y[n]$ is itself periodic with period N and therefore has N coefficients — a number only a sum over one period can deliver.

Symmetry

- x real gives $a_{-k} = a_k^*$, so $|a_k|$ is even in k and $\angle a_k$ is odd.
- x real and even gives a_k real and even.
- x real and odd gives a_k purely imaginary and odd.
- For real x : $\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{a_k\}$ and $\mathcal{O}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{a_k\}$.

The last line is the general statement and the two before it are its special cases. $\mathcal{E}\mathbf{v}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ has coefficients $\frac{1}{2}(a_k + a_{-k})$ by linearity and time reversal, and $a_{-k} = a_k^*$ for a real signal, so that is $\text{Re}\{a_k\}$. The odd part gives $\frac{1}{2}(a_k - a_k^*) = j \text{Im}\{a_k\}$. Splitting a real signal into its even and odd parts splits its coefficients into their real and imaginary parts, and that is what tells you which half of the coefficients you actually have to compute.

The third follows in one line. Real gives $a_{-k} = a_k^*$; odd gives $a_{-k} = -a_k$; together $a_k^* = -a_k$, so $\text{Re}\{a_k\} = 0$. Both sawtooths of this chapter obey it.

PARSEVAL'S RELATION

$$\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

$$\frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

The left side is the average power over one period, with the normalised convention $R = 1 \Omega$. The right side splits it among the harmonics, and $|a_k|^2$ is the power carried by the k -th harmonic alone.

4.8 A periodic signal through an LTI system

ONE MULTIPLICATION PER HARMONIC

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} \underbrace{a_k H(jk\omega_0)}_{b_k} e^{jk\omega_0 t}$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \longrightarrow y[n] = \sum_{k=\langle N \rangle} a_k H(e^{jk\omega_0}) e^{jk\omega_0 n}$$

The frequency response H describes the system. For a periodic input, calculate one product at each harmonic instead of evaluating a convolution.

Putting the output back together

The result above is a sum of complex terms. For real x and real h , $a_{-k} = a_k^*$ and $H(-j\omega) = H^*(j\omega)$, so $b_{-k} = b_k^*$. Pair each k with $-k$:

CONJUGATE-PAIR REASSEMBLY

$$b_k e^{jk\omega_0 t} + b_{-k} e^{-jk\omega_0 t} = 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

$$y(t) = b_0 + \sum_{k=1}^{\infty} 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

b_0 has no partner, so it carries no factor of two.

THE FACTOR OF TWO, AND WHOSE PHASE

The amplitude of the k -th cosine is $2|b_k|$, because both members of the pair contribute. The phase is $\angle b_k$, the one belonging to **positive** k ; taking $\angle b_{-k}$ flips the sign of that cosine's phase. Test any reassembly by setting $H = 1$: it must return the input exactly. A version missing the factor of two returns half the input plus its full average.

EXAMPLE 4.9 – LOW-PASS FILTERING

GIVEN $x(t) = 1 + \cos(\pi t) + \sin(2\pi t) + \cos(3\pi t + \frac{\pi}{3})$ into a system with $h(t) = e^{-t}u(t)$.

FIND The output signal.

METHOD The input is periodic and the system is LTI, so harmonic multiplication applies. Calculate $H(j\omega)$ from h , find T_0 and a_k from x , form $b_k = a_k H(jk\omega_0)$, and combine each conjugate pair.

SOLUTION $H(j\omega) = 1/(1 + j\omega)$, so $|H| = 1/\sqrt{1 + \omega^2}$ and $\angle H = -\arctan \omega$. The component periods 2, 1 and $2/3$ s give $T_0 = 2$ s and $\omega_0 = \pi$. With $a_0 = 1$, $a_{\pm 1} = \frac{1}{2}$, $a_2 = \frac{1}{2j}$, $a_3 = \frac{1}{2}e^{j\pi/3}$:

$$b_0 = 1, \quad b_1 = 0.1517e^{-j1.2626}, \quad b_2 = 0.0786e^{-j2.9838}, \quad b_3 = 0.0528e^{-j0.4178},$$

and therefore

$$y(t) = 1 + 0.303 \cos(\pi t - 1.263) + 0.157 \cos(2\pi t - 2.984) + 0.106 \cos(3\pi t - 0.418).$$

CHECK Each amplitude is $2|b_k|$: 0.303, 0.157, 0.106. The output ranges from about 0.615 to 1.417; using half the required amplitudes would give only 0.81 to 1.21. The average is unchanged because $|H(0)| = 1$, and the third-harmonic amplitude is reduced to about one tenth of its input value.

EXAMPLE 4.10 – HIGH-PASS FILTERING

GIVEN The same $x(t)$, now into a system with $H(j\omega) = \frac{j\omega}{1 + j\omega}$.

FIND The output signal.

METHOD The input is periodic and the system is LTI, so use the same harmonic-multiplication method as in Example 4.9. Replace the low-pass response by the given H , calculate each b_k , and combine the conjugate pairs.

SOLUTION $H(j0) = 0$, so $b_0 = 0$ and the output has zero average. Then $b_1 = 0.4764e^{+j0.3082}$, $b_2 = 0.4938e^{-j1.4130}$, $b_3 = 0.4972e^{+j1.1530}$, and

$$y(t) = 0.953 \cos(\pi t + 0.308) + 0.988 \cos(2\pi t - 1.413) + 0.994 \cos(3\pi t + 1.153).$$

CHECK $|H(jk\pi)|$ is 0.953, 0.988, 0.994 for $k = 1, 2, 3$. These values are close to one, so unit-amplitude input harmonics must produce output amplitudes close to one. Values close to 0.5 would show that the factor of two is missing. Use the phases for positive k : +0.308, -1.413, +1.153. Using b_{-k} instead changes the signs of the first and third phases.

EXAMPLE 4.11 – FILTERING A DISCRETE IMPULSE TRAIN

GIVEN $x[n] = \sum_{m=-\infty}^{\infty} \delta[n - 4m]$, and the systems $h_1[n] = 0.5\delta[n] - 0.5\delta[n - 1]$ and $h_2[n] = 0.5\delta[n] + 0.5\delta[n - 1]$.

FIND Both output signals.

METHOD The input is periodic and both systems are LTI, so discrete-time harmonic multiplication applies. Find a_k for the impulse train, calculate $b_k = a_k H(e^{jk\omega_0})$ with $\omega_0 = \pi/2$, and then combine the conjugate pairs.

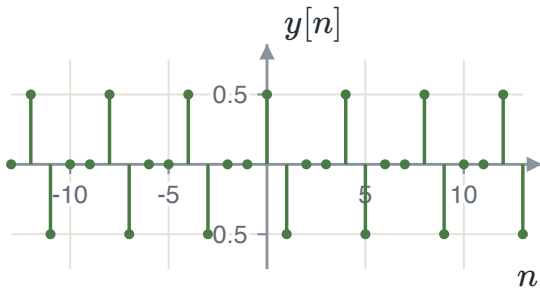
SOLUTION $a_k = \frac{1}{4}$ for every k . For h_1 , $H_1(e^{j\omega}) = 0.5(1 - e^{-j\omega})$, giving $b_0 = 0$, $b_1 = 0.1768e^{j\pi/4}$ and $b_2 = 0.25$, so

$$y[n] = 0.36 \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + 0.25(-1)^n.$$

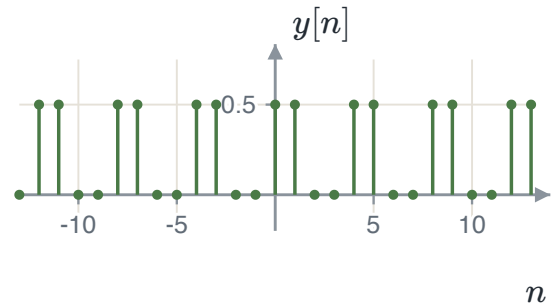
For h_2 , $H_2(e^{j\omega}) = 0.5(1 + e^{-j\omega})$, giving $b_0 = 0.25$, $b_1 = 0.1768e^{-j\pi/4}$ and $b_2 = 0$, so

$$y[n] = 0.25 + 0.36 \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right).$$

CHECK Apply the two systems directly: $0.5x[n] \mp 0.5x[n - 1]$ reproduces both answers sample by sample. Within one coefficient period, the $k = 2$ term has no separate conjugate partner. It therefore appears alone as $b_2(-1)^n$ and has no factor of two. The high-pass system removes the average and retains b_2 ; the low-pass system retains the average and gives $b_2 = 0$.



First difference: zero average.



Two-point average: the average remains.

4.9 Checklist, and what comes next

1. Confirm the signal is periodic and find T_0 or N before anything else.
2. Set $\omega_0 = 2\pi/T_0$ or $2\pi/N$ and write every component frequency as a multiple of it. That multiple is k .
3. If the signal is already a sum of sinusoids, read the coefficients off with Euler's relations. Otherwise integrate or sum, treating $k = 0$ separately.
4. Compare a_0 with the average calculated from the signal, and check the symmetry: a real signal gives $a_{-k} = a_k^*$.
5. Through an LTI system, $b_k = a_k H(jk\omega_0)$ — one product per harmonic.
6. To return to a real signal, pair k with $-k$: amplitude $2|b_k|$, phase $\angle b_k$. The term b_0 stands alone.

Everything in this chapter needed the signal to repeat. A single pulse does not repeat, has no fundamental period, and has no harmonics to carry coefficients. The rectangular wave already showed the way out: its coefficients were samples of one envelope, taken every $\omega_0 = 2\pi/T_0$. Lengthen the period and the envelope stays put while the samples crowd together and shrink as $1/T_0$.

WHERE CHAPTER 5 BEGINS

$$T_0 a_k = E(k\omega_0) \xrightarrow{T_0 \rightarrow \infty} X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

In the limit the stems merge into the curve they were always sampling. That curve is the Fourier transform.

4.10 Exercises

4.1 Two periodic signals have fundamental periods $2/9$ s and $8/21$ s. Find the fundamental period of their sum, and verify it by division.

Answer: $T_0 = \text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$ s. Check: $(8/3)/(2/9) = 12$ and $(8/3)/(8/21) = 7$, both whole, and $\text{gcd}(12, 7) = 1$.

4.2 For $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$, state a_0 and explain in one sentence why it is not zero.

Answer: $a_0 = 1$: the constant term is the $k = 0$ harmonic, and it is also the average of the signal, since both sinusoids complete whole cycles in one period.

4.3 A rectangular wave has $T_0 = 8T_1$. Find a_0 and a_2 .

Answer: $a_0 = 2T_1/T_0 = 1/4$ and $a_2 = \sin(\pi/2)/(2\pi) = 1/(2\pi) \approx 0.159$.

4.4 Explain why the finite geometric sum used for the discrete-time square wave requires $r \neq 1$ and not $|r| < 1$, and say what happens at the excluded value.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1 - r$ vanishes. Here $|r| = 1$ at every k , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is k a multiple of N , every term is 1 and the sum is $2N_1 + 1$.

4.5 A filtered signal has $b_1 = 0.1517e^{-j1.263}$ and $\omega_0 = \pi$. Write the contribution of the pair $k = \pm 1$ to the real output.

Answer: $2|b_1| \cos(\pi t + \angle b_1) = 0.303 \cos(\pi t - 1.263)$.

4.6 A square wave and a triangular wave are truncated to the same number of harmonics. Which is approximated better, and which shows a Gibbs overshoot?

Answer: The triangular wave: its coefficients decay like $1/k^2$ against $1/k$, so its discarded power falls like $1/N^3$ rather than $1/N$. Only the square wave has a jump, so only it shows a Gibbs overshoot.

The continuous-time Fourier transform

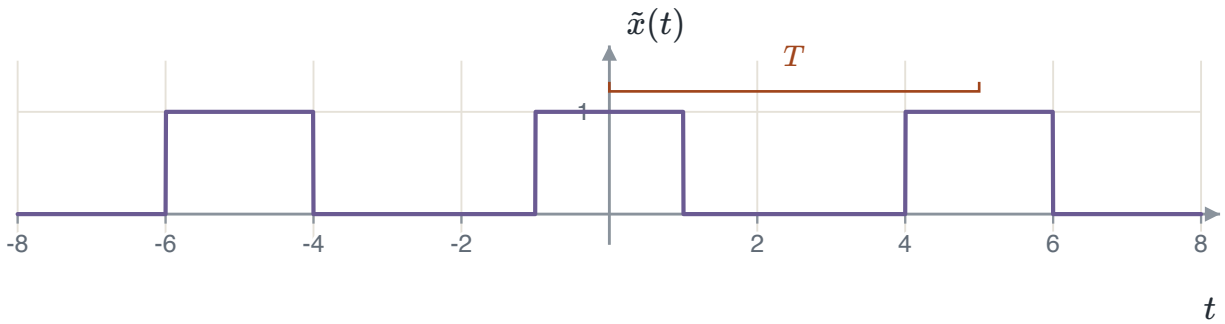
The continuous-time Fourier transform represents an aperiodic signal by a continuous function of frequency. To derive it, construct a periodic extension of a finite-duration signal and increase the repetition period without bound. The harmonic spacing then approaches zero, and the Fourier-series sum approaches an integral.

5.1 From a series to a transform

Let $x(t)$ be zero for $|t| > T_1$. The number T_1 is the half-width of the support and belongs to the signal alone. Build a periodic signal $\tilde{x}$ by repeating the pulse every T seconds, with $T > 2T_1$ so that the copies do not touch:

$$\tilde{x}(t) = \sum_{m=-\infty}^{\infty} x(t - mT), \quad \tilde{x}(t) = \tilde{x}(t + T), \quad T > 2T_1.$$

Inside one period, $\tilde{x}$ equals the original pulse, so its Fourier series can be used. Increase T while keeping the central pulse fixed. The neighbouring copies move to larger values of $|t|$, and $\tilde{x}(t) \rightarrow x(t)$ for every t .



The periodic extension with $T = 5T_1$. The condition $T > 2T_1$ is what keeps the copies apart.

The coefficients are samples of one curve

Apply the analysis equation of Chapter 4 to $\tilde{x}$ over one period. Inside that period $\tilde{x} = x$, and outside $|t| < T_1$ the integrand is zero, so the limits may be opened to all of time:

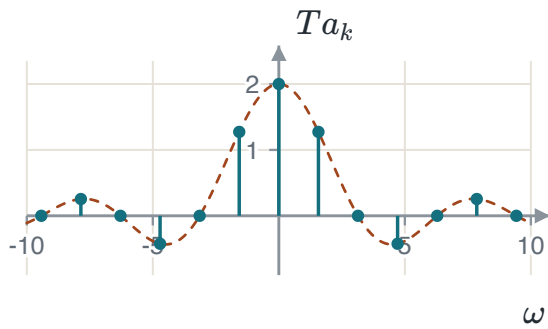
$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T}.$$

The right-hand integral has the same form for every k ; only the sampled frequency $k\omega_0$ changes. Replace that frequency by the continuous variable ω and define the transform.

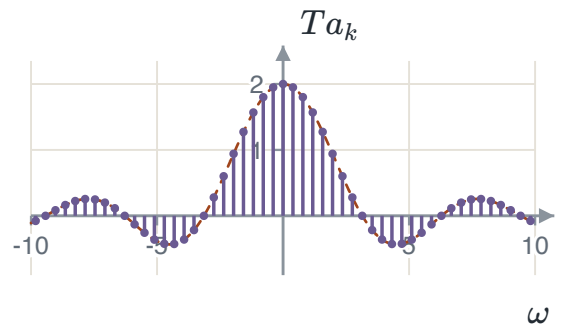
DEFINITION OF THE TRANSFORM

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Then $a_k = \frac{1}{T} X(jk\omega_0)$, so every coefficient of the periodic extension is one point of the curve X , scaled by $1/T$. The curve was built from the pulse alone, so lengthening T cannot move it. What lengthening T does is shrink the spacing $\omega_0 = 2\pi/T$ and shrink each coefficient by the same factor.



$$T = 4T_1.$$



$$T = 16T_1: \text{ the same curve, sampled four times as finely.}$$

The sum becomes an integral

Put $a_k = \frac{1}{T} X(jk\omega_0)$ back into the synthesis equation and replace $1/T$ by $\omega_0/2\pi$. Nothing has been approximated yet.

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t} \omega_0.$$

Every term now includes the spacing ω_0 , so the expression is a Riemann sum. Let $T \rightarrow \infty$. Then $\tilde{x}(t) \rightarrow x(t)$, $\omega_0 \rightarrow d\omega$, the sample frequencies cover the frequency axis, and the sum approaches an integral.

THE CONTINUOUS-TIME FOURIER TRANSFORM PAIR

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{analysis})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (\text{synthesis})$$

Analysis integrates over t and leaves a function of ω : it takes the signal apart. Synthesis integrates over ω and leaves a function of t : it puts the signal back together. The factor $1/2\pi$ arrived as $\omega_0/2\pi$ when the spacing was substituted, and it stays on the synthesis side throughout. The pair is written $x(t) \leftrightarrow X(j\omega)$, with the argument $j\omega$ and never ω alone.

WHICH EQUATION IS WHICH

The variable of integration settles it, and nothing else needs to be remembered. Integrating time away produces a spectrum, and that is analysis. Integrating frequency away produces a signal, and that is synthesis. The two exponents differ by one minus sign, so the pair is as easy to write backwards as forwards, and the names are the only part that can be exchanged by mistake.

5.2 When the transform exists

Two conditions each guarantee that $X(j\omega)$ exists. They are alternatives, joined by "or", and neither implies the other.

1. **Finite energy.** $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$.

2. **The Dirichlet conditions.** x is absolutely integrable, $\int_{-\infty}^{\infty} |x(t)| dt < \infty$, and has finitely many maxima, minima and finite jumps in any finite interval.

That they are independent is shown by one example each way. The signal $\sin(t)/t$ has finite energy, and its modulus decays only like $1/|t|$, so its area diverges. The signal $1/\sqrt{t}$ on $0 < t < 1$ has area 2, and its square is $1/t$, whose integral over the same interval diverges.

Neither condition is necessary. A constant, a complex exponential and every periodic signal fail both and still have spectra, in the limiting sense: the transform is an impulse, defined by what it does inside an integral, exactly as $\delta(t)$ is. Taking $e^{-a|t|} \leftrightarrow 2a/(a^2 + \omega^2)$ and letting $a \rightarrow 0$ gives $1 \leftrightarrow 2\pi\delta(\omega)$, with the curve growing tall and narrow while its area stays 2π .

5.3 Basic transform pairs

EXAMPLE 5.1 – THE IMPULSE AND THE SHIFTED IMPULSE

GIVEN $x(t) = \delta(t)$, and then $x(t) = \delta(t - t_0)$.

FIND $X(j\omega)$ in both cases, with magnitude and phase.

METHOD The signal is an impulse, so the sifting property evaluates the analysis integral directly. Substitute the impulse into the analysis equation.

SOLUTION $\mathcal{F}\{\delta(t)\} = \int \delta(t)e^{-j\omega t} dt = 1$: every frequency present, equally, with no phase. For the shifted impulse the sifting happens at $t = t_0$, giving

$$\mathcal{F}\{\delta(t - t_0)\} = e^{-j\omega t_0}, \quad |X(j\omega)| = 1, \quad \angle X(j\omega) = -\omega t_0.$$

CHECK Push $e^{-j\omega t_0}$ back through the synthesis equation and the impulse returns. Reading: moving a signal in time never changes the size of any frequency component; it rotates each one by an amount proportional to its frequency.

EXAMPLE 5.2 – ONE IMPULSE IN FREQUENCY

GIVEN $X(j\omega) = 2\pi\delta(\omega - \omega_0)$.

FIND $x(t)$.

METHOD The spectrum is an impulse, so use the synthesis equation and apply sifting in the variable ω .

SOLUTION $x(t) = \frac{1}{2\pi} \int 2\pi\delta(\omega - \omega_0)e^{j\omega t} d\omega = e^{j\omega_0 t}$. The 2π of the impulse weight and the $1/2\pi$ of the synthesis equation cancel exactly, which is why the weight is written as 2π and not as 1.

CHECK $|x(t)| = 1$ and $\angle x(t) = \omega_0 t$. With an impulse of weight 1 the answer would be $e^{j\omega_0 t}/2\pi$, which is not a unit-amplitude exponential.

TWO CONSEQUENCES OF THE COMPLEX-EXPONENTIAL PAIR

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

The second is the first at $\omega_0 = 0$.

EXAMPLE 5.3 – THE ONE-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-at}u(t)$, with a real.

FIND $X(j\omega)$, the condition on a , and the magnitude and phase.

METHOD The unit step makes the signal zero for negative time. Apply the analysis equation from 0 to ∞ .

SOLUTION $X(j\omega) = \int_0^\infty e^{-(a+j\omega)t} dt$. The upper limit vanishes only when $a > 0$, because $|e^{-j\omega t}| = 1$ and the decay is carried by e^{-at} alone. So

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a + j\omega}, \quad a > 0,$$

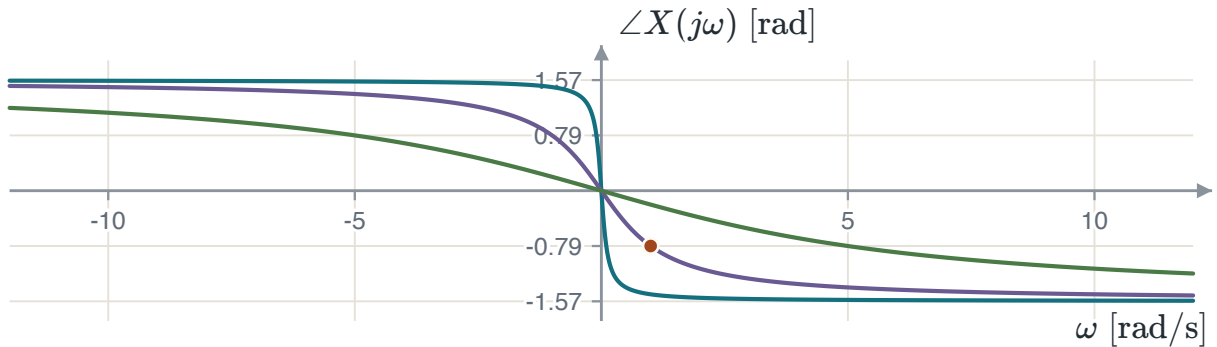
with $|X(j\omega)| = 1/\sqrt{a^2 + \omega^2}$ and, writing the angle of a quotient as a subtraction,

$$\angle X(j\omega) = \angle 1 - \angle(a + j\omega) = 0 - \tan^{-1}(\omega/a) = -\tan^{-1}(\omega/a).$$

CHECK $|X(j0)| = 1/a$ gives 10, 1 and 0.2 for $a = 0.1, 1, 5$. At $a = 1, \omega = 1$ the phase is -0.785398 rad; at $a = 0.1, \omega = 3$ it is -1.537475 ; at $a = 5, \omega = 2$ it is -0.380506 . Every one is negative, and the phase curve falls from $+\pi/2$ to $-\pi/2$. A curve that rises belongs to $+\tan^{-1}(\omega/a)$, which is the angle of the reciprocal.

WHERE THAT MINUS SIGN IS LOST

The numerator is the real number 1, so its angle is zero and it is tempting to skip that term and copy $\tan^{-1}(\omega/a)$ straight out of the denominator. What is then reported is $\angle(a + j\omega)$, the angle of the reciprocal of the answer. Write the subtraction out, including the term that is zero.



The phase of $1/(a + j\omega)$ for $a = 0.1, 1$ and 5 , with the marked value -0.785398 at $a = \omega = 1$.

EXAMPLE 5.4 – THE TWO-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-a|t|}$ with $a > 0$.

FIND $X(j\omega)$.

METHOD The absolute value gives different exponential formulas for negative and positive time. Split the analysis integral at $t = 0$; on the negative-time interval the exponent is $+at$.

SOLUTION $X(j\omega) = \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{2a}{a^2 + \omega^2}$. The imaginary parts cancelled when the fractions were added.

CHECK $X(j0) = 2/a$ gives 4, 2 and 0.4 for $a = 0.5, 1, 5$. The signal is real and even and the transform came out real and even, which is the general rule proved in Section 5.6. Note also that the transform is never zero: at $a = 1$ and $\omega = 10^6$ it is still 2×10^{-12} .

5.4 The rectangular pulse, the sinc, and the inverse relation

EXAMPLE 5.5 – THE RECTANGULAR PULSE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 otherwise.

FIND $X(j\omega)$, its value at the origin, and its zeros.

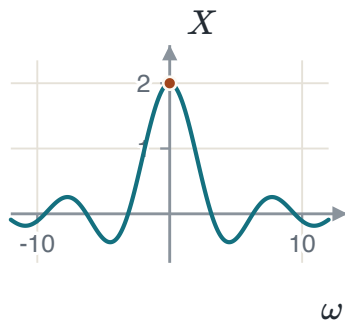
METHOD The signal is non-zero only on $-T_1 < t < T_1$, so restrict the analysis integral to that support and integrate $e^{-j\omega t}$.

SOLUTION $X(j\omega) = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T_1}^{T_1} = \frac{e^{j\omega T_1} - e^{-j\omega T_1}}{j\omega} = \frac{2 \sin(\omega T_1)}{\omega}$, using $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$. The j cancels, so the answer is real.

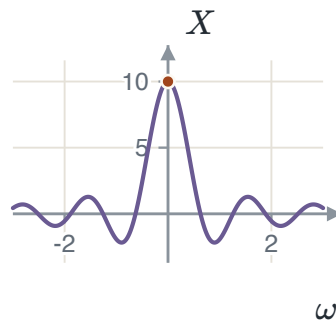
CHECK $X(j0) = 2T_1$ by l'Hôpital, which is the area under the pulse: 2, 10 and 20 for $T_1 = 1, 5, 10$. The zeros are at $\omega = \pm k\pi/T_1$ for $k = 1, 2, 3, \dots$ — the origin is excluded, because there the expression is $0/0$ and its limit is the peak.

THE SINC CONVENTION USED THROUGHOUT

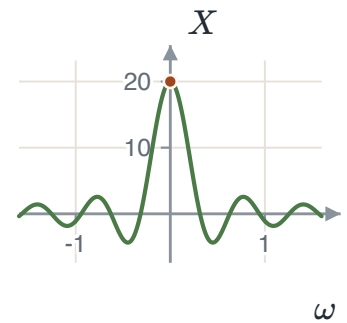
This course writes $\text{sinc}(\theta) = \frac{\sin \theta}{\theta}$, the **unnormalised** sinc, with $\text{sinc}(0) = 1$ and zeros at $\pm\pi, \pm2\pi, \dots$. In that convention the result above reads $X(j\omega) = 2T_1 \text{sinc}(\omega T_1)$. Signal-processing software and many communications texts use the **normalised** sinc, $\text{sinc}_n(\theta) = \sin(\pi\theta)/(\pi\theta)$, whose zeros are at the integers; there the same result is $2T_1 \text{sinc}_n(\omega T_1/\pi)$. The two expressions are equal, and the argument is not: it differs by a factor of π . An argument copied between the conventions agrees at the origin and is wrong everywhere else, so the convention is stated at every point of use.



$T_1 = 1$: peak 2.



$T_1 = 5$: peak 10.



$T_1 = 10$: peak 20.

EXAMPLE 5.6 – THE IDEAL LOW-PASS BAND

GIVEN $X(j\omega) = 1$ for $|\omega| < W$ and 0 otherwise.

FIND $x(t)$, its peak, and its zeros.

METHOD The spectrum is given and is non-zero only on $-W < \omega < W$, so apply the synthesis equation over those limits and retain the factor $1/2\pi$.

SOLUTION

$$x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega = \frac{e^{jWt} - e^{-jWt}}{2\pi jt} = \frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt).$$

CHECK $x(0) = W/\pi$ by l'Hôpital: 0.5, 1 and 2 for $W = 0.5\pi, \pi, 2\pi$. Zeros at $t = \pm k\pi/W$, $k = 1, 2, \dots$, again with the origin excluded. This signal is not a pulse: it rings on both sides for ever, alternating in sign.

The two examples are one statement seen twice: a rectangle in either domain is a sinc in the other. Section 5.7 gives that symmetry a name and a proof.

Narrow in time, wide in frequency

The pulse of half-width T_1 has its first zero at $\omega = \pi/T_1$. Halve T_1 and that zero doubles. Taking the duration as the full width $T = 2T_1$ and the bandwidth as the distance to the first null,

$$T \times \text{BW} = 2T_1 \cdot \frac{\pi}{T_1} = 2\pi \quad \text{at every width.}$$

THE VALUE DEPENDS ON THE WAVEFORM

The product is invariant for scaled versions of one waveform because time scaling changes duration and frequency width by reciprocal factors. A triangular pulse of the same total duration has its first null at $2\pi/T_1$, so its product is 4π . In general, narrowing a signal in time widens its spectrum; both widths cannot be made arbitrarily small.

One implication about band limitation is a theorem and its converse is not. A signal of finite duration cannot be band-limited: the pulse above has a sinc transform, which is non-zero on stretches reaching out to every frequency. But infinite duration guarantees nothing. The signal $e^{-a|t|}$ lasts for ever and its transform $2a/(a^2 + \omega^2)$ is strictly positive at every finite frequency. Small is not zero, and Chapter 7 depends on the difference.

5.5 Periodic signals, sinusoids and the impulse train

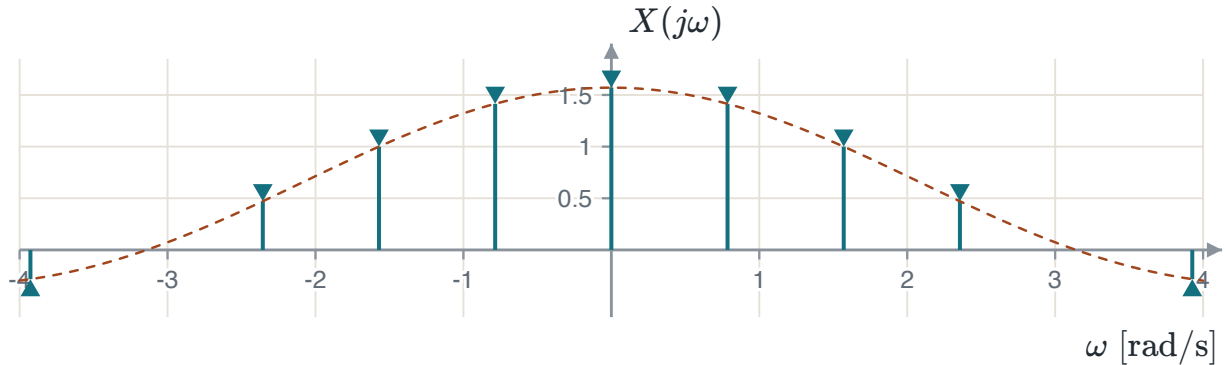
A periodic signal has a Fourier series, and the series can be transformed term by term with $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$.

TRANSFORM OF A PERIODIC SIGNAL

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

The spectrum is a train of impulses at the harmonics, and the impulse at $k\omega_0$ carries weight $2\pi a_k$. The coefficient and the weight are different objects: a_k multiplies a unit exponential, while $2\pi a_k$ is an area under a spectrum. Reporting the coefficients as the transform loses a factor of 2π at every harmonic.

For the rectangular wave of Chapter 4 with $T_1 = 1$, the weights are $2\pi a_k = 2 \sin(k\omega_0 T_1)/k$ and the impulse at the origin carries $2\pi a_0 = 4\pi T_1/T$. Three periods make the sequence visible: $T = 8T_1$ gives $\omega_0 = \pi/4 = 0.392699$ and weight 1.5708; $T = 16T_1$ gives $\omega_0 = \pi/8 = 0.196350$ and weight 0.7854; $T = 32T_1$ gives $\omega_0 = \pi/16 = 0.098175$ and weight 0.3927. Each doubling of the period halves the spacing and halves every weight, while the envelope does not move at all — which is the derivation of Section 5.1, carried out three steps at a time.



The square wave with $T = 8T_1$: impulses of weight $2\pi a_k$ sample the dashed envelope. A magnitude plot would not show which weights are negative.

EXAMPLE 5.7 – A CONSTANT, A COSINE AND A SINE

GIVEN $x(t) = 5 + 4 \cos(3\pi t) + 6 \sin(4\pi t)$.

FIND $X(j\omega)$, drawn so that both size and phase can be read.

METHOD Each sinusoid is a sum of complex exponentials, whose transform pair is known. Expand with Euler's relations and apply linearity term by term.

SOLUTION $4 \cos(3\pi t) = 2e^{j3\pi t} + 2e^{-j3\pi t}$, so its transform is $4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi)$: one impulse at each of $\pm 3\pi$. Likewise $6 \sin(4\pi t) = \frac{6}{2j} [e^{j4\pi t} - e^{-j4\pi t}]$ gives $\frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi)$, with imaginary weights of modulus $6\pi = 18.849556$. The constant contributes $10\pi\delta(\omega)$. Altogether

$$X(j\omega) = 10\pi\delta(\omega) + 4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi) + \frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi).$$

CHECK The signal is real, so the magnitude must be even in ω and the phase odd. Magnitudes: 31.4159 at 0, 12.5664 at $\pm 3\pi$, 18.8496 at $\pm 4\pi$. Phases: 0 except at $\pm 4\pi$, where they are $\mp\pi/2$ because $1/j = -j$. A spectrum with only the positive-frequency impulse violates conjugate symmetry and describes a complex signal.

DRAWING A COMPLEX SPECTRUM

Three weights are real and two are imaginary. One vertical axis cannot represent both signed real values and signed imaginary values. Plot magnitude and phase, or plot real part and imaginary part, and identify the chosen pair in the caption.

EXAMPLE 5.8 – THE IMPULSE TRAIN

GIVEN $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$.

FIND $X(j\omega)$.

METHOD The signal is periodic, so its transform is an impulse at each harmonic. Find its Fourier-series coefficients a_k , then give each impulse the weight $2\pi a_k$.

SOLUTION Over one period exactly one impulse is enclosed, and sifting evaluates the exponential at $t = 0$, so $a_k = 1/T$ for every k . Hence

$$\sum_k \delta(t - kT) \longleftrightarrow \frac{2\pi}{T} \sum_k \delta\left(\omega - \frac{2\pi k}{T}\right).$$

CHECK Spacing and weight are the same number, $2\pi/T$: 6.2832 for $T = 1$ and 3.1416 for $T = 2$. Crowding the impulses in time spreads them in frequency. This one pair is the mechanism behind sampling, in Chapter 7. Note that the limits must enclose exactly one impulse; writing both as $-T/2$ encloses none.

5.6 Properties

PROPERTY	STATEMENT
Linearity	$ax_1 + bx_2 \leftrightarrow aX_1 + bX_2$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Real signal	$X(-j\omega) = X^*(j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Differentiation in time	$d^n x/dt^n \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j dX(j\omega)/d\omega$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Convolution	$x * h \leftrightarrow XH$
Multiplication	$xy \leftrightarrow \frac{1}{2\pi} X * Y$
Real and even	$X(j\omega)$ real and even
Real and odd	$X(j\omega)$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}$, $\mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\int x ^2 dt = \frac{1}{2\pi} \int X ^2 d\omega$

Shifts

For the time shift, substitute $\tau = t - t_0$; the limits are infinite and do not move, and $e^{-j\omega t_0}$ comes out of the integral. Since its modulus is 1, no magnitude changes at any frequency: a delay of t_0 seconds is exactly a linear phase of slope $-t_0$.

For the frequency shift, start from the expression the property states, $e^{+j\omega_0 t} x(t)$, and combine the two exponentials before anything else. The exponent becomes $-j(\omega - \omega_0)t$, which is the analysis integral read at $\omega - \omega_0$, and no substitution is needed. The operand is a complex exponential in **time** with ω_0 fixed; the time-shift kernel $e^{-j\omega t_0}$ has a fixed **time** and the opposite sign, and opening this proof with it proves a different statement.

Conjugation and the symmetry of a real signal

Conjugating the analysis integral flips the sign of j everywhere, which gives $x^*(t) \leftrightarrow X^*(-j\omega)$. If x is real then $x^* = x$, so the two sides have the same transform and $X(-j\omega) = X^*(j\omega)$. Everything follows from that one line: the real part of X is even, the imaginary part is odd, the magnitude is even and the phase is odd. A real and even signal has a real and even transform, since its imaginary part is both odd and even; a real and odd signal has a purely imaginary and odd transform.

REAL DOES NOT MEAN ZERO PHASE

A real transform can be negative. Where $X(j\omega) < 0$ the magnitude is $-X$ and the phase is π , not 0. Only a real and non-negative transform has zero phase everywhere, and the sinc of the rectangular pulse is the standing counterexample: it is real, and its side lobes are negative.

Differentiation

Differentiate the **synthesis equation**, not the signal. On the right the only factor depending on t is $e^{j\omega t}$, and ω is the variable of integration, so it is held fixed:

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [j\omega X(j\omega)] e^{j\omega t} d\omega \quad \implies \quad \frac{d^n x}{dt^n} \longleftrightarrow (j\omega)^n X(j\omega).$$

DO NOT MIX THE TWO DOMAINS

The equation $dx/dt = j\omega x(t)$ is false. The frequency ω labels the transform output; it is not a constant in the time-domain signal. For $x(t) = e^{-t^2}$, dx/dt at $t = 1$ is -0.735759 , while $j\omega x(t)$ at $t = 1$, $\omega = 3$ is $1.103638j$. The differentiation property is correct only when derived from the transform integral.

Integration, and the impulse it leaves behind

The running integral is a convolution with the unit step, $\int_{-\infty}^t x(\tau) d\tau = x(t) * u(t)$, and the step transforms to $\frac{1}{j\omega} + \pi\delta(\omega)$. The convolution property multiplies the transforms, and $X(j\omega)\delta(\omega) = X(0)\delta(\omega)$ by the sampling property of the impulse:

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega).$$

INCLUDE THE IMPULSE TERM WHEN THE AREA IS NON-ZERO

$X(0) = \int x(t) dt$ is the total signal area. If this area is non-zero, the running integral approaches a non-zero constant, whose transform includes an impulse at $\omega = 0$. Writing only $X(j\omega)/(j\omega)$ omits that constant component. Calculate the area before applying the integration property.

Differentiation in frequency

Differentiate the analysis integral with respect to ω . The integrand gains the factor $-jt$, so $dX/d\omega$ is the transform of $-jt x(t)$. Therefore $t x(t) \leftrightarrow j dX/d\omega$. This property obtains the pair for $te^{-at}u(t)$ from the pair for $e^{-at}u(t)$ without another transform integral.

Even and odd parts

For a real signal, $\mathcal{E}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ transforms to $\frac{1}{2}[X(j\omega) + X(-j\omega)] = \frac{1}{2}[X + X^*] = \text{Re}\{X\}$, and the odd part gives $j \text{Im}\{X\}$. Splitting a real signal into even and odd parts splits its spectrum into real and imaginary parts, and the two symmetry rules above are the cases in which one part is absent. On $e^{-at}u(t)$ the even part is $\frac{1}{2}e^{-a|t|}$, whose transform $a/(a^2 + \omega^2)$ is exactly $\text{Re}\{1/(a + j\omega)\}$; the odd part is $\frac{1}{2}\text{sgn}(t)e^{-a|t|}$, whose transform $-j\omega/(a^2 + \omega^2)$ is $j \text{Im}\{1/(a + j\omega)\}$.

Time scaling

Substitute $\tau = at$, so $t = \tau/a$ and $dt = d\tau/a$. For $a > 0$ the limits keep their order and the coefficient is $1/a = 1/|a|$. For $a < 0$ the substitution sends $t \rightarrow -\infty$ to $\tau \rightarrow +\infty$, so the limits arrive reversed; swapping them back costs one minus sign, and $-1/a = 1/|a|$ for negative a . That is the only sign in the calculation.

COUNTING THE FLIP TWICE

Writing the reversed limits **and** an explicit -1 in front applies the same correction twice and gives $-\frac{1}{|a|} X(j\omega/a)$, contradicting the property. Do the flip once: reverse the limits, or write the minus, never both. The case $a = -1$ is time reversal, $x(-t) \leftrightarrow X(-j\omega)$, with coefficient 1.

Worked instance. If $X(j\omega) = 1$ on $|\omega| < 2\pi$, then $x(0.5t) \leftrightarrow 2X(j2\omega)$, which is 2 on $|\omega| < \pi$, and $x(2t) \leftrightarrow 0.5X(j\omega/2)$, which is 0.5 on $|\omega| < 4\pi$. The three areas are $2 \cdot 2\pi$, $1 \cdot 4\pi$ and $0.5 \cdot 8\pi$, all 4π ; since $\int X d\omega = 2\pi x(0)$, the value at $t = 0$ is the same for all three, which is what a time scaling cannot change.

5.7 Duality

The two equations of the pair differ only by a sign and a factor, so any pair can be read a second time with the domains exchanged.

DUALITY

$$x(t) \longleftrightarrow X(j\omega) \implies X(t) \longleftrightarrow 2\pi x(-\omega)$$

The argument on the right is $-\omega$, a real number, and not $-j\omega$: the letter x names a signal and a signal takes a real argument. The j belongs to the frequency-domain function only, and duality is the point in the chapter where that distinction is most easily lost.

Proof. Rename the variables of the synthesis equation, swapping t and ω , to get $2\pi x(\omega) = \int X(jt)e^{j\omega t} dt$. Now replace ω by $-\omega$; the exponent becomes the analysis exponent and the left side becomes $2\pi x(-\omega)$, so the right side is $\mathcal{F}\{X(t)\}$.

EXAMPLE 5.9 – DUALITY ON THE RECTANGULAR PULSE

- GIVEN** $x_1(t) = 1$ on $|t| < W$, so $X_1(j\omega) = 2 \sin(W\omega)/\omega$.
- FIND** The transform of $x_2(t) = 2 \sin(Wt)/t$.
- METHOD** x_2 has the same formula as X_1 with the independent variable changed, so duality applies. Use duality first, then verify the result with synthesis.
- SOLUTION** Duality gives $X_2(j\omega) = 2\pi x_1(-\omega)$, and x_1 is even, so

$$X_2(j\omega) = 2\pi \text{ on } |\omega| < W, \quad 0 \text{ beyond.}$$

- CHECK** The second route works backwards: putting a band of height 2π on $|\omega| < W$ through the synthesis equation gives $\frac{1}{2\pi} \int_{-W}^W 2\pi e^{j\omega t} d\omega = 2 \sin(Wt)/t$, which is x_2 . The two routes use different equations, so their agreement is a real check. At the origin, $X_2(j0) = 2\pi = 6.283185$ for any W , and $x_2(0) = 2W$ by l'Hôpital.
-

Duality also carries properties across: the time-shift rule becomes the frequency-shift rule, and differentiation in time becomes differentiation in frequency. Every pair already derived gives a second one for nothing.

5.8 Parseval's relation

Energy and power in this course are normalised, with $R = 1 \Omega$, so the instantaneous power is $|x(t)|^2$ and the energy is its integral, in joules.

PARSEVAL'S RELATION

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

The proof writes $|x|^2 = x x^*$, replaces x^* by the conjugate of the synthesis equation and exchanges the two integrals; the inner one is then $X(j\omega)$. The step actually used is $|x(t)|^2 = x^2(t)$ for a real signal, not $|x(t)| = x(t)$, which is a stronger claim and fails wherever a real signal is negative. The quantity $|X(j\omega)|^2$ is the energy spectral density: $\frac{1}{2\pi} |X|^2 d\omega$ is the energy in a narrow band.

EXAMPLE 5.10 – ENERGY FROM A TWO-BAND SPECTRUM

- GIVEN** $X_3(j\omega) = 2$ for $|\omega| < 2\pi$, 1 for $2\pi < |\omega| < 4\pi$, and 0 beyond.
- FIND** The total energy.
- METHOD** The spectrum is piecewise constant, so the frequency-domain energy integral is simpler than the time-domain integral. Apply Parseval, square each height, and multiply by the corresponding width.
- SOLUTION** $E_\infty = \frac{1}{2\pi} [2\pi + 16\pi + 2\pi] = \frac{20\pi}{2\pi} = 10 \text{ J}$. The inner band contributes $2^2 \cdot 4\pi = 16\pi$ and the two outer bands $1^2 \cdot 2\pi$ each.
- CHECK** Synthesis gives $x_3(t) = [\sin(2\pi t) + \sin(4\pi t)]/(\pi t)$, and integrating x_3^2 over all time returns 10 J as well. Note that $\frac{1}{2\pi} \int X_3 d\omega = 6$, which is the peak $x_3(0)$ and not the energy: using the heights unsquared computes the wrong quantity correctly.
-

The same relation on $e^{-at}u(t)$ is worth doing both ways. In time, $E_\infty = \int_0^\infty e^{-2at} dt = 1/(2a)$. In frequency, $\frac{1}{2\pi} \int d\omega/(a^2 + \omega^2) = \frac{1}{2\pi} \cdot \frac{\pi}{a} = 1/(2a)$, using $\int du/(1+u^2) = \pi$ after $u = \omega/a$. Without the $1/2\pi$ the second route reports 2π times too much.

5.9 Convolution and multiplication

THE TWO DUAL PROPERTIES

$$y(t) = x(t) * h(t) \longleftrightarrow Y(j\omega) = X(j\omega)H(j\omega)$$

$$z(t) = x(t)y(t) \longleftrightarrow Z(j\omega) = \frac{1}{2\pi}X(j\omega) * Y(j\omega)$$

The premise is stated with one set of symbols: $x \leftrightarrow X$ is the input, $h \leftrightarrow H$ the impulse response, y the output. Convolution in time carries no factor; convolution in frequency carries $1/2\pi$. $H(j\omega)$ exists when h is absolutely integrable, which for an LTI system is exactly bounded-input bounded-output stability, so an unstable system has no frequency response to plot.

The proof of the first writes the convolution inside the analysis integral and exchanges the order; the inner bracket is then the time-shift property applied to h , and $H(j\omega)$, not depending on τ , leaves the outer integral.

EXAMPLE 5.11 – TWO EXPONENTIALS IN CASCADE

GIVEN $x(t) = e^{-at}u(t)$ and $h(t) = e^{-bt}u(t)$, with $a, b > 0$ and $a \neq b$.

FIND $y(t) = x * h$.

METHOD The output is a convolution in time, so it becomes multiplication in frequency. Multiply the transforms, expand the product into simple fractions, and invert each term.

SOLUTION $Y = \frac{1}{(a + j\omega)(b + j\omega)} = \frac{A}{a + j\omega} + \frac{B}{b + j\omega}$ with $A = \frac{1}{b - a}$ and $B = -A$, so

$$y(t) = \frac{e^{-at} - e^{-bt}}{b - a} u(t).$$

CHECK $y(0) = 0$, as a convolution of two causal signals must be. For $a = 1, b = 2$ the peak is at $t = \ln 2 = 0.693147$ with value exactly $1/4$. At $\omega = 0$: $|X| = 1, |H| = 0.5$ and $|Y| = 0.5$, and the product of the first two is the third – the property itself at one frequency.

EXAMPLE 5.12 – IDEAL FILTERS IN CASCADE

GIVEN $X(j\omega) = 2$ on $|\omega| \leq 4\pi$; an ideal low-pass system with $H(j\omega) = 3$ on $|\omega| \leq 2\pi$.

FIND $Y(j\omega), y(t)$ and the three time-domain peaks.

METHOD An LTI system gives $Y = XH$. Multiply the input spectrum and frequency response at each frequency, then apply the inverse transform.

SOLUTION Beyond 2π the system contributes zero, so $Y = 6$ on $|\omega| \leq 2\pi$ and zero elsewhere; the narrower band decides. Inverting, $y(t) = 6 \sin(2\pi t)/(\pi t)$.

CHECK Each peak is the area of its own band divided by 2π : $x(0) = 8, h(0) = 6, y(0) = 12$. The output peak is the largest because a peak counts area, and $6 \times 4\pi$ exceeds $2 \times 8\pi$.

5.10 Amplitude modulation

A cosine of frequency ω_c is called a carrier. Multiplying by it convolves the signal spectrum with two impulses. The factor $1/2\pi$ in the multiplication property and the weight π of each impulse give a factor $\frac{1}{2}$ for each shifted copy.

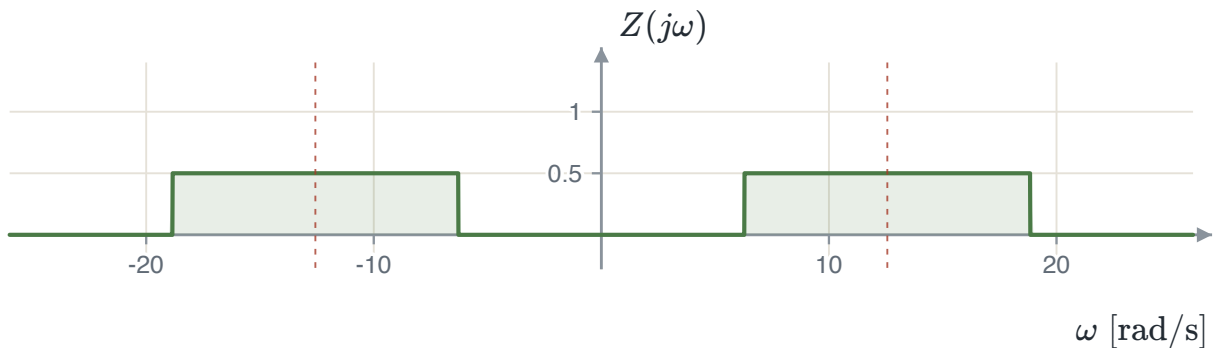
DOUBLE-SIDEBAND SUPPRESSED-CARRIER MODULATION

$$z(t) = x(t) \cos(\omega_c t) \longleftrightarrow Z(j\omega) = \frac{1}{2}X(j(\omega - \omega_c)) + \frac{1}{2}X(j(\omega + \omega_c))$$

The spectrum is not moved to ω_c ; it is **uplicated**, one copy at $+\omega_c$ and one at $-\omega_c$, each at half height. A description that says "the signal moved up to the carrier" loses the negative-frequency copy and the factor of one half in the same sentence.

With $x(t) = \cos(\pi t)$ and carrier $\cos(4\pi t)$, the product-to-sum identity gives $\frac{1}{2} \cos(5\pi t) + \frac{1}{2} \cos(3\pi t)$, whose transform has impulses of weight $\pi/2 = 1.570796$ at $\pm 3\pi$ and $\pm 5\pi$. The property gives the same four positions, $\pm(\omega_c \pm \omega_1)$. Nothing sits at $\omega_c = 4\pi$ itself, which is what "suppressed carrier" records.

With a band-limited signal the picture is cleaner. If $X = 1$ on $|\omega| < 2\pi$ and the carrier is $\cos(4\pi t)$, the copies occupy $2\pi \leq |\omega| \leq 6\pi$ at height 0.5: the same width as the original band, at half the height, and twice as much of the axis occupied because there are now two of them.



Two half-height copies on $2\pi \leq |\omega| \leq 6\pi$. The dashed lines mark $\pm\omega_c$, where nothing sits.

TWO DIFFERENT EVENTS

Copies appear whenever a signal is multiplied by a carrier. That happens at every carrier frequency and it loses nothing.

Copies overlap only when the carrier is low enough for the shifted bands to reach each other. Overlap is what destroys information, because once two copies have been added there is no way to tell what each contributed. If a band on $\pi \leq |\omega| \leq 3\pi$ is modulated by $\cos(2\pi t)$, the two half-height copies both land on $|\omega| \leq \pi$ and add to 1 there, while only one reaches $3\pi \leq |\omega| \leq 5\pi$, where the height stays 0.5. Chapter 7 asks the same question about the copies that sampling produces.

Products of two band-limited signals follow the same arithmetic. Two rectangles of height A and half-width ω_0 convolve to a triangle of apex $2A^2\omega_0$ on $|\omega| \leq 2\omega_0$; with the $1/2\pi$, the transform of $[\sin(2\pi t)/(\pi t)]^2$ is a triangle of apex 2 on $|\omega| \leq 4\pi$, and the time-domain peak is $2^2 = 4$. With unequal half-widths 2π and 4π the result is a trapezoid of height 2, flat on $|\omega| \leq 2\pi$ and zero beyond 6π , with time-domain peak $2 \times 4 = 8$. The flat top is as wide as the difference of the two half-widths, and it shrinks to a point when they are equal.

5.11 Systems described by a differential equation

The Fourier transform converts a linear differential equation with constant coefficients into an algebraic equation. Apply linearity to the sums and replace each derivative d^k/dt^k by the factor $(j\omega)^k$.

FREQUENCY RESPONSE FROM THE COEFFICIENTS

$$\sum_{k=0}^N a_k \frac{d^k y}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x}{dt^k} \implies H(j\omega) = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k}$$

The differential-equation coefficients form two polynomials in $j\omega$, and their ratio is the frequency response. Next calculate $Y = XH$, expand it into simple fractions, and invert each term with the transform table. During the algebra, define $s = j\omega$ and substitute back at the end.

EXAMPLE 5.13 – SIMPLE POLES

GIVEN $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 3y = \frac{dx}{dt} + 2x$, at rest.

FIND $H(j\omega)$ and $h(t)$.

METHOD The differential equation has constant coefficients, so the differentiation property turns it into algebra. Form the frequency-response ratio, factor it, expand it into simple fractions, and invert each term.

SOLUTION $H = \frac{s+2}{s^2+4s+3} = \frac{s+2}{(s+1)(s+3)}$ with $s = j\omega$. Covering up $(s+1)$ and setting $s = -1$ gives $A = \frac{1}{2}$; covering up $(s+3)$ and setting $s = -3$ gives $B = \frac{1}{2}$. Hence $h(t) = [\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t}]u(t)$.

CHECK $H(j0) = 2/3$, and $\int_0^\infty h dt = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$. Also $h(0^+) = 1$. The poles are at $s = -1$ and $s = -3$, both with negative real part, so h is absolutely integrable and H was entitled to exist.

Repeated poles

The cover-up rule assumes each factor of the denominator appears once. When a factor is repeated, one term is needed per power, and covering up reaches only the highest of them.

REPEATED-POLE RULE

$$F(s) = \frac{N(s)}{(s - \lambda)^m Q(s)}, \quad c_{m-k} = \frac{1}{k!} \frac{d^k}{ds^k} [(s - \lambda)^m F(s)] \Big|_{s=\lambda}$$

$k = 0$ is the cover-up rule and gives c_m ; each further derivative peels off one more coefficient. The inverse transforms needed are $1/(s + a) \leftrightarrow e^{-at}u(t)$ and $1/(s + a)^2 \leftrightarrow t e^{-at}u(t)$: a repeated pole always brings a factor of t into the time domain.

EXAMPLE 5.14 – A REPEATED POLE, AND THE CHECK THAT CATCHES A SIGN

GIVEN The system of Example 5.13, with input $x(t) = e^{-t}u(t)$.

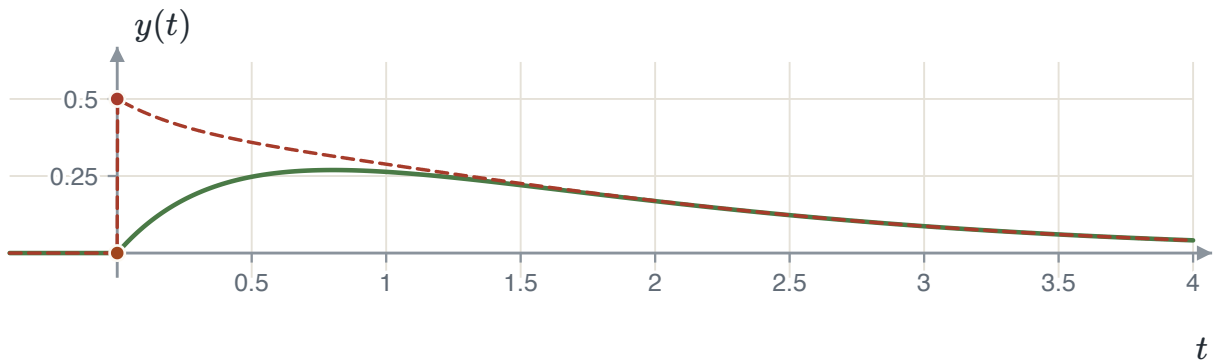
FIND $y(t)$.

METHOD The input passes through an LTI system, so calculate $Y = XH$. Define $s = j\omega$, expand Y into partial fractions, and invert each term.

SOLUTION $Y = \frac{1}{s+1} \cdot \frac{s+2}{(s+1)(s+3)} = \frac{s+2}{(s+1)^2(s+3)}$: the input pole coincides with a system pole, so $s = -1$ is now double. Then $B = [(s+1)^2 Y]_{s=-1} = \frac{1}{2}$, $A = \frac{d}{ds} [(s+1)^2 Y]_{s=-1} = \frac{1}{(s+3)^2} \Big|_{s=-1} = \frac{1}{4}$, and $C = [(s+3)Y]_{s=-3} = -\frac{1}{4}$. Hence

$$y(t) = \left[\frac{1}{4}e^{-t} + \frac{1}{2}te^{-t} - \frac{1}{4}e^{-3t} \right] u(t).$$

CHECK $y(0) = \frac{1}{4} + 0 - \frac{1}{4} = 0$, as a convolution of two causal signals must be. Assembling C as $+\frac{1}{4}$ instead gives $y(0) = \frac{1}{2}$, which no causal convolution can. The two candidates agree to three decimals past $t = 2 - 0.168549$ against 0.169789 — so only the value at the origin separates them. Convolution directly returns the same three terms with the same signs.



The output with $C = -\frac{1}{4}$ and, dashed, the version with the sign lost. They are indistinguishable past $t = 2$ and differ by 0.5 at the origin.

5.12 Summary

- Analysis integrates over t and returns $X(j\omega)$; synthesis integrates over ω , carries $1/2\pi$, and returns $x(t)$.
- Finite energy **or** the Dirichlet conditions is enough; neither is necessary, and signals meeting neither have impulse spectra in the limiting sense.
- A periodic signal transforms to impulses of weight $2\pi a_k$ at the harmonics — not to the coefficients themselves.
- The sinc convention here is $\text{sinc}(\theta) = \sin \theta / \theta$, and it is stated at every point of use.
- A real signal has an even magnitude and an odd phase, and both halves of its spectrum are kept.
- Narrowing a signal in time widens its spectrum. Finite duration forbids band limitation; infinite duration guarantees nothing.
- Convolution in time is multiplication in frequency; multiplication in time is convolution in frequency, with $1/2\pi$.
- A carrier makes two copies at half height. Copies always appear; overlap is a separate event and is what loses information.
- A differential equation gives $H(j\omega)$ directly; a repeated pole needs the derivative rule, and a causal convolution must start at zero.

Everything here was continuous time. A computer sees a sequence, not a signal, and the same question has to be asked again for $x[n]$. One thing changes and it changes everything: $e^{-j(\omega+2\pi)n} = e^{-j\omega n}$ for every integer n , so a discrete-time spectrum repeats with period 2π . That is Chapter 6.

The discrete-time Fourier transform

The discrete-time Fourier transform describes the frequency content of a sequence that is not periodic. We derive it in three steps: replicate a finite sequence to make it periodic, apply the series from Chapter 4, and let the period grow without bound. The result is a continuous function of frequency that repeats every 2π . Every figure shows more than one period so that this defining property remains visible.

6.1 From a periodic sequence to an aperiodic one

Let $x[n]$ have finite support: $x[n] = 0$ for $|n| > N_1$. The discrete-time Fourier series applies to periodic sequences only, so build a periodic sequence out of x by laying the same finite sequence down again every N samples.

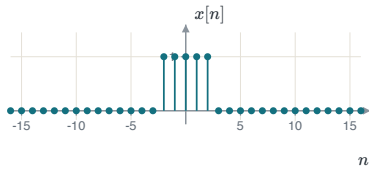
PERIODIC REPLICATION

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - rN]$$

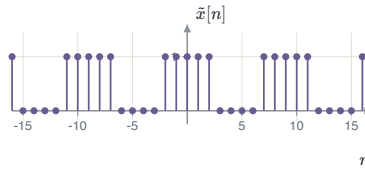
Periodic replication places a copy of the complete sequence every N samples. It does not insert samples within a copy.

THE CONDITION THE CONSTRUCTION NEEDS

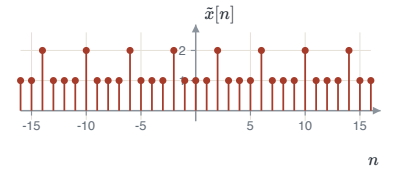
The copies must not overlap, so the period must be longer than the support: $N > 2N_1$. Only then is $\tilde{x}[n] = x[n]$ for $-N_1 \leq n \leq N_1$, a band of $2N_1 + 1$ samples centred on the origin. With $N \leq 2N_1$ the tails of neighbouring copies add and no period of $\tilde{x}$ equals x any more. Both ends of the interval point the same way; reversing the second relation would leave an interval with nothing in it.



The finite-support sequence, $N_1 = 2$.



Replication with $N = 9 > 2N_1$: the copies stand clear.



Replication with $N = 4 \leq 2N_1$: the copies add.

The coefficients are samples of one function

DISCRETE-TIME FOURIER SERIES

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}, \quad \omega_0 = \frac{2\pi}{N}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk\omega_0 n}$$

The analysis exponent is negative. Synthesis builds the sequence up out of exponentials and carries $e^{+jk\omega_0 n}$; analysis takes it apart and carries the conjugate. A plus sign in both places breaks the identity derived immediately below.

Over one period, $\tilde{x}[n] = x[n]$. Because x is zero outside $-N_1 \leq n \leq N_1$, extending the sum to all n adds only zero terms:

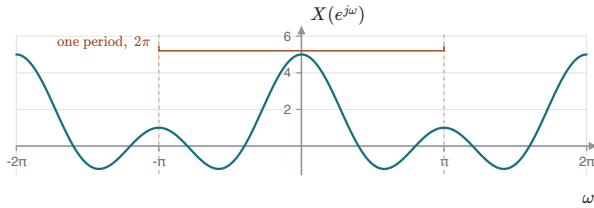
$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_1} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}.$$

The last sum has the same form for every k . Define it as a function of the continuous variable ω , and then evaluate it at $\omega = k\omega_0$.

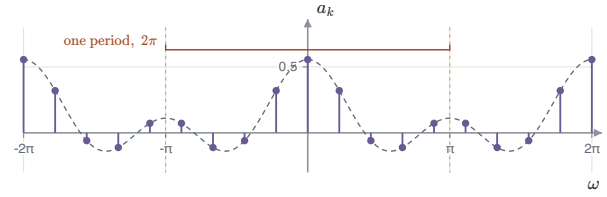
THE ENVELOPE

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \implies a_k = \frac{1}{N}X(e^{jk\omega_0})$$

The series coefficients are samples of the envelope, spaced $\omega_0 = 2\pi/N$ apart and scaled by $1/N$. The identity holds only because the two exponents match, which is the check on the sign.



The envelope of the pulse with $N_1 = 2$, over two periods. Its value at $\omega = 0$ is $2N_1 + 1 = 5$.



The coefficients for $N = 9$: the dashed envelope divided by N , sampled at $\omega = k\omega_0$.

Letting the period grow

Put $a_k = \frac{1}{N}X(e^{jk\omega_0})$ back into the synthesis equation and write $\frac{1}{N}$ as $\frac{\omega_0}{2\pi}$:

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} \frac{1}{N}X(e^{jk\omega_0})e^{jk\omega_0 n} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(e^{jk\omega_0})e^{jk\omega_0 n} \omega_0.$$

Interpret the right-hand side as a sum of rectangles. Each rectangle has height $X(e^{j\omega})e^{j\omega n}$ at a sampled frequency and width ω_0 . The N rectangles cover an interval of length $N\omega_0 = 2\pi$. As N grows without bound, the replicated copies separate and $\tilde{x}[n]$ approaches $x[n]$. At the same time, $\omega_0 = 2\pi/N$ approaches zero, so the rectangle sum approaches an integral over an interval of length 2π .

THE DISCRETE-TIME FOURIER TRANSFORM PAIR

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (\text{analysis})$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{synthesis})$$

Analysis takes the sequence to its spectrum; synthesis rebuilds the sequence from it. The names go by what each equation does, never by which is written first. Three details separate this pair from the continuous-time one: the factor $\frac{1}{2\pi}$ sits on the synthesis side, the integral runs over one period rather than the whole line, and the left-hand side is written $X(e^{j\omega})$ to record that it is a function of $e^{j\omega}$ and therefore repeats.

WHEN THE SUM CONVERGES

A sufficient condition is that x is absolutely summable, $\sum_n |x[n]| < \infty$. Then the analysis sum converges at every ω and $X(e^{j\omega})$ is continuous. Finite-energy sequences also have a transform, in a mean-square sense.

6.2 Why the spectrum repeats

PERIODICITY IN THE FREQUENCY VARIABLE

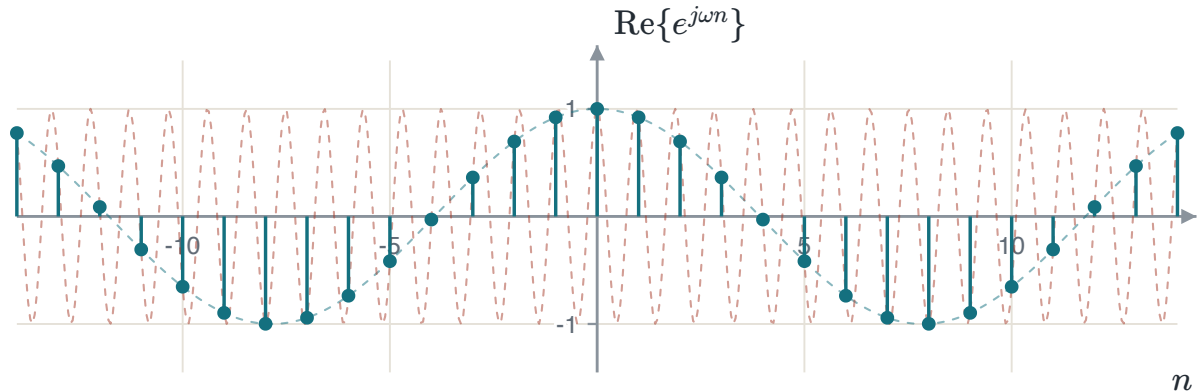
$$X(e^{j(\omega+2\pi)}) = \sum_n x[n]e^{-j(\omega+2\pi)n} = \sum_n x[n]e^{-j\omega n}e^{-j2\pi n} = X(e^{j\omega})$$

The whole argument is that n is an integer, so $e^{-j2\pi n} = 1$ in every term of the sum and the extra factor disappears term by term.

The same step in continuous time gives $\int x(t)e^{-j\omega t}e^{-j2\pi t} dt$, where t runs over the reals. There $e^{-j2\pi t}$ equals 1 only at integer t , so it cannot leave the integral, and nothing forces $X(j\omega)$ to repeat.

WHAT FREQUENCY MEANS IN DISCRETE TIME

A sequence $e^{j\omega n}$ is unchanged when ω moves by 2π , because it is only ever sampled at integers. High frequency therefore means ω near $\pm\pi$, not ω large. The fastest sequence a discrete-time signal can carry is $e^{j\pi n} = (-1)^n$.



Two continuous curves, $\omega = 0.4$ and $\omega = 0.4 + 2\pi$, and the one stem sequence they share. At the integers they agree exactly, so no sequence can tell them apart.

THE HABIT THIS CHAPTER IS BUILT TO PREVENT

A spectrum drawn on $-\pi \leq \omega \leq \pi$ alone is the same information as three periods of it, and it is also the picture a reader later mistakes for a spectrum that stops at $\pm\pi$. Every spectrum here is drawn over more than one period with the period marked.

6.3 Four transforms worked out

EXAMPLE 6.1 – A SHIFTED UNIT SAMPLE

GIVEN $x[n] = \delta[n - n_0]$.

FIND $X(e^{j\omega})$, its magnitude and its phase.

METHOD Use the analysis sum because the sequence is given in time. The unit sample makes every term with $n \neq n_0$ equal to zero, so only one term remains.

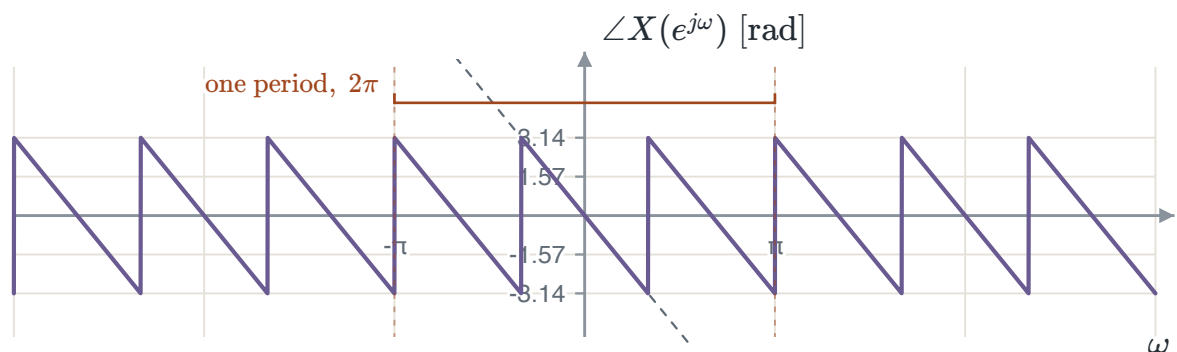
SOLUTION

$$X(e^{j\omega}) = \sum_n \delta[n - n_0] e^{-j\omega n} = e^{-j\omega n_0}.$$

So $|X(e^{j\omega})| = 1$ at every frequency and $\angle X(e^{j\omega}) = -n_0\omega$, a straight line of slope $-n_0$.

CHECK At $n_0 = 0$ the sequence is $\delta[n]$ and the transform is the constant 1, which is what the sum gives directly.

READING A straight line is not periodic, and yet the transform is. What is drawn is the principal value of the phase, which jumps by 2π whenever the line leaves $(-\pi, \pi]$. The sawtooth is that wrapping, and its period is $2\pi/|n_0|$.



The phase for $n_0 = 3$. The dashed line is the unwrapped $-n_0\omega$; the sawtooth is its principal value. Both describe the same transform.

EXAMPLE 6.2 – THE ONE-SIDED EXPONENTIAL

GIVEN $x[n] = a^n u[n]$ with $|a| < 1$.

FIND $X(e^{j\omega})$, and the extremes of its magnitude and phase.

METHOD Use the geometric-series formula because the one-sided sequence produces powers of the same ratio $ae^{-j\omega}$. The stated condition makes the infinite sum converge.

SOLUTION

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}},$$

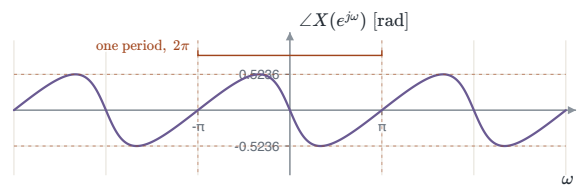
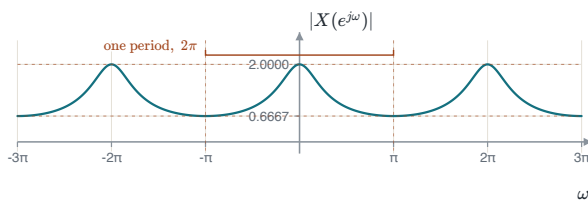
which converges because $|ae^{-j\omega}| = |a| |e^{-j\omega}| = |a| < 1$. Then

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 - 2a \cos \omega + a^2}}, \quad \angle X(e^{j\omega}) = -\arctan \frac{a \sin \omega}{1 - a \cos \omega}.$$

EXTREMES Write the denominator as $1 + ae^{j(\pi-\omega)}$: a circle of radius $|a|$ about the point 1. Its nearest point to the origin gives $|X|_{\max} = 1/(1 - |a|)$, its furthest gives $|X|_{\min} = 1/(1 + |a|)$, and the tangent from the origin gives $\max |\angle X| = \arcsin |a|$, reached where $\cos \omega = a$. For positive a the largest magnitude is at $\omega = 0$ and the smallest at $\omega = \pm\pi$; for negative a the two ends swap.

NUMBERS $a = \frac{1}{2}$: magnitude between $\frac{2}{3}$ and 2, and $\max |\angle X| = \arcsin \frac{1}{2} = \pi/6 = 0.5236$ rad exactly. $a = \frac{1}{8}$: magnitude between $\frac{8}{9} = 0.8889$ and $\frac{8}{7} = 1.1429$, and $\max |\angle X| = 0.1253$ rad.

CHECK Every one of these is a closed form, so none has to be read off the frame of a plot. An extreme that lands exactly on an axis limit is the one place a printed number is likeliest to be the limit rather than the value.



Magnitude for $a = \frac{1}{2}$, touching $1/(1 - a) = 2$ and $1/(1 + a) = 2/3$.

Phase for $a = \frac{1}{2}$, touching $\pm \arcsin \frac{1}{2} = \pm 0.5236$ rad at $\omega = \mp \pi/3$.

EXAMPLE 6.3 – THE TWO-SIDED EXPONENTIAL

GIVEN $x[n] = a^{|n|}$ with $|a| < 1$.

FIND $X(e^{j\omega})$ and its extremes.

METHOD Split the analysis sum at $n = 0$ because the absolute value gives different exponents on the two sides. Each remaining sum is geometric and converges under the stated condition.

SOLUTION

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}} + \frac{ae^{j\omega}}{1 - ae^{j\omega}} = \frac{1 - a^2}{1 - 2a \cos \omega + a^2}.$$

EXTREMES At $\omega = 0$ the value is $(1 + a)/(1 - a)$; at $\omega = \pm\pi$ it is $(1 - a)/(1 + a)$. For $a = \frac{1}{2}$ these are 3 and $\frac{1}{3}$; for $a = \frac{1}{4}$ they are 1.6667 and 0.6.

READING The denominator is $|1 - ae^{-j\omega}|^2$, which is positive, and the numerator is positive for $|a| < 1$. So this spectrum is real and strictly positive, and here $|X| = X$ with $\angle X = 0$ throughout. That is a property of this example, not of real spectra in general.

EXAMPLE 6.4 – THE RECTANGULAR PULSE

GIVEN $x[n] = 1$ for $-N_1 \leq n \leq N_1$ and zero elsewhere.

FIND $X(e^{j\omega})$ in closed form and its value at $\omega = 0$.

METHOD Use the finite geometric-series formula because the pulse has consecutive non-zero samples. Then balance the endpoint exponents so each difference can be written as a sine.

SOLUTION

$$X(e^{j\omega}) = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \frac{e^{j\omega N_1} - e^{-j\omega(N_1+1)}}{1 - e^{-j\omega}}.$$

Multiplying numerator and denominator by $e^{j\omega/2}$ turns each into a difference of conjugate exponentials, $2j \sin(\omega(N_1 + \frac{1}{2}))$ over $2j \sin(\omega/2)$:

$$X(e^{j\omega}) = \frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}.$$

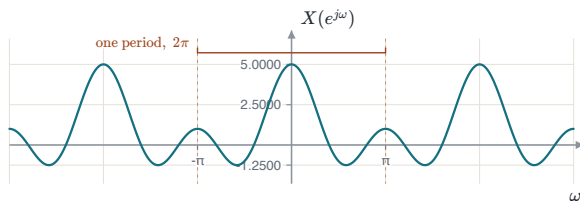
The minus in each bracket is what makes a sine; a plus would give a cosine and a different function.

EXCLUDED POINTS $r = e^{-j\omega} = 1$ at $\omega = 0, \pm 2\pi, \dots$, which is exactly where $\sin(\omega/2)$ vanishes. There every term of the sum is 1, so $X = 2N_1 + 1$: five for $N_1 = 2$, nine for $N_1 = 4$.

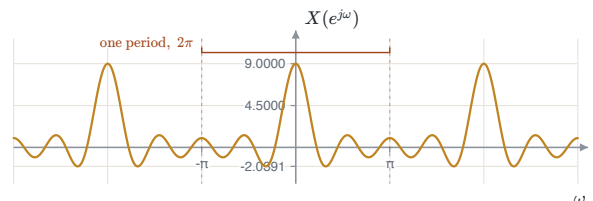
CHECK This is not a sinc. The denominator is $\sin(\omega/2)$, not $\omega/2$, and that is what makes the function periodic: a ratio of two sines repeats, while a sine over a straight line decays.

WHAT A FINITE GEOMETRIC SUM REQUIRES

$\sum_{n=p}^q r^n = (r^p - r^{q+1})/(1 - r)$ needs $r \neq 1$ and nothing else. There are finitely many terms, so nothing has to converge; only the division can fail. Here $|r| = 1$ exactly, so a condition $|r| < 1$ would exclude the sum altogether, while $|r| \leq 1$ would admit the one value that breaks it. The condition $|a| < 1$ of Example 6.2 belongs to the infinite sum, where it makes the tail vanish.



$N_1 = 2$: peak 5, least value -1.2500 .



$N_1 = 4$: peak 9, least value -2.0391 , and a main lobe half as wide.

REAL IS NOT THE SAME AS NON-NEGATIVE

The kernel above is real at every ω and negative on part of every period. Realness fixes the imaginary part and says nothing about the sign, so it gives $\angle X = 0$ where $X > 0$ and $\angle X = \pi$ where $X < 0$; and $|X| = X$ needs the extra hypothesis $X \geq 0$. A real spectrum has an even magnitude and a phase taking only the values 0 and π . Example 6.3 is the contrasting case: there the spectrum is real **and** strictly positive, so the phase really is zero, because of positivity and not because of realness.

6.4 Inverting an ideal low-pass spectrum

EXAMPLE 6.5 – THE IDEAL LOW-PASS SEQUENCE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq W$ and 0 for $W < |\omega| \leq \pi$, repeated with period 2π .

FIND $x[n]$.

METHOD Use the synthesis integral because the spectrum is given and the sequence is required. Choose $-\pi \leq \omega \leq \pi$ so the band where the spectrum is 1 appears once within the integration period.

SOLUTION

$$x[n] = \frac{1}{2\pi} \int_{-W}^W e^{j\omega n} d\omega = \frac{\sin(Wn)}{\pi n}, \quad x[0] = \frac{W}{\pi}.$$

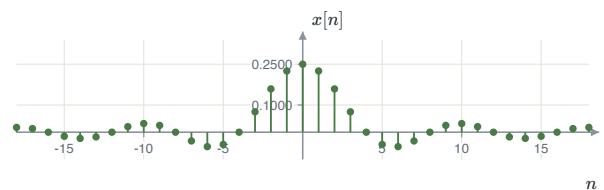
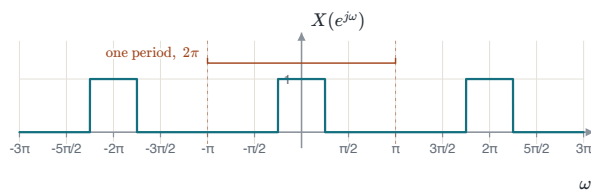
At $n = 0$ the integrand is 1 and the integral is $2W$, which is also the limit of the ratio.

NUMBERS $W = \pi/4$: 0.25, 0.225079, 0.159155, 0.075026 at $n = 0, 1, 2, 3$. $W = \pi/2$: 0.5, 0.318310, 0, -0.106103 .

CHECK $x[0]$ must be the fraction W/π of the period that the band occupies. That single number catches both a lost 2π and a wrong prefactor.

THE SINC CONVENTION

This course uses the unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, with no π inside the argument. In that convention the same sequence reads $x[n] = \frac{W}{\pi} \text{sinc}(Wn)$. The other common definition puts a π inside the argument and moves every zero crossing, so the convention is restated at every point of use. The prefactor is the constant W/π : writing W/n makes the expression depend on n twice and is wrong by a factor π at every index, giving 0.707107 at $n = 1$ where the true value is 0.225079.



The spectrum for $W = \pi/4$, over three periods. The band repeats; it does not stop at $\pm\pi$.

Its inverse transform, with $x[0] = 0.25$.

6.5 Periodic sequences and impulse spectra

A complex exponential is not absolutely summable, so its analysis sum does not converge in the ordinary sense. Its transform exists as a train of impulses, and the definition is that the synthesis equation works.

THE COMPLEX EXPONENTIAL

$$e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi k)$$

Substituting into the synthesis equation and integrating over the period containing ω_0 leaves exactly one impulse, and $\frac{1}{2\pi} \cdot 2\pi e^{j\omega_0 n} = e^{j\omega_0 n}$. The copies must be there: a single impulse is not 2π -periodic, and a sequence cannot distinguish ω_0 from $\omega_0 + 2\pi$.

WEIGHT, NOT HEIGHT

An impulse has no value at a point; it has a weight, the number that comes out when it is integrated. Every figure here draws an impulse as an arrow whose height is its weight, so the two can be read off the same axis. Arrows drawn to a fixed height with the weight written beside them hide exactly the comparison these pictures are for.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n} \longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\cos \omega_0 n \longleftrightarrow \pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$$

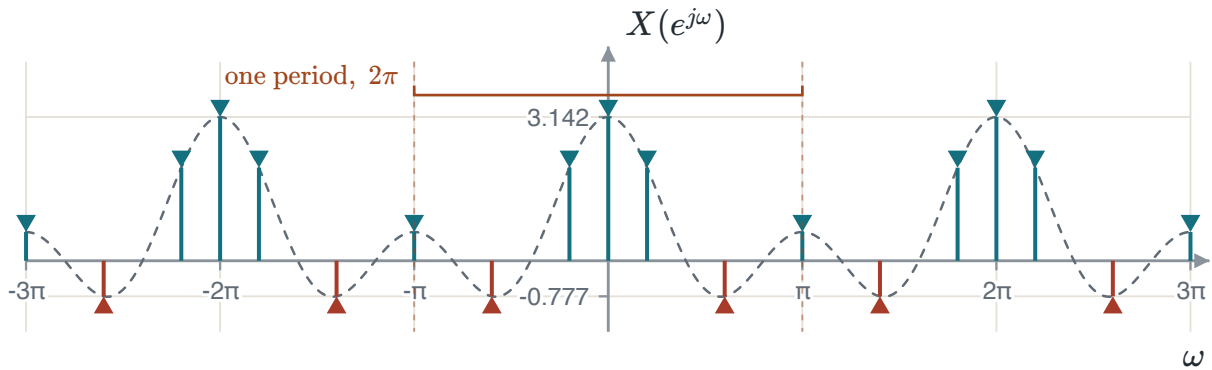
$$\sin \omega_0 n \longleftrightarrow \frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$$

The spacing is $2\pi/N$, so one period holds exactly N impulses. A real sequence always produces the pair at $+\omega_0$ and $-\omega_0$; dropping the negative one rebuilds $\frac{1}{2}e^{j\omega_0 n}$, which is complex, instead of the cosine.

For the periodic square wave, 1 on $|n| \leq N_1$ inside each period of length N , the coefficients are the same kernel again, sampled and divided by N :

$$a_k = \frac{1}{N} \frac{\sin\left(\frac{2\pi k}{N}(N_1 + \frac{1}{2})\right)}{\sin(\pi k/N)}, \quad a_k = \frac{2N_1 + 1}{N} \text{ when } k \equiv 0 \pmod{N}.$$

With $N_1 = 2$: at $N = 10$, $a_0 = 0.5000$ and $a_1 = 0.3236$; at $N = 20$, 0.2500 and 0.2260 ; at $N = 30$, 0.1667 and 0.1594 . The coefficients shrink like $1/N$ while the shape they trace stays put. Some of them are negative — at $N = 10$, $a_3 = -0.1236$ — so the impulses of weight $2\pi a_k$ are of unequal size and some point downwards. Sketching them all the same length hides both facts.



The square wave with $N = 10$, $N_1 = 2$. The dashed line is the envelope; the red arrows are the harmonics whose weight is negative.

EXAMPLE 6.6 – THE IMPULSE TRAIN

GIVEN $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$: a unit sample every N indices. Here n is the sequence index and k labels the copies, so the two roles use different letters.

FIND The coefficients and the transform.

SOLUTION One period holds a single sample, at $n = 0$, so $a_k = \frac{1}{N}$ for every k , and

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right).$$

NUMBERS Weights $2\pi/5 = 1.2566$, $2\pi/10 = 0.6283$, $2\pi/15 = 0.4189$ for $N = 5, 10, 15$.

READING One period always holds N impulses of weight $2\pi/N$, so their weights add to 2π whatever N is. A sparser train in time is a denser train in frequency.

Bringing a frequency into range

EXAMPLE 6.7 – TWO COSINES

GIVEN $x[n] = 2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$.

FIND The spectrum, and the fundamental period of the sequence.

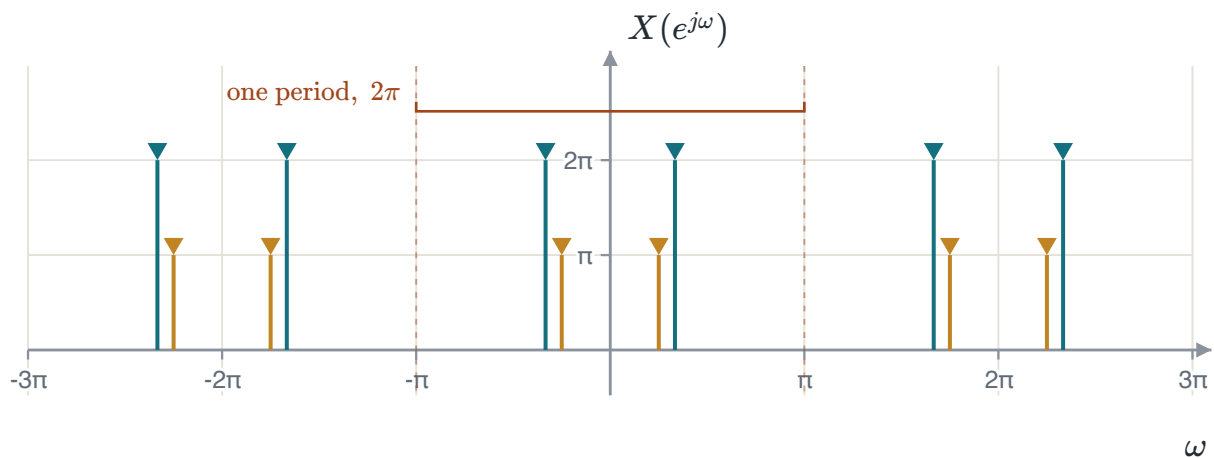
METHOD First reduce each frequency to $-\pi < \omega \leq \pi$ because equivalent discrete-time frequencies describe the same sequence. Then transform each cosine separately and add the results by linearity.

REDUCTION $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$, and $2\pi n$ is a whole number of turns at every integer n , so $\cos\left(\frac{5\pi}{3}n\right) = \cos\left(\frac{\pi}{3}n\right)$. Likewise $\cos\left(\frac{7\pi}{4}n\right) = \cos\left(\frac{\pi}{4}n\right)$. These are not merely similar sequences; they take the same value at every n .

SOLUTION Inside one period there are four impulses: weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$. The amplitude 2 doubles the first pair from π to 2π . Outside the period the same impulses reappear every 2π : weight 2π at $\pm 5\pi/3$ and $\pm 7\pi/3$, weight π at $\pm 7\pi/4$ and $\pm 9\pi/4$.

ORDER On the positive axis $\pi/4 < \pi/3 < 5\pi/3 < 7\pi/4 < 2\pi < 9\pi/4 < 7\pi/3$: the two trains interleave, and the order flips either side of 2π .

PERIOD $N_0 = (2\pi/\omega_0)m$, where m is the smallest positive integer making N_0 an integer. For $\omega_1 = 5\pi/3$: $2\pi/\omega_1 = 6/5$, $m = 5$, $N_0 = 6$. For $\omega_2 = 7\pi/4$: $8/7$, $m = 7$, $N_0 = 8$. The sum repeats after $\text{LCM}(6, 8) = 24$, and $24/6 = 4$, $24/8 = 3$ with $\text{gcd}(4, 3) = 1$, so nothing smaller works.



The spectrum over three periods. Tall arrows carry weight 2π , short ones π , and each pattern repeats every 2π .

6.6 Properties

PROPERTY	PAIR
Linearity	$a x_1[n] + b x_2[n] \leftrightarrow a X_1 + b X_2$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Real and even	$X(e^{j\omega})$ real and even
Real and odd	$X(e^{j\omega})$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{X\}$, $\mathcal{O}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

TWO NAMES THAT MUST NOT BE EXCHANGED

Discrete time has a **difference**, not a derivative: there is nothing between n and $n + 1$ over which to take a limit, and the factor is $1 - e^{-j\omega}$ rather than $j\omega$. The **differentiation** in this chapter is in frequency, and it is a genuine derivative because ω is continuous. Naming the first difference a derivative collapses the continuous and discrete cases on the one page where the distinction is being built.

For a real sequence, conjugation gives $X(e^{-j\omega}) = X^*(e^{j\omega})$, so the real part and the magnitude are even in ω and the imaginary part and the phase are odd. Half a period then determines the whole spectrum — for a real sequence only. A complex sequence needs the full period drawn.

That one line also fixes the last three rows of the table. $\mathcal{E}\mathbf{v}\{x[n]\} = \frac{1}{2}(x[n] + x[-n])$ transforms to $\frac{1}{2}[X(e^{j\omega}) + X(e^{-j\omega})] = \frac{1}{2}[X + X^*] = \text{Re}\{X\}$, and the odd part gives $j \text{Im}\{X\}$; a real and even sequence has no odd part, so its transform is real and even, and a real and odd sequence has no even part, so its transform is purely imaginary and odd. Take $x[n] = a^n u[n]$ with $0 < a < 1$. Its even part is $\frac{1}{2}a^{|n|} + \frac{1}{2}\delta[n]$ — the two halves overlap at $n = 0$ — and transforming term by term gives $(1 - a \cos \omega)/(1 - 2a \cos \omega + a^2)$, which is $\text{Re}\{1/(1 - ae^{-j\omega})\}$ exactly.

Accumulation undoes differencing, so their factors are reciprocals, and the impulse train is the part differencing destroys. A running sum is fixed only up to a constant offset, and the difference cannot see that offset; the weight $\pi X(e^{j0}) = \pi \sum_n x[n]$ is what puts it back. Where the sequence sums to zero the term vanishes on its own. It is the discrete-time counterpart of the $\pi X(0)\delta(\omega)$ in the integration property of Chapter 5, and it fails in the same place for the same reason: $1 - e^{-j\omega}$ is zero at $\omega = 0$.

Time expansion

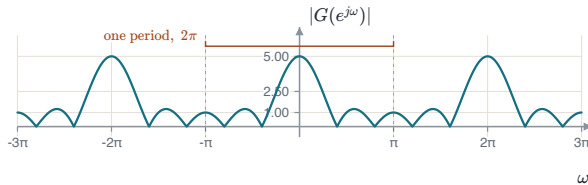
For a positive integer k , $x_{(k)}[n]$ is $x[n/k]$ when n is a multiple of k and zero otherwise: a stretch with zeros inserted, since discrete time has no operation that stretches a sequence without leaving gaps. Only the indices $n = rk$ contribute to the analysis sum, so putting $n = rk$ turns it into $\sum_r x[r]e^{-j\omega kr} = X(e^{jk\omega})$. Replacing ω by $k\omega$ compresses the frequency axis by k , so one period of length 2π now holds k copies of the old picture.

EXAMPLE 6.8 – EXPANSION OF A FIVE-POINT PULSE

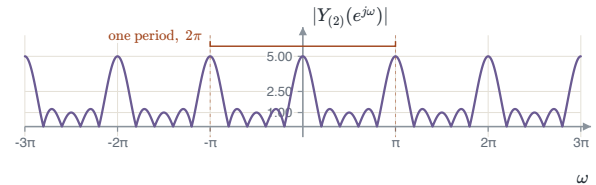
GIVEN $g[n] = 1$ on $|n| \leq 2$, so $G(e^{j\omega}) = \sin(5\omega/2) / \sin(\omega/2)$.

BUILD $y[n] = g[n - 2]$, so $Y(e^{j\omega}) = e^{-j2\omega} G(e^{j\omega})$. Expanding by two gives $Y_{(2)}(e^{j\omega}) = Y(e^{j2\omega}) = e^{-j4\omega} \sin(5\omega) / \sin(\omega)$. Finally $x[n] = y_{(2)}[n] + 2y_{(2)}[n - 1]$, so $X(e^{j\omega}) = (1 + 2e^{-j\omega})Y_{(2)}(e^{j\omega})$, which peaks at $3 \times 5 = 15$.

WHERE THE ARGUMENT HALVED G carries $\sin(5\omega/2) / \sin(\omega/2)$ and $Y_{(2)}$ carries $\sin(5\omega) / \sin(\omega)$: the same expression with ω replaced by 2ω . A denominator $\sin(\omega)$ is right for the expanded sequence and wrong for the original, where it would put a pole at $\omega = \pi - 1000.0$ at $\omega = \pi - 10^{-3}$, against the true $G(e^{j\pi}) = 1$, and -0.7071 at $\pi/2$ against the true -1.0000 .



$|G(e^{j\omega})|$: peak 5, and the finite value 1 at $\omega = \pi$.



$|Y_{(2)}(e^{j\omega})|$: the same picture twice inside every 2π .

Parseval

The quantity $|X(e^{j\omega})|^2$ is the energy-density spectrum. Integrating it over one period and dividing by 2π gives the total energy. Integrating over part of a period gives the energy in that frequency band. Both the single-period range and the factor $\frac{1}{2\pi}$ are required, although neither is visible from the plot shape. Check them numerically: for $a^n u[n]$ with $a = \frac{1}{2}$, the time side gives $\sum_{n \geq 0} (\frac{1}{4})^n = \frac{4}{3}$ and the frequency side gives $1/(1 - a^2) = \frac{4}{3}$; for the rectangular pulse with $N_1 = 2$, both sides give 5, the number of samples.

6.7 Convolution and multiplication

CONVOLUTION

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Both pairs $x \leftrightarrow X$ and $h \leftrightarrow H$ are declared before the conclusion is written, and y names the output and nothing else. A product of complex numbers gives two separate statements, $|Y| = |X| \cdot |H|$ and $\angle Y = \angle X + \angle H$. Putting modulus bars around a whole equation is not an operation on anything.

EXAMPLE 6.9 – CONVOLVING TWO EXPONENTIALS

GIVEN $x[n] = a^n u[n]$ and $h[n] = b^n u[n]$, with $|a| < 1$, $|b| < 1$ and $a \neq b$. Every one-sided exponential carries its $u[n]$: without it the sequence is defined for negative n too, where it grows without bound.

METHOD Use the convolution property because the required output is a time convolution. Multiply the transforms, then use partial fractions so each term matches the known one-sided exponential pair. Write $z = e^{-j\omega}$ as an algebraic variable; then $Y = 1/((1 - az)(1 - bz))$ is a rational function of z , and substituting $z = 1/a$ is algebra in z , not a claim that $e^{-j\omega}$ takes that value.

SOLUTION Cover-up gives $A = a/(a - b)$ and $B = -b/(a - b)$, so

$$y[n] = \frac{1}{a - b} [a^{n+1} - b^{n+1}] u[n].$$

CONDITION Both coefficients divide by $a - b$, so the route requires $a \neq b$. At $a = b$ the two poles merge and the answer takes a different form, worked out in Section 6.8.

CHECK $a = \frac{1}{2}$, $b = \frac{1}{4}$: $y[0] = 1$, $y[1] = 0.75$, $y[2] = 0.4375$, $y[3] = 0.234375$. Direct convolution gives $y[0] = x[0]h[0] = 1$ and $y[1] = \frac{1}{4} + \frac{1}{2} = 0.75$; the first value must be $x[0]h[0]$ for any two causal sequences. The magnitude $|Y|$ runs between 0.5333 and 2.6667.

The product of two ideal low-pass responses is the narrower of the two, so a cascade keeps only the band both filters pass. With cutoffs $\pi/2$ and $\pi/4$, the cascade has cutoff $\pi/4$ and impulse response $\sin(\pi n/4)/(\pi n)$. In writing such a spectrum, close one

branch and leave the other open — 1 for $|\omega| \leq \pi/4$ and 0 for $\pi/4 < |\omega| \leq \pi$ — since for an ideal filter the band edge is precisely where a convention is needed rather than assumed.

EXAMPLE 6.10 – A STEPPED SPECTRUM THROUGH A FILTER

GIVEN $X(e^{j\omega}) = 2$ for $|\omega| \leq \pi/4$ and 1 for $\pi/4 < |\omega| \leq 3\pi/4$, zero to π ; H is ideal low-pass with cutoff $\pi/2$.

PRODUCT $Y(e^{j\omega}) = 2$ for $|\omega| \leq \pi/4$ and 1 for $\pi/4 < |\omega| \leq \pi/2$, zero on the rest of the period.

SOLUTION Split Y into two stacked ideal bands, one of height 1 out to $\pi/2$ and one more out to $\pi/4$:

$$y[n] = \frac{\sin(\pi n/2)}{\pi n} + \frac{\sin(\pi n/4)}{\pi n}, \quad y[0] = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}.$$

CHECK $y[0]$ must be the area of one period of Y divided by 2π . A sketch of a spectrum is not a solution; the closed form costs one line once the stack is seen.

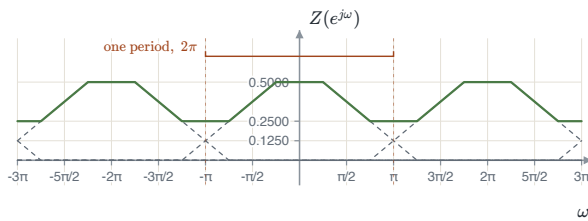
Multiplication is a periodic convolution

MULTIPLICATION

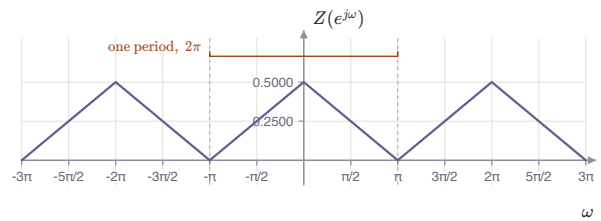
$$z[n] = x[n]y[n] \longleftrightarrow Z(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

The integral runs over one period and both factors are 2π -periodic. That operation is a periodic convolution, and it differs from the ordinary convolution of two functions on the line whenever the result is wider than one period.

Take $X = 1$ on $|\omega| \leq 3\pi/4$ and $Y = 1$ on $|\omega| \leq \pi/2$, both repeated every 2π . First convolve the rectangles on the line. This gives a trapezoid of height $\frac{1}{2\pi} \cdot 2 \cdot \frac{\pi}{2} = \frac{1}{2}$, flat on $|\omega| \leq \pi/4$ and reaching zero at $|\omega| = 5\pi/4$. Since $5\pi/4 > \pi$, the result extends beyond one period. A periodic copy therefore overlaps it near $\omega = \pm\pi$, and the two values add.



The dashed trapezoids are the copies; the solid curve is their sum. At $\omega = \pi$ the value doubles from 0.125 to 0.25.



Two equal bands of half-width $\pi/2$: the triangle fills one period and reaches zero at both boundaries, so repeated copies do not overlap.

The values are $Z(e^{j0}) = Z(e^{j\pi/4}) = 0.5$ on the flat top, $Z(e^{j3\pi/4}) = 0.25$ on the slope, and $Z(e^{j\pi}) = \frac{1}{8} + \frac{1}{8} = 0.25$ at the edge. The check is the synthesis equation: $\frac{1}{2\pi} \int_{2\pi} Z d\omega$ must equal $z[0] = x[0]y[0] = \frac{3}{4} \cdot \frac{1}{2} = 0.375$, and it does.

EXAMPLE 6.11 – MULTIPLYING BY A COSINE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq \pi/4$, repeated every 2π , and $y[n] = \cos(\omega_0 n)$ with $\omega_0 = \pi/3$.

METHOD Write Y as its impulse train, $\pi \sum_l [\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)]$. The periodic convolution integrates over a single period, so choosing $-\pi < \theta \leq \pi$ leaves exactly two impulses, at $\theta = \pm\pi/3$, and only those contribute. Taking the whole train into an ordinary convolution would count every copy and diverge.

SOLUTION

$$Z(e^{j\omega}) = \frac{1}{2} X(e^{j(\omega-\omega_0)}) + \frac{1}{2} X(e^{j(\omega+\omega_0)}),$$

where the height $\frac{1}{2}$ comes from $\frac{1}{2\pi} \cdot \pi$.

BAND EDGES

The band moves to $\omega_0 \mp \pi/4$, so with $\omega_0 = \pi/3$ the edges are $\pi/12$ and $7\pi/12$, and the mirror image occupies $-7\pi/12 \leq \omega \leq -\pi/12$. Both numbers are computed from ω_0 and the bandwidth, so they cannot drift apart from the drawn figure; with ω_0 below $\pi/4$ the two bands would overlap through the origin instead.

THIS OVERLAP IS NOT ALIASING

This overlap occurs between periodic copies during frequency-domain convolution. Chapter 7 studies a different overlap between replicas produced by sampling. Only the sampling overlap is called aliasing.

6.8 Difference equations

Transform both sides of a linear constant-coefficient difference equation. Linearity acts term by term and each shift brings out one factor $e^{-j\omega k}$:

FREQUENCY RESPONSE OF A DIFFERENCE EQUATION

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

The impulse response of the recursion is now available without running the recursion: factor the denominator, split into partial fractions, and invert each piece with the exponential pair. Every step is algebra.

EXAMPLE 6.12 – A SECOND-ORDER SYSTEM

GIVEN $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$.

FIND The frequency response and the impulse response.

SOLUTION

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}} = \frac{2}{\left(1 - \frac{1}{2}e^{-j\omega}\right)\left(1 - \frac{1}{4}e^{-j\omega}\right)},$$

checked by expansion: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ and $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. With $z = e^{-j\omega}$, cover-up at $z = 2$ gives $A = 4$ and at $z = 4$ gives $B = -2$, so

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n].$$

CHECK $h[0] = 2$, $h[1] = 1.5$, $h[2] = 0.875$. Substituting into the equation: $h[1] - \frac{3}{4}h[0] = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0$, as required for $n \geq 1$. Both poles have modulus below 1, so the system is stable, and $|H|$ runs between 1.0667 and 5.3333: this recursion is a low-pass filter.

The repeated-pole pair

Example 6.9 excluded $a = b$ because its partial fractions divide by $a - b$. When $a = b$, the poles coincide and require a separate pair. Derive it from the geometric pair by differentiating both sides with respect to a while holding ω fixed:

$$\sum_{n=1}^{\infty} n a^{n-1} e^{-j\omega n} = \frac{e^{-j\omega}}{(1 - a e^{-j\omega})^2}.$$

REPEATED POLE

$$(n+1)a^n u[n] \longleftrightarrow \frac{1}{(1 - a e^{-j\omega})^2}, \quad |a| < 1$$

Putting $m = n - 1$ and cancelling one factor $e^{-j\omega}$ gives the pair. The exponent 2 comes from the differentiation and the minus sign is inherited unchanged, so neither has to be remembered. One number settles any doubt: at $\omega = 0$ with $a = \frac{1}{4}$ the value is $(1 - \frac{1}{4})^{-2} = \frac{16}{9} = 1.7778$, while a plus sign would give $(\frac{5}{4})^{-2} = 0.6400$.

EXAMPLE 6.13 – THE OUTPUT OF THAT SYSTEM

GIVEN The system of Example 6.12 driven by $x[n] = (\frac{1}{4})^n u[n]$.

METHOD Use partial fractions because the output transform is rational and each fraction can be inverted with a known pair. The input pole coincides with one system pole, so the expansion must include both a simple and a squared term for that pole:

$$Y = \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)^2} = \frac{A}{1 - \frac{1}{4}z} + \frac{B}{(1 - \frac{1}{4}z)^2} + \frac{C}{1 - \frac{1}{2}z}.$$

COEFFICIENTS Cover-up at $z = 2$ gives $C = 2/(1 - \frac{1}{2})^2 = 8$; at $z = 4$ it gives $B = 2/(1 - 2) = -2$; setting $z = 0$ in the identity gives $2 = A + B + C$, so $A = -4$.

SOLUTION

$$y[n] = -4\left(\frac{1}{4}\right)^n u[n] - 2(n+1)\left(\frac{1}{4}\right)^n u[n] + 8\left(\frac{1}{2}\right)^n u[n].$$

CHECK $y[0] = 2, y[1] = 2, y[2] = 1.375, y[3] = 0.8125$, and $y[0]$ must equal $h[0]x[0] = 2$ for two causal sequences. Direct convolution of h and x reproduces the whole sequence. Without the repeated-pole pair the middle term could not be inverted at all.

6.9 Duality

There is no duality inside the discrete-time transform pair. The analysis equation is a sum over an integer and the synthesis equation an integral over a continuous variable, so no relabelling turns one into the other. Two dualities do hold.

1. In the discrete-time Fourier series, where both domains are discrete and both objects are periodic with the same period N : if $x[n] \leftrightarrow a_k$, then reading the coefficients as a sequence gives $a[n] \leftrightarrow \frac{1}{N}x[-k]$. The impulse train with $N = 21$ is the example: its coefficients are all $1/21 = 0.0476$, and reading those constants back as a sequence returns an impulse train.
2. Between the discrete-time transform and the continuous-time series. $X(e^{j\omega})$ is a 2π -periodic function of a continuous variable, so it has a continuous-time Fourier series, and comparing the two synthesis equations identifies its coefficients as $x[-k]$: the spectrum of a sequence is a periodic signal whose series coefficients are the sequence itself, reversed.

A worked check on the second. Take the 2π -periodic square wave in ω , equal to 1 for $|\omega| \leq \pi/2$ and zero over the rest of the period. Its series coefficients are $a_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{-jk\omega} d\omega = \sin(k\pi/2)/(k\pi)$, that is 0.5, $1/\pi = 0.3183$, 0, $-1/(3\pi) = -0.1061$. Its inverse transform is $x[n] = \sin(\pi n/2)/(\pi n)$, which is the same list of numbers, so $a_k = x[-k]$ – and here x is even, so $a_k = x[k]$ as well.

6.10 What to carry into Chapter 7

- The pair, with analysis as the sum and synthesis as the integral over one period, carrying $\frac{1}{2\pi}$.
- The 2π -periodicity and its one-line proof, and that the same line fails in continuous time.
- Real means the phase is 0 or π , not that the phase is zero.
- Multiplication in time is a **periodic** convolution over 2π , so anything reaching across a period boundary folds back and adds.
- The unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, restated at every point of use.
- Discrete time has a difference, not a derivative; the derivative in this chapter is in frequency.

A sequence can be formed by reading a continuous-time signal at regular instants. The next chapter studies how this sampling operation changes the spectrum. Sampling produces repeated spectral copies separated by the sampling frequency, so the periodic-overlap ideas from this chapter will be used again.

Exercises

6.1 Prove that $X(e^{j\omega})$ is 2π -periodic, and say precisely where the same argument fails for $X(j\omega)$.

Answer: Substituting $\omega + 2\pi$ into the analysis sum produces a factor $e^{-j2\pi n}$, which is 1 in every term because n is an integer. In continuous time the corresponding factor is $e^{-j2\pi t}$ with t real, which equals 1 only at integer t and cannot leave the integral.

6.2 For $x[n] = a^n u[n]$ with $a = \frac{1}{8}$, give $|X|_{\max}$, $|X|_{\min}$ and $\max |\angle X|$ in closed form and as numbers.

Answer: $1/(1 - a) = 8/7 = 1.1429$, $1/(1 + a) = 8/9 = 0.8889$, and $\arcsin \frac{1}{8} = 0.1253$ rad.

6.3 Explain why the finite geometric sum behind the Dirichlet kernel needs $r \neq 1$ rather than $|r| < 1$, and give the value of the kernel at the excluded points.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1 - r$ vanishes. Here $|r| = 1$ at every ω , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is ω a multiple of 2π , every term is 1 and the sum is $2N_1 + 1$.

6.4 The Dirichlet kernel is real at every ω . Does it follow that $|X| = X$ and $\angle X = 0$?

Answer: No. Realness fixes the imaginary part and not the sign. The kernel is negative on part of every period — least value -1.2500 for $N_1 = 2$ — and there $|X| = -X$ and $\angle X = \pi$. The conclusion needs the extra hypothesis $X \geq 0$, which holds for $a^{|n|}$ but not here.

6.5 Invert the ideal low-pass spectrum with $W = \pi/2$ and give $x[0]$ and $x[2]$. State the sinc convention you use.

Answer: $x[n] = \sin(Wn)/(\pi n)$ with $x[0] = W/\pi = 0.5$, and $x[2] = \sin(\pi)/(2\pi) = 0$. In the unnormalised convention $\text{sinc } \theta = \sin \theta/\theta$, the same sequence is $\frac{W}{\pi} \text{sinc}(Wn)$.

6.6 Where do the impulses of $2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$ lie inside $-\pi < \omega \leq \pi$, and what is the fundamental period of the sequence?

Answer: Weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$, after reducing $5\pi/3$ to $\pi/3$ and $7\pi/4$ to $\pi/4$. The two components have periods 6 and 8, so the sum has period $\text{LCM}(6, 8) = 24$.

6.7 Two spectra, 1 on $|\omega| \leq 3\pi/4$ and 1 on $|\omega| \leq \pi/2$, belong to sequences that are multiplied together. Why is $Z(e^{j\pi}) = \frac{1}{4}$ and not $\frac{1}{8}$?

Answer: The multiplication property gives a periodic convolution over one period. The ordinary convolution of the two rectangles is a trapezoid reaching to $|\omega| = 5\pi/4$, which is wider than one period, so near $\omega = \pi$ a second copy arrives from the other side and the two contributions of $\frac{1}{8}$ add.

6.8 Derive the pair for $(n+1)a^n u[n]$ and say what fixes the exponent and the sign.

Answer: Differentiate $\sum_{n \geq 0} a^n e^{-j\omega n} = 1/(1 - ae^{-j\omega})$ with respect to a , shift the index and cancel one factor $e^{-j\omega}$, giving $1/(1 - ae^{-j\omega})^2$. One differentiation of a first power produces the exponent 2, and the minus sign is inherited from the pair being differentiated.

6.9 Find the impulse response of $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$ and check it against the equation.

Answer: $h[n] = 4(\frac{1}{2})^n u[n] - 2(\frac{1}{4})^n u[n]$, so $h[0] = 2$, $h[1] = 1.5$, $h[2] = 0.875$. Substituting: $h[1] - \frac{3}{4}h[0] = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0$.

6.10 Which properties of the continuous-time transform carry over unchanged, and which change form?

Answer: Linearity, the time shift, conjugate symmetry, Parseval and the convolution property carry over unchanged. The synthesis equation changes its range to one period; the multiplication property becomes a periodic convolution; the time-domain operator is a difference and not a derivative; and the transform pairs themselves differ, since the discrete-time ones are 2π -periodic and carry their own convergence conditions.

Sampling and aliasing

Sampling keeps the value of a signal every T seconds and discards the values between those instants. This chapter gives the condition for exact recovery, derives the required reconstruction filter, and shows what changes when the condition fails. Sampling always replicates the spectrum. **Aliasing** occurs only when those copies overlap.

7.1 Impulse-train sampling

Sampling is modelled as a multiplication. The sampling function is an impulse train of period T , and the sampled signal is the product of the signal with it.

IMPULSE-TRAIN SAMPLING

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \quad x_p(t) = x(t)p(t)$$

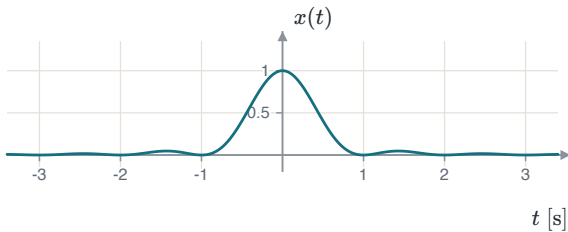
$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$

The second line follows from the sifting property $x(t)\delta(t - nT) = x(nT)\delta(t - nT)$: an impulse is zero everywhere except at its own instant, so only the value there survives.

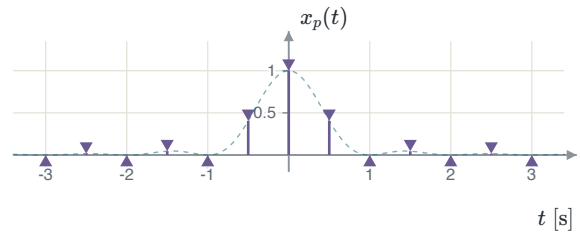
HOW AN IMPULSE IS DRAWN

An impulse is drawn as an arrow whose **height is its weight**. In a picture of $x_p(t)$ the arrow at $t = nT$ therefore reaches $x(nT)$, and the outline traced by the arrowheads is the signal itself. The arrows are weights, not values of a function.

The sampler keeps only the numbers $x(nT)$. Written as a sequence, they are $x_p[n] = x(nT)$. These sample values form the discrete-time representation of the signal.



A band-limited signal, $x(t) = (\sin \pi t / \pi t)^2$.



The same signal sampled with $T = 0.5$ s. Each arrow reaches $x(nT)$.

7.2 The rate, in radians per second and in hertz

A sampling rate can be stated as an angular frequency or in hertz. Define both forms before using them so that their units remain distinct.

SAMPLING RATE, BOTH READINGS

$$\omega_s = \frac{2\pi}{T} \left[\frac{\text{rad}}{\text{s}} \right], \quad f_s = \frac{1}{T} [\text{Hz}], \quad \omega_s = 2\pi f_s$$

ω_s is the sampling **angular** frequency. f_s is the sampling frequency in hertz, the number of samples taken per second. Every spectrum in this chapter is drawn against ω in rad/s, so ω_s is the working symbol; the hertz reading appears only where a physical rate is quoted.

THE FACTOR OF 2π IS NOT DECORATION

Calling $2\pi/T$ the sampling frequency and then reading it in hertz is the most expensive slip in this chapter. With $T = 0.25$ ms the two readings are $\omega_s = 8000\pi$ rad/s and $f_s = 4000$ Hz. They differ by 2π , and an answer that mixes them is wrong by that factor everywhere it is used. The same applies to the signal: a bandwidth of ω_M rad/s is $f_M = \omega_M/2\pi$ hertz.

One multiplication tests any pair of values: $\omega_s T = 2\pi$ always, and $f_s T = 1$ always. Every sampling period computed in this chapter is followed by that check.

7.3 The spectrum of a sampled signal

Multiplication in time is convolution in frequency. The transform of an impulse train of period T is an impulse train of spacing $\omega_s = 2\pi/T$, and convolving with a shifted impulse shifts the function. Three steps:

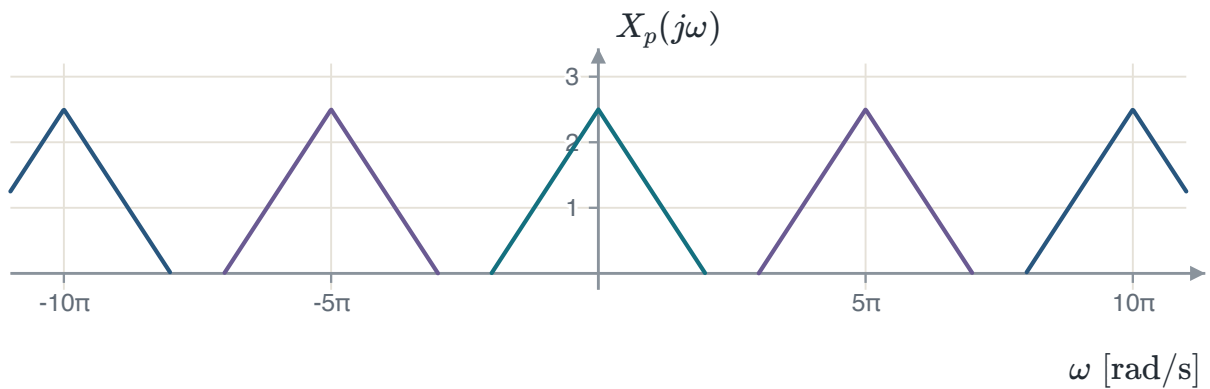
$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)], \quad P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s).$$

SPECTRUM OF THE SAMPLED SIGNAL

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

The $1/2\pi$ of the first step and the $2\pi/T$ of the second combine, and what is left in front is $1/T$. Every copy is $X(j\omega)$ scaled by $1/T$, not by 1. The reconstruction filter later carries gain T , exactly the inverse, so the two must be kept together.

Spacing T in time gives spacing $2\pi/T$ in frequency: the closer the samples, the further apart the copies, and the taller each one is.



The sampled spectrum of the signal above at $T = 0.4$ s, so $\omega_s = 5\pi$ rad/s. One copy per multiple of ω_s , each of height $1/T = 2.5$.

7.4 Replication and aliasing are not the same thing

The sum for $X_p(j\omega)$ runs over every integer k for every value of T . There is no rate at which the copies fail to appear and no rate at which they disappear. Two things, and only two, change with the rate: the copies sit ω_s apart, and each is scaled by $1/T$.

NAMING THE PIECES

The term $k = 0$ is $X(j\omega)/T$, the spectrum of the signal itself, sitting where it always sat. It is the **baseband** and it is not a copy of anything. The terms $k = \pm 1, \pm 2, \dots$ are the **copies**, centred at $\pm\omega_s, \pm 2\omega_s, \dots$. Calling the centre group the first replica shifts every later index by one.

The baseband occupies $|\omega| \leq \omega_M$, and the copy at $k = 1$ occupies $|\omega - \omega_s| \leq \omega_M$, so it begins at $\omega_s - \omega_M$. The gap between them is the quantity that decides everything.

GUARD BAND

$$(\omega_s - \omega_M) - \omega_M = \omega_s - 2\omega_M$$

Positive: the copies stand clear and a filter can cut the baseband out untouched. Zero: they touch. Negative: they have moved into one another and add.

REPLICATION IS UNCONDITIONAL

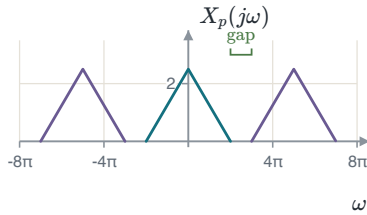
It is sometimes said that below the Nyquist rate the spectrum is no longer replicated. That is false. The formula for $X_p(j\omega)$ carries no condition, so the copies are there at every rate; what changes below the Nyquist rate is that they **overlap**.

Replication is what sampling does. Aliasing is the overlap of what sampling made.

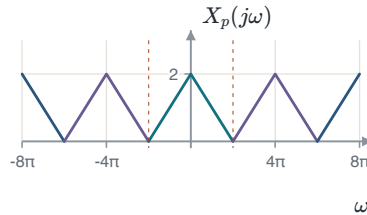
The overlap cannot be repaired afterwards. Where two copies meet, the sampler stores one number that is the sum of two contributions,

$$X_p(j\omega) = \frac{1}{T} \left[\underbrace{X(j\omega)}_{\text{wanted}} + \underbrace{X(j(\omega - \omega_s))}_{\text{intruder}} \right].$$

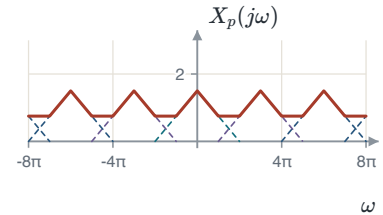
A filter multiplies $X_p(j\omega)$ frequency by frequency, including multiplying by zero. It cannot look at one value and report the two numbers that were added to make it. So the only cures are to raise the rate, or to remove the offending frequencies from $x(t)$ before the sampler.



$\omega_s = 5\pi$: a gap of π rad/s.



$\omega_s = 4\pi = 2\omega_M$: the gap is zero.



$\omega_s = 3\pi$: the copies add. The solid curve is the sum.

7.5 The sampling theorem

SAMPLING THEOREM

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M \text{ and } \omega_s > 2\omega_M$$

$$\implies x(t) \text{ is determined uniquely by } x(nT), n = 0, \pm 1, \pm 2, \dots$$

The signal is recovered by passing $x_p(t)$ through an ideal lowpass filter of gain T and cutoff ω_c with $\omega_M < \omega_c < \omega_s - \omega_M$. The quantity $2\omega_M$ is the **Nyquist rate** of the signal.

WHY THE INEQUALITY HAS TO BE STRICT

Put $\omega_s = 2\omega_M$ into the cutoff condition. It becomes $\omega_M < \omega_c < \omega_M$, an interval with nothing in it. There is no admissible cutoff at the Nyquist rate, so the theorem cannot be satisfied at its own boundary. Writing $\omega_s \geq 2\omega_M$ promises a filter that the same sentence then rules out. In practice the rate is chosen well above $2\omega_M$, so that the guard band is wide enough for a real filter, whose transition from pass to stop is not vertical.

EXAMPLE 7.1 – A SIGNAL THE NYQUIST RATE THROWS AWAY

GIVEN $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$, sampled at exactly $\omega_s = 2\omega_M = 8000\pi$ rad/s.

FIND What happens to the term $\sin(4000\pi t)$.

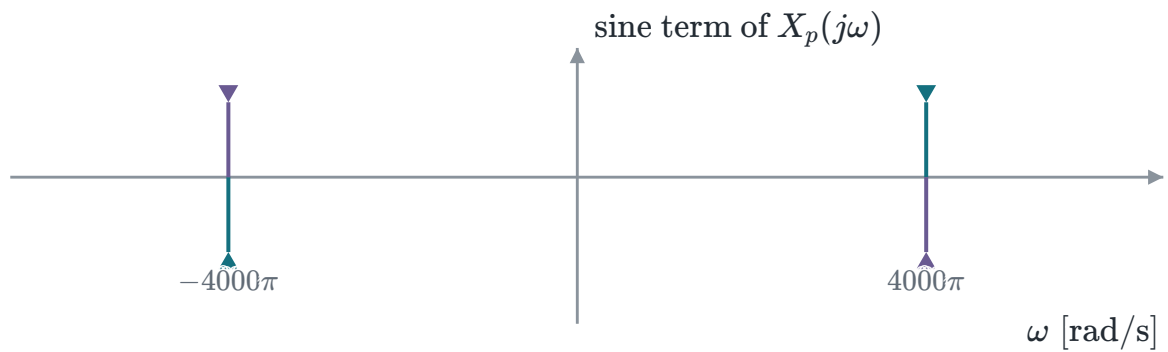
METHOD Use the sampled-spectrum sum because cancellation can occur between the baseband and a copy. Locate every contribution to X_p at $\omega = +4000\pi$, then verify the result from the sample values in time.

SOLUTION The bandwidth is $\omega_M = 4000\pi$ rad/s, so $T = 2\pi/\omega_s = 1/4000$ s = 0.25 ms. The sine transforms to $\frac{\pi}{j}[\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi)]$. At $\omega = +4000\pi$ two terms of the sum contribute: the baseband gives $\frac{1}{T} \cdot \frac{\pi}{j} = +\frac{4000\pi}{j}$, and the $k = 1$ copy carries the impulse from -4000π up by ω_s and gives $-\frac{4000\pi}{j}$. Their sum is exactly zero, and the same happens at -4000π . In time,

$$\sin(4000\pi nT) = \sin(\pi n) = 0 \text{ for every integer } n.$$

CHECK Every sample of the sine term is taken at a zero crossing, so the sampler never sees it. The component is absent, not merely marginal.

REPAIR Add a guard band. With $\omega_g = 1000\pi$ rad/s the rate becomes $\omega_s = 9000\pi$ rad/s, so $T = 1/4500$ s, about 222.2 μ s, and the admissible cutoff interval $4000\pi < \omega_c < 5000\pi$ rad/s is no longer empty.



At $\omega = \pm 4000\pi$ the baseband arrow and the arrow the neighbouring copy brings there have the same length and point in opposite directions.

7.6 Worked examples on the rate

EXAMPLE 7.2 – ONE SIGNAL AT THREE RATES

GIVEN $x(t) = (\sin \pi t / \pi t)^2$, whose transform is a triangle of peak 1 reaching zero at $\pm 2\pi$. So $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s.

FIND For $T_1 = 0.40$ s, $T_2 = 0.50$ s and $T_3 = 2/3$ s: the rate, the guard band and the height of each copy.

METHOD Use the sampling definitions because the question asks for the rate, separation, and scale of the copies: $\omega_s = 2\pi/T$, guard band $\omega_s - 2\omega_M$, and copy height $1/T$ times the original peak.

SOLUTION

$$\begin{aligned} T_1 = 0.40 : \quad \omega_s &= 5\pi, \quad 5\pi > 4\pi, \quad \text{guard} = +\pi, \quad 1/T_1 = 2.5 \\ T_2 = 0.50 : \quad \omega_s &= 4\pi, \quad 4\pi = 4\pi, \quad \text{guard} = 0, \quad 1/T_2 = 2.0 \\ T_3 = 2/3 : \quad \omega_s &= 3\pi, \quad 3\pi < 4\pi, \quad \text{guard} = -\pi, \quad 1/T_3 = 1.5 \end{aligned}$$

CHECK $\omega_s T = 2\pi$ in each row: $5\pi \times 0.4 = 2\pi$, $4\pi \times 0.5 = 2\pi$, $3\pi \times 2/3 = 2\pi$. Only the first row is safe; the second sits on the boundary and the third overlaps by π rad/s on each side.

EXAMPLE 7.3 – A LINE SPECTRUM, AND ITS SAMPLING PERIOD

GIVEN $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$.

FIND $X(j\omega)$, the bandwidth, and the sampling period at the Nyquist rate.

METHOD Use linearity because the signal is a sum of standard terms. Transform each term separately, identify the largest frequency, convert the resulting rate to a period, and check the units three ways.

SOLUTION

$$X(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 2000\pi) + \pi\delta(\omega + 2000\pi) + \frac{\pi}{j}\delta(\omega - 4000\pi) - \frac{\pi}{j}\delta(\omega + 4000\pi).$$

Five impulses, each written as a function of ω so that its position can be read. The furthest is at 4000π , so $\omega_M = 4000\pi$ rad/s, that is $f_M = 2000$ Hz, and the Nyquist rate is 8000π rad/s, that is 4000 Hz. Then

$$T = \frac{2\pi}{8000\pi} = \frac{1}{4000} \text{ s} = 2.5 \times 10^{-4} \text{ s} = 0.25 \text{ ms}.$$

CHECK Three independent tests. $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. The copies are scaled by $1/T = 4000$. And doubling the rate must halve the period: at $\omega_s = 16000\pi$ the period is $125 \mu\text{s}$, and 0.25 ms divided by $125 \mu\text{s}$ is exactly 2.

WARNING The answer is a quarter of a millisecond. Cancelling the π but not the thousand gives 0.25 s, and every later number in the problem is then wrong by a factor of 1000. A quarter of a second between samples would be four samples per second for a signal carrying 2000 Hz.

EXAMPLE 7.4 – AREA, PEAK, AND THE HEIGHT OF A COPY

GIVEN	$x(t) = (\sin(4000\pi t)/\pi t)^2$, the square of a signal whose transform is a rectangle of height 1 on $ \omega \leq 4000\pi$.
FIND	The peak of $X(j\omega)$, the bandwidth, the Nyquist period, and the height of one copy after sampling.
METHOD	Use the multiplication property because the signal is squared in time. The spectrum therefore convolves with itself and divides by 2π .
SOLUTION	$A = [R * R](0) = \int_{-4000\pi}^{4000\pi} (1)(1) d\tau = 8000\pi, \quad X_{\max} = X(0) = \frac{A}{2\pi} = 4000.$ The triangle reaches zero at twice the rectangle half-width, so $\omega_M = 8000\pi$ rad/s and $T = 2\pi/16000\pi = 1.25 \times 10^{-4}$ s, that is 125 μs. The copy height is $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$.
CHECK	This bandwidth is twice the one in Example 7.3, so this period is half of it. And $\omega_s T = 16000\pi \times 1.25 \times 10^{-4} = 2\pi$.
WARNING	A is an area and $X_{\max}$ is the peak of a spectrum. They differ by 2π , so substituting one where the other belongs inflates every later height by that factor. Give the peak its own symbol.

7.7 Reconstruction and band-limited interpolation

If the copies stand clear, recovering $x(t)$ means keeping the one at the origin and discarding the rest.

IDEAL RECONSTRUCTION FILTER

$$H_r(j\omega) = \begin{cases} T, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

$$X_r(j\omega) = T \cdot \frac{1}{T} X(j\omega) = X(j\omega) \implies x_r(t) = x(t)$$

The gain is T because the baseband is $X(j\omega)/T$. A filter of gain 1 returns a signal T times too small, and since T changes with the rate the error changes with it. The second line is true only when the copies do not overlap.

Filtering is convolution, and x_p is a train of impulses, so the output is a sum of shifted copies of the filter's impulse response. Each sample is replaced by a curve of its own height, and the curves are added.

BAND-LIMITED INTERPOLATION

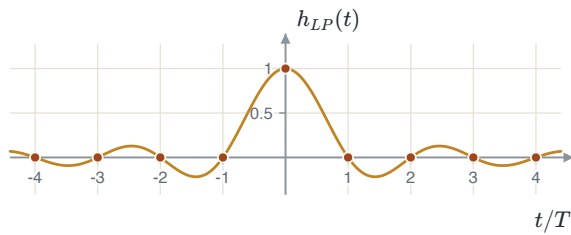
$$h_{LP}(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} T e^{j\omega t} d\omega = \frac{T \sin(\omega_c t)}{\pi t}$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT) \frac{T}{\pi} \cdot \frac{\sin(\omega_c(t - nT))}{t - nT}$$

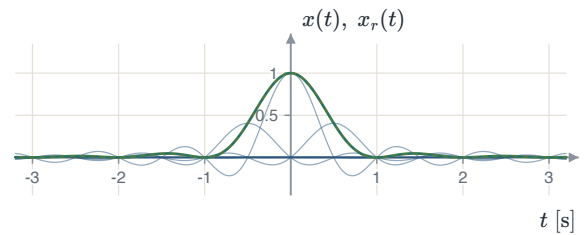
Throughout this course $\text{sinc}(\theta) = \sin \theta / \theta$, **unnormalised**. With the choice $\omega_c = \pi/T$ the kernel is $h_{LP}(t) = \sin(\pi t/T)/(\pi t/T) = \text{sinc}(\pi t/T)$. Anything written as $\text{sinc}(t/T)$ belongs to the other convention and is a different function.

KEEP THE FACTOR OF π INSIDE THE SINE

The kernel must equal 1 at $t = 0$ and 0 at every other sample instant. Then each sample sets its own value without changing the others. Removing π breaks this condition: with $T = 1$ the correct kernel at $t = 1$ is $\sin(\pi)/\pi = 0$, while $\sin(1)/\pi = 0.267849$. Also, $\omega_c = \pi/T$ equals $\omega_s/2$, which is an endpoint of the admissible interval rather than an interior point. State this cutoff as a choice.



The kernel with $\omega_c = \pi/T$: one at its own instant, zero at every other.



One kernel per sample, in grey, and their sum. It lands on the dashed original.

7.8 The zero-order and first-order holds

The ideal filter is not buildable: its impulse response starts before $t = 0$ and never ends. What hardware does instead is hold.

ZERO-ORDER HOLD

$$h_0(t) = \begin{cases} 1, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$$

$$H_0(j\omega) = \int_0^T e^{-j\omega t} dt = \frac{1 - e^{-j\omega T}}{j\omega} = e^{-j\omega T/2} \frac{2 \sin(\omega T/2)}{\omega}$$

The magnitude is $|2 \sin(\omega T/2)/\omega|$ and the phase is $-\omega T/2$, a delay of half a sampling period. At the origin $H_0(0) = T$, so the hold already supplies the gain the ideal filter needed.

The zero-order hold differs from the ideal filter in three ways. Its magnitude decreases below T as ω grows, so it changes the wanted spectrum. Its response is non-zero beyond the cutoff, so parts of neighbouring copies remain. Its phase also adds a delay. Correcting these effects would require the compensator

$$H_r(j\omega) = \frac{H(j\omega)}{H_0(j\omega)} = e^{j\omega T/2} H(j\omega) \frac{\omega}{2 \sin(\omega T/2)},$$

which has to advance in time and to grow without bound where $\sin(\omega T/2)$ vanishes. It is approximated in practice, not built.

FIRST-ORDER HOLD

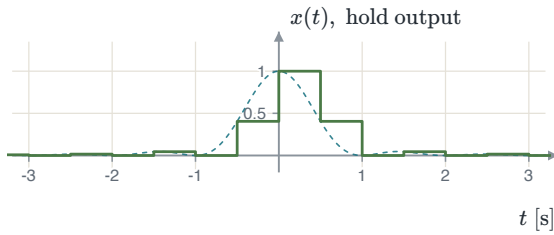
$$h_1(t) = \frac{1}{T} [g * g](t), \quad g(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$H_1(j\omega) = \frac{1}{T} G^2(j\omega) = \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$

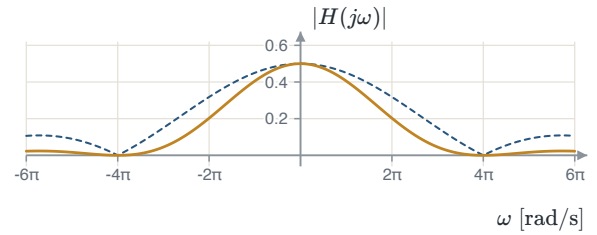
The convolution of the rectangle with itself is a triangle of peak T on $|t| \leq T$, so h_1 is a triangle of peak 1 on the same interval and the output joins consecutive samples by straight lines. Every one of these is a function of t ; the independent variable of all three sketches is time. H_1 is real and non-negative, $H_1(0) = T$, and it falls off as $1/\omega^2$ instead of $1/\omega$.

A HOLD IS NOT A RECONSTRUCTION

Both holds are filters with a real, non-flat, non-band-limited response. The staircase is a different signal from $x(t)$, and no argument about small steps makes them equal: raising the rate makes the error smaller but never zero at any finite rate. The sampling theorem guarantees that the samples *determine* $x(t)$ and names the filter that recovers it. Feeding the same samples to a different filter gives a different output.



The zero-order hold: each sample held for one period.



With $T = 0.5$ s: the first-order hold, solid, against the zero-order hold, dashed. Both start at T .

7.9 What comes back, and where from

Take $x(t) = \cos(\omega_0 t)$, so $\omega_M = \omega_0$, and reconstruct with the cutoff at $\omega_c = \omega_s/2$. Sampling puts a line at every $k\omega_s \pm \omega_0$, and the filter keeps whatever lies in $|\omega| < \omega_c$.

$$\begin{aligned} \omega_s = 6\omega_0 : \quad \omega_c = 3\omega_0, \quad \text{inside} : \pm\omega_0 &\Rightarrow x_r(t) = \cos(\omega_0 t) \\ \omega_s = 3\omega_0 : \quad \omega_c = 1.5\omega_0, \quad \text{inside} : \pm\omega_0 &\Rightarrow x_r(t) = \cos(\omega_0 t) \\ \omega_s = 1.5\omega_0 : \quad \omega_c = 0.75\omega_0, \quad \text{inside} : \pm 0.5\omega_0 &\Rightarrow x_r(t) = \cos(0.5\omega_0 t) \end{aligned}$$

THE ALIAS FREQUENCY

$$X_r(j\omega) = \pi\delta(\omega - (\omega_s - \omega_0)) + \pi\delta(\omega + (\omega_s - \omega_0)) \xrightarrow{\mathcal{F}^{-1}} x_r(t) = \cos((\omega_s - \omega_0)t)$$

In the third case the line at ω_0 lies outside the filter and is discarded, while the line the $k = 1$ copy contributes at $\omega_s - \omega_0$ lies inside and is kept. The signal takes the identity of a lower frequency, $|\omega_s - \omega_0|$. Each impulse is written as a function of ω ; an expression such as $\pi\delta(\omega_s - \omega_0)$ is a constant and locates nothing. The arrow is the inverse transform, going from a spectrum back to a signal.

EXAMPLE 7.5 – THREE PERIODS FOR ONE COSINE

GIVEN $x(t) = \cos(2\pi t)$, so $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s. Take $\omega_c = \omega_s/2$.

FIND Whether $T_1 = 1/4$ s, $T_2 = 1/3$ s and $T_3 = 2/3$ s recover the signal, and what is recovered when one does not.

METHOD Use the sampled line spectrum because the filter selects individual lines. Compute $\omega_s = 2\pi/T$, compare it with 4π , list all line positions, and keep only those inside the cutoff.

SOLUTION

$$T_1 = 1/4 : \quad \omega_s = 8\pi > 4\pi, \quad \omega_c = 4\pi, \quad x_r(t) = \cos(2\pi t)$$

$$T_2 = 1/3 : \quad \omega_s = 6\pi > 4\pi, \quad \omega_c = 3\pi, \quad x_r(t) = \cos(2\pi t)$$

$$T_3 = 2/3 : \quad \omega_s = 3\pi < 4\pi, \quad \omega_c = 1.5\pi, \quad x_r(t) = \cos(\pi t)$$

CHECK $\cos(2\pi n T_3) = \cos(4\pi n/3)$ and $\cos(\pi n T_3) = \cos(2\pi n/3)$. Since $4\pi n/3 = 2\pi n - 2\pi n/3$ and the cosine is even, the two sample sequences are identical, so no reconstruction could prefer one over the other.

WARNING The whole verdict rests on the number the rate is compared with, and for this signal that number is $2\omega_M = 4\pi$. Comparing against a neighbouring problem's bandwidth of 6π makes the middle row read $6\pi \geq 6\pi$, an equality, so a comfortable margin is turned into the boundary case the theorem excludes.

EXAMPLE 7.6 – TWO COMPONENTS, ONE RATE

GIVEN $x(t) = \cos(\pi t) + \cos(3\pi t)$, sampled with $T = 2/5$ s.

FIND $x_r(t)$, with $\omega_c = \omega_s/2$.

METHOD Treat the components separately because one can alias while the other remains unchanged. First test the rate against this signal's Nyquist rate, then track each baseband and copy line through the filter.

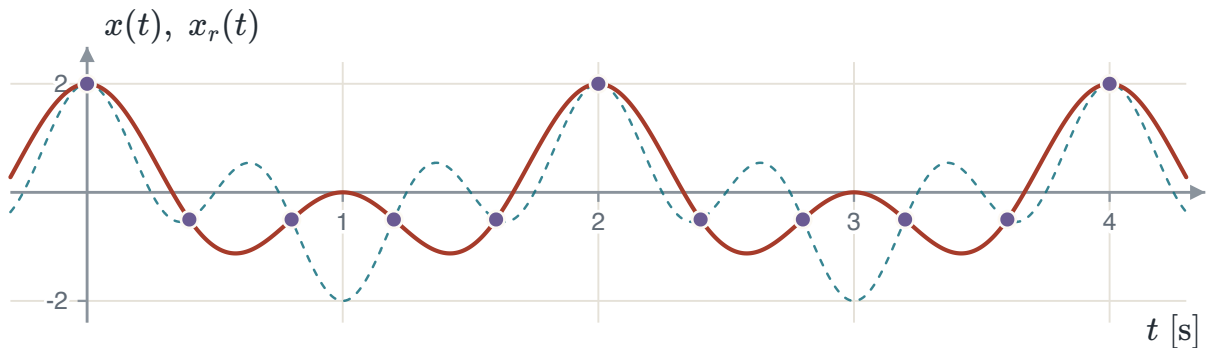
SOLUTION $\omega_M = 3\pi$, so the Nyquist rate is 6π rad/s, while $\omega_s = 2\pi/T = 5\pi$ rad/s and $\omega_c = 2.5\pi$ rad/s. Since $5\pi < 6\pi$, aliasing is expected. From the component at π : the baseband line at π is inside and survives, the copy line at 4π is outside. From the component at 3π : the baseband line is outside, the copy line at $5\pi - 3\pi = 2\pi$ is inside. Hence

$$X_r(j\omega) = \pi[\delta(\omega - \pi) + \delta(\omega + \pi)] + \pi[\delta(\omega - 2\pi) + \delta(\omega + 2\pi)]$$

and $x_r(t) = \cos(\pi t) + \cos(2\pi t)$.

CHECK $\cos(3\pi nT) = \cos(6\pi n/5)$ and $\cos(2\pi nT) = \cos(4\pi n/5)$, and $6\pi n/5 = 2\pi n - 4\pi n/5$, so the two sequences agree.

INTERPRETATION The component at π remains unchanged. The component at 3π appears at 2π , a frequency that was not present in the original signal.



Both curves pass through every sample. The solid one is what the filter returns.

7.10 The anti-aliasing filter

A real signal is not band-limited, and a sampler at a fixed rate cannot be helped after the fact. The repair is a lowpass filter placed **before** the sampler, which removes everything above $\omega_s/2$ while it can still be removed cleanly.



The chain is in the only order that works. The sampler output is $x_p(t)$; the name $x_r(t)$ belongs to the output of the reconstruction filter and to nothing before it.

PLACE THE FILTER BEFORE THE SAMPLER

After sampling, aliased high-frequency content has already been added to wanted content at the same frequency. A later filter cannot separate those contributions. The anti-aliasing filter must remove the high frequencies before the sampler.

The comparison is quantitative. Take $x(t) = \cos(\pi t) + \cos(3\pi t)$ at $T = 2/5$ s, as in Example 7.6. Without the filter, $x_r(t) = \cos(\pi t) + \cos(2\pi t)$ and the error is $e(t) = \cos(3\pi t) - \cos(2\pi t)$. With a lowpass at $\omega_c = 2.5\pi$ ahead of the sampler, the component at 3π is removed first, what is sampled is $\cos(\pi t)$ alone, it is recovered exactly, and the error is $e(t) = \cos(3\pi t)$.

$$\overline{e^2}\bigg|_{\text{no filter}} = \frac{1}{2} + \frac{1}{2} = 1, \quad \overline{e^2}\bigg|_{\text{filtered}} = \frac{1}{2}$$

The two cosines in the unfiltered error sit at different harmonics of π rad/s, so their cross term averages to zero and the powers add. Filtering first halves the error power. It costs the component at 3π , which was lost either way; what it buys is the removal of an impostor at 2π , a frequency the signal never contained. Losing content is a known, bounded error. Adding content that was never there is not, because nothing downstream can tell it apart from the signal.

NO FINITE-DURATION SIGNAL IS BAND-LIMITED

A rectangular pulse of width $2T_1$ has transform $2 \sin(\omega T_1)/\omega$, which crosses zero again and again but is never zero on a whole interval. There is no ω_M beyond which it vanishes, so no rate makes the copies disjoint and some aliasing is present at every rate. What is done in practice is to impose the band limit with a filter, and then to reconstruct the *filtered* signal exactly. The difference from the original is a design decision with a known size, not an accident.

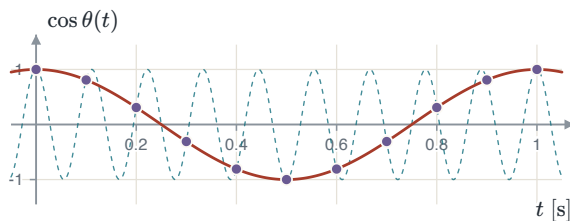
7.11 Aliasing in time and in space

A camera is a sampler with $T = 1/f_s$, and a strobe lamp is the same device built from light. Nothing about the rule changes.

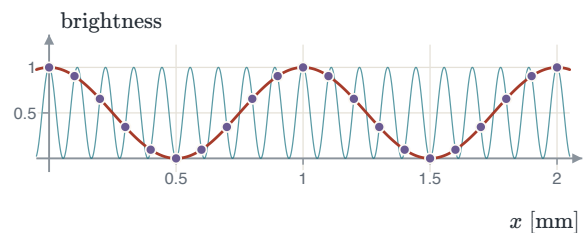
EXAMPLE 7.7 – A WHEEL THAT TURNS BACKWARDS

GIVEN	A marked spoke turns at 9 revolutions per second. A camera records 10 frames per second.
FIND	The rotation the recording appears to show.
METHOD	Model the spoke angle as a sinusoid because the camera records it at regular time instants. The angle has $\omega_0 = 18\pi$ rad/s and the camera has $\omega_s = 20\pi$ rad/s, so compare these rates and locate the surviving line.
SOLUTION	$\omega_s = 20\pi < 2\omega_0 = 36\pi$, so the rate is below the Nyquist rate and the surviving line sits at $ \omega_s - \omega_0 = 2\pi$ rad/s, that is 1 revolution per second. The surviving contribution comes from the copy of the negative-frequency component, so the apparent rotation is in the opposite direction.
CHECK	$\cos(18\pi n/10) = \cos(2\pi n - 2\pi n/10) = \cos(2\pi n/10)$: the frames are exactly those a 1 Hz rotation would produce. Between frames the spoke advances 9/10 of a turn, which the eye reads as 1/10 of a turn backwards. At 20 frames per second, $40\pi > 36\pi$ and the recording is correct.

Position takes the place of time without any change to the arithmetic. A striped pattern of 9 cycles per millimetre recorded on a grid of 10 sample points per millimetre satisfies $2 \times 9 > 10$, so it is recorded as a pattern of $|10 - 9| = 1$ cycle per millimetre. The broad bands the grid produces are smooth, regular and convincing, and nothing in the samples marks them as false. Two regular patterns laid over one another give the same effect without any sampler: stripes at 1.0 mm and at 1.1 mm have spatial frequencies 1.000 and 0.909 cycles per millimetre, so the pattern they make together repeats every 11 mm.



Nine revolutions per second, dashed, ten frames per second as dots, and the one revolution per second the frames describe.



The same arithmetic in space: nine cycles per millimetre recorded as one.

7.12 Summary

1. Find ω_M , the highest angular frequency the signal carries. If there is none, the signal is not band-limited and must be filtered before anything else.
2. Choose ω_s strictly above $2\omega_M$, leaving a guard band $\omega_s - 2\omega_M$ wide enough for a real filter.
3. Compute $T = 2\pi/\omega_s$ and check $\omega_s T = 2\pi$. Quote $f_s = 1/T$ in hertz only when hertz was asked for.
4. Draw the copies at every multiple of ω_s , each $1/T$ times the original height, and mark the baseband $k = 0$.

5. Place the reconstruction filter: gain T , cutoff strictly inside $\omega_M < \omega_c < \omega_s - \omega_M$.
6. If the copies overlap, say which line came from which copy and give the alias frequency $|\omega_s - \omega_0|$ for each component that moved.

THE FOUR TRAPS, IN ONE PLACE

Saying the copies disappear below the Nyquist rate — they never do, they overlap. Treating $\omega_s = 2\omega_M$ as safe — the admissible cutoff interval is empty there. Reading $2\pi/T$ in hertz — that is rad/s, and the error is a factor of 2π . Calling a hold output the reconstructed signal — a hold is an approximation with a measurable error.

When the signal is band-limited and the sampling rate satisfies the strict condition, its samples determine it uniquely. This result allows the earlier continuous-time operations to be represented and processed with discrete samples.

7.13 Exercises

7.1 A signal is sampled at $\omega_s = 8000\pi$ rad/s. Give T and f_s , and verify the period with a unit check.

Answer: $T = 2\pi/\omega_s = 1/4000$ s = 0.25 ms and $f_s = 1/T = 4000$ Hz. Check: $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. Note that 0.25 s fails the same check by a factor of 1000.

7.2 State in one sentence each what sampling does at every rate, and what happens only below the Nyquist rate.

Answer: At every rate sampling puts a copy of $X(j\omega)$ at every multiple of ω_s , scaled by $1/T$. Below the Nyquist rate those copies overlap and add, and that overlap is aliasing.

7.3 Show that the sampling theorem cannot be satisfied at $\omega_s = 2\omega_M$, and give a signal that demonstrates it.

Answer: The cutoff condition $\omega_M < \omega_c < \omega_s - \omega_M$ becomes $\omega_M < \omega_c < \omega_M$, which is empty. For $x(t) = \sin(\omega_M t)$ sampled at $\omega_s = 2\omega_M$ every sample is $\sin(\pi n) = 0$: the baseband and the $k = 1$ copy contribute equal and opposite weights at $\pm\omega_M$.

7.4 A triangular spectrum has peak $X_{\max} = 4000$ and is sampled with $T = 125$ μ s. Find the height of one copy, and say why the area $A = 8000\pi$ is not the number to use.

Answer: $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$. The area of the convolution and the peak of the spectrum differ by 2π , since squaring in time convolves in frequency and divides by 2π ; using A inflates every height by that factor.

7.5 $x(t) = \cos(\pi t) + \cos(3\pi t)$ is sampled with $T = 2/5$ s and reconstructed with $\omega_c = \omega_s/2$. Find $x_r(t)$ and name the component that moved.

Answer: $\omega_s = 5\pi < 6\pi$, so aliasing occurs. Surviving lines: π from the baseband and $5\pi - 3\pi = 2\pi$ from the $k = 1$ copy, giving $x_r(t) = \cos(\pi t) + \cos(2\pi t)$. The component at 3π moved to 2π ; the one at π passed untouched.

7.6 Explain why the anti-aliasing filter must precede the sampler, and state what the output of the whole chain is equal to.

Answer: After the sampler the intruding content has been added to the wanted content at the same frequency, and a filter cannot recover two numbers from their sum. Placed first, the filter removes that content while it is still at its own frequency. The output then equals the filtered signal exactly, not the original, and the difference is set by the chosen cutoff.

APPENDIX A

Summary of formulas

A.1 Energy and power

QUANTITY	CONTINUOUS TIME	DISCRETE TIME
Total energy	$E_\infty = \int_{-\infty}^{\infty} x(t) ^2 dt$	$E_\infty = \sum_{n=-\infty}^{\infty} x[n] ^2$
Average power	$P_\infty = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt$	$P_\infty = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x[n] ^2$
Energy signal	$E_\infty < \infty, P_\infty = 0$	same
Power signal	$E_\infty \rightarrow \infty, 0 < P_\infty < \infty$	same

A.2 Operations and periodicity

ITEM	STATEMENT
Shift	$x(t - t_0)$: delay if $t_0 > 0$, advance if $t_0 < 0$
Reversal	$x(-t)$, $x[-n]$
Scaling	$y(t) = x(at)$: compressed if $a > 1$, stretched if $0 < a < 1$; support divided by a
Combination	$x(at - b)$: shift by b first, then scale by a
Periodicity	$x(t) = x(t + T)$; $x[n] = x[n + N]$ with N an integer
Fundamental frequency	$\omega_0 = 2\pi/T_0 = 2\pi/N_0$
Even and odd	$\mathcal{E}\mathbf{v}\{x\} = \frac{1}{2}[x(t) + x(-t)]$, $\mathcal{O}\mathbf{d}\{x\} = \frac{1}{2}[x(t) - x(-t)]$

A.3 Impulses and steps

ITEM	CONTINUOUS TIME	DISCRETE TIME
Step and impulse	$\delta(t) = \frac{d}{dt}u(t)$, $u(t) = \int_{-\infty}^t \delta(\tau)d\tau$	$\delta[n] = u[n] - u[n - 1]$, $u[n] = \sum_{k=0}^{\infty} \delta[n - k]$
Sampling	$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$	$x[n]\delta[n - n_0] = x[n_0]\delta[n - n_0]$
Sifting	$x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt$	$x[n_0] = \sum_n x[n]\delta[n - n_0]$
Representation	$x(t) = \int x(\tau)\delta(t - \tau)d\tau$	$x[n] = \sum_k x[k]\delta[n - k]$

A.4 Complex exponentials

ITEM	STATEMENT
Continuous time	$x(t) = Ce^{at}$; with $C = Ae^{j\theta}$ and $a = r + j\omega_0$: $x(t) = Ae^{rt}e^{j(\omega_0 t + \theta)}$
Period	$T_0 = 2\pi/\omega_0$, always periodic for $\omega_0 \neq 0$
Discrete time	$x[n] = C\alpha^n$ with $\alpha = e^{\beta}$; growth boundary at $ \alpha = 1$
Period	$N = \frac{2\pi}{\omega_0}k$; periodic only if $\omega_0/2\pi \in \mathbb{Q}$
Frequency wrap-around	$e^{j(\omega_0 + 2\pi)n} = e^{j\omega_0 n}$

A.5 Systems and convolution

PROPERTY	GENERAL CRITERION	LTI CRITERION IN TERMS OF h
Memoryless	output at t uses only input at t	$h(t) = a\delta(t)$ or $h[n] = a\delta[n]$
Invertible	distinct inputs give distinct outputs	$h * g = \delta$ for some g
Causal	output uses only $\tau \leq t$	$h(t) = 0$ for $t < 0$; $h[n] = 0$ for $n < 0$
BIBO stable	bounded input gives bounded output	$\int h dt < \infty$; $\sum_k h[k] < \infty$
Time invariant	$x(t - t_0) \rightarrow y(t - t_0)$	—
Linear	$ax_1 + bx_2 \rightarrow ay_1 + by_2$	—

CONVOLUTION

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) \, d\tau$$

Convolution is commutative, distributive, and associative. The output support is the sum of the input supports. The total areas, or the total sums in discrete time, multiply.

A.7 Fourier series properties

A periodic signal of period T_0 with $\omega_0 = 2\pi/T_0$, or a periodic sequence of period N with $\omega_0 = 2\pi/N$. In both columns $x \leftrightarrow a_k$ and $y \leftrightarrow b_k$.

PROPERTY	CONTINUOUS TIME	DISCRETE TIME
Linearity	$Ax(t) + By(t) \leftrightarrow Aa_k + Bb_k$	$Ax[n] + By[n] \leftrightarrow Aa_k + Bb_k$
Time shift	$x(t - t_0) \leftrightarrow a_k e^{-jk\omega_0 t_0}$	$x[n - n_0] \leftrightarrow a_k e^{-jk(2\pi/N)n_0}$
Frequency shift	$e^{jM\omega_0 t} x(t) \leftrightarrow a_{k-M}$	$e^{jM(2\pi/N)n} x[n] \leftrightarrow a_{k-M}$
Conjugation	$x^*(t) \leftrightarrow a_{-k}^*$	$x^*[n] \leftrightarrow a_{-k}^*$
Time reversal	$x(-t) \leftrightarrow a_{-k}$	$x[-n] \leftrightarrow a_{-k}$
Time scaling	$x(\alpha t), \alpha > 0: a_k, \text{ period } T_0/\alpha$	$x_{(m)}[n]: \frac{1}{m}a_k, \text{ period } mN$
Periodic convolution	$\int_{T_0} x(\tau)y(t - \tau)d\tau \leftrightarrow T_0 a_k b_k$	$\sum_{r=\langle N \rangle} x[r]y[n - r] \leftrightarrow N a_k b_k$
Multiplication	$x(t)y(t) \leftrightarrow \sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$	$x[n]y[n] \leftrightarrow \sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Derivative or difference	$dx/dt \leftrightarrow jk\omega_0 a_k$	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-jk(2\pi/N)})a_k$
Integral or running sum	$\int_{-\infty}^t x d\tau \leftrightarrow \frac{a_k}{jk\omega_0}, \text{ only if } a_0 = 0$	$\sum_{r=-\infty}^n x[r] \leftrightarrow \frac{a_k}{1 - e^{-jk(2\pi/N)}}, \text{ only if } a_0 = 0$
Real signal	$a_k = a_{-k}^*; a_k \text{ even, } \angle a_k \text{ odd}$	same
Real and even	a_k real and even	same
Real and odd	a_k purely imaginary and odd	same
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{a_k\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{a_k\}$	same
Parseval	$\frac{1}{T_0} \int_{T_0} x ^2 dt = \sum_k a_k ^2$	$\frac{1}{N} \sum_{n=\langle N \rangle} x ^2 = \sum_{k=\langle N \rangle} a_k ^2$

The two columns differ in three rows because discrete-time coefficients repeat, $a_k = a_{k+N}$. For multiplication, the discrete-time sum therefore covers one period instead of all integers. Time expansion carries the factor $\frac{1}{m}$ because the average uses m times as many samples. Discrete time also uses a first difference where continuous time uses a derivative. At $k = 0$, integration or accumulation requires zero mean; otherwise the result grows without bound.

A.8 Continuous-time Fourier transform

PROPERTY	STATEMENT
Linearity	$ax_1(t) + bx_2(t) \leftrightarrow aX_1(j\omega) + bX_2(j\omega)$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time and frequency scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Convolution	$x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$
Multiplication	$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Differentiation in time	$\frac{d^n x}{dt^n} \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j \frac{dX(j\omega)}{d\omega}$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Real signal	$X(-j\omega) = X^*(j\omega); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(j\omega) \text{ real and even}$
Real and odd	$X(j\omega) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$
SIGNAL	TRANSFORM
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}$
$te^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^2}$
$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^n}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
1 on $ t < T_1$, 0 beyond	$\frac{2 \sin(\omega T_1)}{\omega} = 2T_1 \text{sinc}(\omega T_1)$
$\frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt)$	1 on $ \omega < W$, 0 beyond
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$
$\sin \omega_0 t$	$\frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$
$\sum_k a_k e^{jk\omega_0 t}$	$\sum_k 2\pi a_k \delta(\omega - k\omega_0)$
Square wave: 1 on $ t < T_1$, 0 on $T_1 < t \leq T/2$	$\sum_k \frac{2 \sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$
$\sum_k \delta(t - kT)$	$\frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$

This table uses the unnormalised definition $\text{sinc}(\theta) = \sin \theta / \theta$. The normalised convention divides the argument by π . Convert the argument when moving between conventions; otherwise the result loses a factor of π . The exponential pairs require $a > 0$.

A.9 Discrete-time Fourier transform

PROPERTY	STATEMENT
Linearity	$ax_1[n] + bx_2[n] \leftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$
Real sequence	$X(e^{-j\omega}) = X^*(e^{j\omega}); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(e^{j\omega}) \text{ real and even}$
Real and odd	$X(e^{j\omega}) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$
SEQUENCE	TRANSFORM
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \pi \sum_k \delta(\omega - 2\pi k)$
$a^n u[n], a < 1$	$\frac{1}{1 - ae^{-j\omega}}$
$(n + 1)a^n u[n], a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{(n+r-1)!}{n!(r-1)!} a^n u[n], a < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$
$a^{ n }, a < 1$	$\frac{1 - a^2}{1 - 2a \cos \omega + a^2}$
1 on $ n \leq N_1$, 0 beyond	$\frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}$
$\frac{\sin(Wn)}{\pi n}, 0 < W < \pi$	1 on $ \omega \leq W$, 0 on $W < \omega \leq \pi$
1	$2\pi \sum_k \delta(\omega - 2\pi k)$
$e^{j\omega_0 n}$	$2\pi \sum_k \delta(\omega - \omega_0 - 2\pi k)$
$\cos \omega_0 n$	$\pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$
$\sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_k a_k \delta(\omega - \frac{2\pi k}{N})$
$\sum_k \delta[n - kN]$	$\frac{2\pi}{N} \sum_k \delta(\omega - \frac{2\pi k}{N})$

Every transform in this section is periodic in ω with period 2π because the time index is discrete. A periodic sequence has discrete spectral coefficients, while an aperiodic sequence has a continuous spectrum. In each case, discreteness in one domain produces periodicity in the other.

A.10 Symbols

SYMBOL	MEANING
$x(t), x[n]$	input signal, continuous and discrete time
$y(t), y[n]$	output signal
$h(t), h[n]$	impulse response of a linear time-invariant system
$\delta(t), \delta[n]$	unit impulse
$u(t), u[n]$	unit step
E_∞, P_∞	total energy and average power over all time
T_0, N_0	fundamental period
ω_0	fundamental angular frequency, rad/s or rad/sample
$*$	convolution
j	imaginary unit, $j^2 = -1$